

Bi-conformal scalar-tensor gravity: exact PPN equivalence with general relativity and falsifiable predictions

(Complete theory with appendices)

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Abstract

This document presents the complete Induced Scalar-Pressure Gravity (ISPG) theory, including the main theoretical framework and all supporting appendices. The theory is defined by a scalar-tensor action with bi-conformal metric ansatz, yielding exact post-Newtonian equivalence with general relativity at 1PN order while producing falsifiable deviations at 2PN order. The emergent MOND phenomenology is developed in a companion paper.

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Part I

Main Theory

1 Introduction

The nature of gravity remains one of the central open problems in theoretical physics. General relativity (GR), formulated by Einstein in 1915 [1], describes gravitation as the curvature of spacetime and has been confirmed to extraordinary precision in the solar system [2, 3] and in strong-field regimes through gravitational-wave observations [4, 5]. Yet GR faces persistent challenges on both the largest and the smallest scales.

On galactic scales, the observed flatness of rotation curves and the tight baryonic Tully–Fisher relation [6, 7] require either a dominant component of non-baryonic dark matter or a modification of gravitational dynamics. Milgrom’s Modified Newtonian Dynamics (MOND) [8] successfully captures this phenomenology through a single empirical acceleration scale $a_0 \approx 1.2 \times 10^{-10} \text{ m s}^{-2}$, whose numerical coincidence with cH_0 has remained unexplained [9, 10]. Relativistic extensions of MOND—notably Bekenstein’s Tensor-Vector-Scalar theory (TeVeS) [12] and various $f(R)$ formulations [13]—have encountered difficulties with gravitational-wave speed, CMB compatibility, or internal consistency [14].

On the theoretical side, GR predicts singularities under generic conditions [15, 16], and the reconciliation of gravity with quantum mechanics remains incomplete [17, 18].

Scalar-tensor theories constitute the most natural and best-studied extension of GR. The general framework was established by Brans and Dicke [19] and classified in full generality by Horndeski [20], who identified the most general four-dimensional scalar-tensor action yielding second-order field equations. The recent Bellini–Sawicki parameterization [21] provides a unified language for confronting any Horndeski theory with cosmological observations through four time-dependent functions $\{\alpha_K, \alpha_B, \alpha_M, \alpha_T\}$.

In this paper we present a scalar-tensor theory that is unusually constrained: it is defined by a single action with no free parameters beyond G , and a metric ansatz that leaves no room for tuning. Despite this rigidity, the theory simultaneously:

1. reproduces all post-Newtonian solar-system tests ($\gamma = \beta = 1$) at 1PN order, and predicts a structurally distinct 2PN Shapiro-delay signature (arising from the exponential bi-conformal refractive index); the exact 2PN coefficient awaits a full finite-endpoint derivation and is flagged as future work in Sec. 4;
2. is fully compatible with the gravitational-wave speed constraint $|c_g - c|/c < 10^{-15}$ from GW170817 [5], predicting two tensor polarizations plus a suppressed scalar breathing mode;
3. resolves the Schwarzschild singularity at the classical level, yielding a singularity-free, horizonless compact object with the innermost stable circular orbit at $r_{\text{ISCO}} = \varphi_{\text{gr}}^2 r_s$, where φ_{gr} is the golden ratio;
4. produces an identical CMB angular power spectrum to Λ CDM at linear order, with all Bellini–Sawicki parameters vanishing on the cosmological background;

5. proves the scaling $a_0 \propto c H_0$ from the damped-oscillator coherence length, and within the mature spiral-galaxy closure of the companion MOND paper [46] fixes the normalization $1/(2\pi)$ together with the interpolating function $\mu(x) = x/(1+x)$: the local Bernoulli pressure-deficit channel plus the vortex-transport channel. The baryonic Tully–Fisher relation, the prediction $a_0(z) = c H(z)/(2\pi)$, and a unique vortex-lag mechanism that resolves the 15% a_0 offset and predicts galaxy-age-dependent a_0 all follow from this derivation.

The theory belongs to the Horndeski class and is defined by

$$S = \frac{1}{16\pi G} \int d^4x \sqrt{-g} \left[R + \frac{1}{2} g^{\mu\nu} \partial_\mu \varphi \partial_\nu \varphi \right] + S_m, \quad (1)$$

where φ is a dimensionless scalar field and S_m is the matter action. The static metric takes the bi-conformal form $ds^2 = -e^\varphi c^2 dt^2 + e^{-\varphi} (dx^2 + dy^2 + dz^2)$, which uniquely determines the phantom kinetic sign $(+\frac{1}{2})$. The theory grew out of a physical picture in which the scalar field is interpreted as a medium—an “information substance”—whose pressure gradients generate gravitational, inertial, and galactic-rotation phenomena. The original working name, *Information Substance Pressure Gradient* (ISPG) [45], reflects this intuition. In the present paper every result is derived within the standard scalar-tensor formalism; the name ISPG is retained only as an identifier. The quantum extension [47] makes the intuition precise: the Bernoulli identity of the scalar-metric medium produces a static pressure $P_{\text{static}} = -e^\varphi \varphi^4 / (32\pi G r_s^2)$ whose gradient is identically the gravitational force—answering *why* mass curves spacetime.

The paper is organized as follows. Section 2 derives the action, the field equations, and the Horndeski classification. Section 3 establishes weak-field correspondence with GR through the PPN formalism, light deflection, perihelion precession, and frame dragging. Section 4 presents the Shapiro time delay at 1PN order (where ISPG matches GR exactly) and sketches the structural 2PN signature induced by the bi-conformal exponential refractive index, while flagging the complete 2PN coefficient as open (future work). Section 5 analyzes gravitational-wave propagation, polarization content, scalar dipole radiation from mixed-compactness binaries, and observational constraints. Section 6 addresses the stability of the phantom kinetic term. Section 7 derives the singularity-free compact object solution, the ISCO, photon sphere, and geodesic structure. Section 8 computes the Bellini–Sawicki parameters and establishes CMB compatibility. Section 9 derives the MOND phenomenology from cosmological perturbations. Section 10 develops the physical interpretation of inertia as scalar-field backreaction, derives the identity of gravitational and inertial mass from the field equations, establishes the position-dependent effective mass $m_{\text{eff}} = m_0 e^{\varphi/2}$ and its local undetectability, and explains the local invariance of the speed of light through the bi-conformal structure. Section 11 discusses the strong equivalence principle, limitations, open questions, and future directions. Section 12 summarizes the results.

Throughout we use the metric signature $(-, +, +, +)$ and define the reduced Planck mass as $M_{\text{Pl}}^2 \equiv (8\pi G)^{-1}$. Factors of c are retained explicitly in all equations.

2 The action and field equations

2.1 Bi-conformal metric ansatz

We begin with the physical requirement that the scalar field φ controls both the rate of local clocks and the length of local rods in a reciprocal manner: a region of lower φ experiences slower

time flow and correspondingly expanded spatial scales. The most general static, isotropic line element satisfying this reciprocity with a single scalar degree of freedom is the *bi-conformal* metric

$$ds^2 = -e^\varphi c^2 dt^2 + e^{-\varphi} (dx^2 + dy^2 + dz^2), \quad (2)$$

where the temporal and spatial conformal factors are inverse powers of e^φ . This structure is distinguished from a purely conformal rescaling $g_{\mu\nu} = e^\varphi \eta_{\mu\nu}$, which would yield $\gamma_{\text{PPN}} \neq 1$ and conflict with the Cassini bound $|\gamma - 1| < 2.3 \times 10^{-5}$ [22]. The bi-conformal form enforces equal but opposite contributions to the temporal and spatial sectors, which—as we show in Sec. 3—is the unique choice producing $\gamma = \beta = 1$.

2.2 The action

Given the metric (2), the gravitational dynamics are governed by the Einstein–Hilbert term supplemented by a kinetic term for φ . The action is

$$S = \frac{1}{16\pi G} \int d^4x \sqrt{-g} \left[R + \frac{1}{2} g^{\mu\nu} \partial_\mu \varphi \partial_\nu \varphi \right] + S_{\text{m}}[g_{\mu\nu}, \psi], \quad (3)$$

where R is the Ricci scalar of $g_{\mu\nu}$, S_{m} is the matter action with matter fields ψ minimally coupled to the metric, and the factor $1/(16\pi G)$ sets the overall normalization.

The kinetic sign $+\frac{1}{2}$ (phantom) is not a free choice. One might naturally begin with the canonical sign $-\frac{1}{2}$, as the author initially did; however, substituting the ansatz (2) into the Einstein tensor and comparing the resulting $G_{\mu\nu}$ component by component with the scalar energy-momentum tensor yields a unique value $\kappa = -1/2$ for the coupling constant—the phantom sign. The canonical alternative $\kappa = +1/2$ produces an overdetermined, inconsistent system. This is verified component by component in the repository script `tools/check_kappa_consistency.py`, which for a general static spherically symmetric $\varphi(r)$ shows that the $\kappa = -1/2$ system reduces to the single ODE $r\varphi''(r) + 2\varphi'(r) = 0$ (the vacuum equation 105 of Appendix 1), while the $\kappa = +1/2$ system leaves three mutually incompatible conditions on $\varphi(r)$. The stability implications of the phantom sign are the subject of Sec. 6.

2.3 Field equations

Variation of (3) with respect to $g^{\mu\nu}$ yields the modified Einstein equations

$$G_{\mu\nu} = -\frac{1}{2} \tau_{\mu\nu}^{(\varphi)} + 8\pi G T_{\mu\nu}^{(\text{m})}, \quad (4)$$

where the scalar-field energy-momentum tensor is

$$\tau_{\mu\nu}^{(\varphi)} = \partial_\mu \varphi \partial_\nu \varphi - \frac{1}{2} g_{\mu\nu} g^{\alpha\beta} \partial_\alpha \varphi \partial_\beta \varphi, \quad (5)$$

and $T_{\mu\nu}^{(\text{m})}$ is the matter energy-momentum tensor.

Direct variation of (3) with respect to φ , with matter minimally coupled to $g_{\mu\nu}$ so that $\delta S_{\text{m}}/\delta\varphi \equiv 0$, gives the vacuum scalar equation

$$\square\varphi = 0, \quad (6)$$

where $\square \equiv g^{\mu\nu} \nabla_\mu \nabla_\nu$ is the covariant d'Alembertian (derivation in Appendix 1, eq. 105). The sourced form used throughout the main theory,

$$\square\varphi = -\frac{8\pi G}{c^4} T, \quad (7)$$

where $T \equiv g^{\mu\nu} T_{\mu\nu}^{(m)}$ is the trace of the matter tensor, is *not* an independent Euler–Lagrange equation. It arises as a consistency condition obtained by tracing (4) after substituting the bi-conformal ansatz $g_{\mu\nu}[\varphi]$ (Appendix 3); it is the reduced-sector statement that the ten components of (4) collapse to a single scalar equation sourced by T . In matter-free regions ($T = 0$) Eq. (7) coincides with Eq. (6), so the weak-field exterior used in PPN, Shapiro, and lensing analyses (Appendices 4, 5, 7) is unaffected by the distinction.

The system (4)–(7) propagates $2 + 1 = 3$ degrees of freedom: two tensor (TT, transverse-traceless) polarizations of the metric and one scalar mode from φ .

2.4 Horndeski classification

The action (3) belongs to the Horndeski class [20], the most general scalar-tensor theory with second-order field equations in four dimensions. The Horndeski action is specified by four free functions $G_2(\varphi, X)$, $G_3(\varphi, X)$, $G_4(\varphi, X)$, $G_5(\varphi, X)$, where $X \equiv -\frac{1}{2} g^{\mu\nu} \partial_\mu \varphi \partial_\nu \varphi$.

For the action (3), with $\frac{1}{2}(\partial\varphi)^2 = -X$, the identification is

$$\boxed{\begin{aligned} G_2 &= -\frac{M_{\text{Pl}}^2}{2} X, & G_3 &= 0, \\ G_4 &= \frac{M_{\text{Pl}}^2}{2}, & G_5 &= 0, \end{aligned}} \quad (8)$$

where $M_{\text{Pl}}^2 = (8\pi G)^{-1}$. The theory is therefore a *minimally coupled phantom scalar field* with constant G_4 (no running of the Planck mass) and vanishing G_3 , G_5 (no derivative coupling or braiding). Within the Horndeski landscape it occupies the k -essence subclass with $G_2 = G_2(X)$, distinct from Brans–Dicke theory (which has $G_4 = G_4(\varphi)$) and from quintessence (which has $G_2 = +X - V(\varphi)$ with the canonical sign).

2.5 Degrees of freedom

In the unconstrained formulation the theory has three configuration-space degrees of freedom: two from the transverse-traceless part of the metric (TT polarizations h_+ , $h_\times$) and one from the scalar field φ (breathing mode h_b). This matches the standard count for a Horndeski theory with $G_4 = \text{const}$ and is consistent with the gravitational-wave polarization analysis of Sec. 5.

In the bi-conformal formulation (2), the full metric is determined by the single function φ , reducing the dynamical content to **one** scalar degree of freedom *on the background*. General perturbations around this background (e.g. gravitational waves) are not required to be bi-conformal and therefore retain the full 3 DOF count of the unconstrained theory. Classical perturbative stability is guaranteed by the hyperbolicity of the principal symbol (Sec. 6); the constraint plays an additional role in suppressing quantum vacuum decay (see Sec. 6.3 and Appendix 11 for details).

3 Weak-field limit and solar-system tests

3.1 Static spherically symmetric solution

For a static, spherically symmetric source of mass M we seek a solution of the vacuum equation $\square\varphi = 0$ with the boundary condition $\varphi \rightarrow 0$ as $r \rightarrow \infty$. In the weak-field regime $|\varphi| \ll 1$ the d'Alembertian reduces to the flat-space Laplacian, and the unique regular solution is

$$\varphi(r) = -\frac{r_s}{r}, \quad r_s \equiv \frac{2GM}{c^2}, \quad (9)$$

where r_s is the Schwarzschild radius. The bi-conformal metric (2) then becomes

$$ds^2 = -e^{-r_s/r} c^2 dt^2 + e^{r_s/r} (dr^2 + r^2 d\Omega^2). \quad (10)$$

The bi-conformal property $g_{tt} \cdot g_{rr} = -c^2$ is not an *ad hoc* ansatz: for a static radial scalar field, the mixed stress-energy components satisfy $\tau^t_t = -\tau^r_r$, hence $\tau^t_t + \tau^r_r = 0$; via the vacuum Einstein equation this forces $G^t_t + G^r_r = 0$, which for a general static isotropic metric implies $(\alpha + \beta)' = 0$ for the metric exponents. Asymptotic flatness then gives $\alpha = -\beta$, i.e. $g_{tt} \cdot g_{rr} = -c^2$ (the full proof is given in [45], Appendix 2, Sec. 4).

In the weak-field limit ($r_s/r \ll 1$) the exponentials expand as

$$g_{tt} \approx -\left(1 - \frac{r_s}{r} + \frac{1}{2} \frac{r_s^2}{r^2} - \dots\right), \quad (11)$$

$$g_{rr} \approx 1 + \frac{r_s}{r} + \frac{1}{2} \frac{r_s^2}{r^2} + \dots. \quad (12)$$

3.2 PPN parameters

The parametrized post-Newtonian (PPN) formalism [2, 3] provides the standard framework for comparing weak-field predictions of gravity theories. The isotropic PPN metric for a static spherical source is

$$g_{tt} = -1 + 2U - 2\beta U^2 + \dots, \quad (13)$$

$$g_{ij} = (1 + 2\gamma U) \delta_{ij} + \dots, \quad (14)$$

where $U = GM/(c^2 r) = r_s/(2r)$ is the Newtonian potential. Comparing with the expansion (11)–(12):

Determination of γ .—The spatial metric gives $g_{rr} \approx 1 + r_s/r = 1 + 2U$, so $2\gamma U = 2U$, hence

$$\boxed{\gamma = 1.} \quad (15)$$

Determination of β .—The temporal metric gives $g_{tt} \approx -1 + r_s/r - r_s^2/(2r^2) = -1 + 2U - 2U^2$, so $2\beta U^2 = 2U^2$, hence

$$\boxed{\beta = 1.} \quad (16)$$

These values are identical to GR and consistent with the tightest experimental bounds: $|\gamma - 1| < 2.3 \times 10^{-5}$ (Cassini spacecraft [22]) and $|\beta - 1| < 3 \times 10^{-4}$ (lunar laser ranging [23]).

3.3 Light deflection

The deflection of light by a point mass M at impact parameter b is determined by the transverse gradient of the effective refractive index $n(r) = c/c_{\text{local}} = e^{-\varphi} \approx 1 + r_s/r$ in the weak-field limit. The deflection angle is

$$\delta = \int_{-\infty}^{+\infty} \nabla_{\perp}(n-1) dl = \frac{2GM}{c^2} \int_{-\infty}^{+\infty} \frac{b dx}{(x^2 + b^2)^{3/2}}. \quad (17)$$

Evaluating the standard integral gives

$$\delta = \frac{4GM}{c^2 b}. \quad (18)$$

This is identical to the GR result [1] and consistent with solar deflection measurements from very-long-baseline interferometry ($\delta_{\odot} = 1.7505'' \pm 0.0002''$). The agreement follows directly from $\gamma = 1$.

Extension to massive particles: gravity as matter-wave refraction. The refractive picture extends to matter. An oscillon moving through the medium carries a de Broglie wavelength $\lambda = h/(m_{\text{eff}} v)$; its trajectory is governed by the same medium-property gradients that deflect light.

The Hamilton–Jacobi equation for a particle of rest mass m_0 , $g^{\mu\nu} \partial_{\mu} S \partial_{\nu} S + m_0^2 c^2 = 0$, yields the local coordinate momentum

$$p^2(r) = e^{-\varphi} \left(\frac{e^{-\varphi} E^2}{c^2} - m_0^2 c^2 \right), \quad (19)$$

with E the conserved energy and $p_{\infty}^2 = E^2/c^2 - m_0^2 c^2$ at spatial infinity. Define the *matter-wave refractive index*

$$n_{\text{m}}(r) \equiv \frac{p(r)}{p_{\infty}}. \quad (20)$$

Two limits are instructive:

- *Ultra-relativistic* ($m_0 \rightarrow 0$, $v \rightarrow c$): $n_{\text{m}} \rightarrow e^{-\varphi} = n(r)$, recovering the photon refractive index identically.
- *Non-relativistic* ($v_{\infty} \ll c$):

$$n_{\text{m}}(r) \approx 1 - \frac{c^2 \varphi}{2 v_{\infty}^2} = 1 + \frac{GM}{v_{\infty}^2 r}, \quad (21)$$

where the last equality uses $\varphi = -r_s/r$.

The trajectory satisfies the optical–mechanical analogue of Fermat’s principle (Maupertuis’ principle):

$$\delta \int n_{\text{m}} dl = 0. \quad (22)$$

In the non-relativistic weak-field limit the deflection integral gives the Newtonian gravitational scattering angle

$$\delta_{\text{m}} = \frac{2GM}{v_{\infty}^2 b}, \quad (23)$$

and the corresponding equation of motion is

$$\mathbf{a} = -\frac{c^2}{2} \nabla \varphi = -\nabla \Phi_N, \quad (24)$$

with $\Phi_N = \frac{1}{2}c^2\varphi$: Newton's law. Gravity on massive particles is wave refraction in the same medium that deflects light; the two cases differ only in their dispersion relation, while the mechanism—turning toward regions of lower c_{coord} (higher n)—is identical. The continuous interpolation $n_m \rightarrow n_{\text{photon}}$ as $v \rightarrow c$ confirms that photon deflection is a limiting case of the universal matter-wave refraction, not a separate phenomenon.

3.4 Perihelion precession

The anomalous perihelion advance per orbit in the PPN framework is [2]

$$\Delta\phi = \frac{6\pi GM}{c^2 a(1-e^2)} \cdot \frac{2+2\gamma-\beta}{3}, \quad (25)$$

where a is the semi-major axis and e the eccentricity. Substituting $\gamma = \beta = 1$:

$$\Delta\phi = \frac{6\pi GM}{c^2 a(1-e^2)}. \quad (26)$$

For Mercury ($a = 5.79 \times 10^{10}$ m, $e = 0.2056$) this yields $\Delta\phi \approx 42.98''/\text{century}$, in agreement with the observed value $42.98'' \pm 0.04''/\text{century}$ and identical to GR.

3.5 Frame dragging

A rotating source with angular momentum $\mathbf{S} = I\boldsymbol{\omega}$ induces a Lense–Thirring precession of a gyroscope at distance r :

$$\boldsymbol{\Omega}_{\text{LT}} = \frac{G}{c^2 r^3} [3(\mathbf{S} \cdot \hat{\mathbf{r}})\hat{\mathbf{r}} - \mathbf{S}]. \quad (27)$$

This expression is derived from the off-diagonal (gravitomagnetic) part of the metric, which in the present theory takes the same form as in GR at leading order ($\gamma = 1$). For the Earth, the predicted precession rate is $\Omega_{\text{LT}} = 39.2 \text{ mas yr}^{-1}$, consistent with the Gravity Probe B measurement [24] $37.2 \pm 7.2 \text{ mas yr}^{-1}$.

3.6 Gravitational redshift

A photon emitted at radial coordinate r_e in the static metric (10) and received at r_o undergoes a frequency shift determined by the ratio of proper-time rates:

$$\frac{\nu_o}{\nu_e} = \sqrt{\frac{g_{tt}(r_e)}{g_{tt}(r_o)}} = e^{[\varphi(r_e) - \varphi(r_o)]/2}. \quad (28)$$

In the weak-field limit $|\varphi| \ll 1$,

$$\frac{\Delta\nu}{\nu} \equiv \frac{\nu_o - \nu_e}{\nu_e} \approx \frac{\varphi(r_e) - \varphi(r_o)}{2} = -\frac{\Phi(r_e) - \Phi(r_o)}{c^2}, \quad (29)$$

where $\Phi = -GM/r$ is the Newtonian potential.

This result admits a refractive interpretation consistent with the framework of Sec. 4. The effective refractive index $n(r) = e^{-\varphi}$ increases toward the source. A photon propagating into a

region of higher n has its leading wavefront decelerated before the trailing one; the wavelength compresses and a blueshift results. A photon climbing out of the field enters a region of lower n , the wavefronts stretch, and a redshift follows.

For a vertical path of height h at the Earth's surface,

$$\frac{\Delta\nu}{\nu} = \frac{gh}{c^2}, \quad (30)$$

where $g = GM_{\oplus}/R_{\oplus}^2$. The Pound–Rebka experiment [55] ($h = 22.6$ m) measured $\Delta\nu/\nu = (2.57 \pm 0.26) \times 10^{-15}$; the predicted value is 2.46×10^{-15} , in agreement within experimental error. Modern hydrogen-maser tests [56] have confirmed the prediction to 7×10^{-5} fractional accuracy. Both results are identical to GR at 1PN order, as guaranteed by $\gamma = \beta = 1$.

3.7 Summary of solar-system tests

Table 1 collects the solar-system predictions. At first post-Newtonian order the theory is indistinguishable from GR.

Table 1: Solar-system tests: theory versus observation.

Test	This work	GR	Observation
γ	1	1	$1 \pm 2.3 \times 10^{-5}$
β	1	1	$1 \pm 3 \times 10^{-4}$
Light deflection	$4GM/c^2b$	same	$1.7505''$
Mercury precession	$42.98''/\text{cen}$	same	$42.98''/\text{cen}$
Lense–Thirring	39.2 mas/yr	same	37.2 ± 7.2
Grav. redshift	gh/c^2	same	$(1.00 \pm 0.007) \times \text{pred.}$

4 Shapiro time delay at 1PN and 2PN

4.1 First post-Newtonian order

The additional travel time of a light signal passing through the gravitational field of a mass M is obtained from the effective refractive index $n(r) = e^{-\varphi} \approx 1 + r_s/r$ in the weak-field limit. The Shapiro delay is

$$\Delta t = \frac{1}{c} \int (n(\mathbf{r}) - 1) dl = -\frac{2}{c^3} \int \Phi(\mathbf{r}) dl, \quad (31)$$

where $\Phi = -GM/r$ is the Newtonian potential and the integration is along the unperturbed straight-line trajectory. For a signal from source at distance r_0 to receiver at r_1 , passing at closest approach b from the mass:

$$\Delta t_{1\text{PN}} = \frac{2GM}{c^3} \ln \left[\frac{(r_0 + \ell_0)(r_1 + \ell_1)}{b^2} \right], \quad (32)$$

where $\ell_i = \sqrt{r_i^2 - b^2}$. This is identical to the standard GR result [25], as required by $\gamma = 1$.

4.2 Second post-Newtonian order

The bi-conformal metric relates the local speed of light to the scalar field through an exponential: $c_{\text{local}} = c e^{\varphi}$, giving the exact refractive index $n = e^{-\varphi}$. Expanding to second order in $\varphi = -r_s/r$:

$$n(r) = e^{r_s/r} \approx 1 + \frac{r_s}{r} + \frac{1}{2} \frac{r_s^2}{r^2} + \dots \quad (33)$$

The observable Shapiro delay is the excess over the flat-space straight-line travel time between the same asymptotic endpoints:

$$\Delta t \equiv T - T_{\text{flat}} = \frac{1}{c} \int_{\text{ray}} (n - 1) dl + \frac{L_{\text{ray}} - L_0}{c}, \quad (34)$$

where L_{ray} is the Euclidean length of the bent spatial path and L_0 the corresponding straight-line length.

Refractive contribution at 2PN order (derived).—For a straight-line path at impact parameter b , the second-order term in the expansion (33) produces a refractive integral that evaluates exactly to

$$\Delta t_2^{(\text{refr})} = \frac{2G^2 M^2}{c^5} \int_{-\infty}^{+\infty} \frac{dx}{x^2 + b^2} = \frac{2\pi G^2 M^2}{c^5 b}. \quad (35)$$

This contribution is an exact, analytic consequence of the *exponential* bi-conformal structure $n = e^{-\varphi}$: the $O(\varphi^2)$ piece $\frac{1}{2}(r_s/r)^2$ in (33) integrates to π/b along the straight line, producing the clean 2π coefficient. This is structurally distinct from GR, where the analogous integrand (derived from the Schwarzschild/isotropic refractive index) is a rational function of r_s/r and generates a different coefficient. The refractive contribution (35) has been verified numerically (`tools/check_shapiro_born.py`).

Bent-ray optical and excess path-length contributions (in progress).—At 2PN order the photon follows a bent trajectory corrected at $O(r_g)$, and the bending produces both (i) a shift of the $O(r_g)$ refractive integrand along the corrected path, and (ii) an excess Euclidean length of the bent spatial path. A *rigorous* derivation of these two contributions requires (a) a consistent convention for the asymptotic impact parameter b versus the geometric closest approach, and (b) a careful treatment of the finite emitter/receiver endpoints r_1, r_2 , because individually each piece is infra-red sensitive (log/linear in the endpoint radii) and only the full combination is finite. A self-contained such derivation in the bi-conformal ISPG geometry is deferred to future work.

Summary (epistemic status). The *refractive 2PN piece* (35) of the ISPG Shapiro delay is established analytically and verified numerically. A *complete* 2PN coefficient—including the bent-ray and path-length contributions—requires the finite-endpoint analysis outlined above and is flagged here as an open task (see Appendix 6 for the current derivation sketch). What is theoretically secure at this stage is the *structural* claim that the bi-conformal exponential $n = e^{-\varphi}$ produces a 2PN Shapiro signature distinct from GR’s rational-polynomial refractive index, even though the overall numerical coefficient and its observational prospects remain to be fixed by the completed derivation.

5 Gravitational waves

5.1 Propagation speed

The tensor speed excess in the Bellini–Sawicki parameterization [21] is

$$\alpha_T = \frac{2X}{M_*^2} (2G_{4,X} - 2G_{5,\varphi} - (\ddot{\varphi} - H\dot{\varphi})G_{5,X}). \quad (36)$$

For the Horndeski map (8), $G_{4,X} = G_5 = 0$, so

$$\alpha_T = 0 \quad \implies \quad c_g = c \quad (\text{exact}). \quad (37)$$

This is consistent with the constraint $|c_g/c - 1| < 10^{-15}$ from the coincident gravitational-wave and gamma-ray observation of GW170817/GRB 170817A [5].

5.2 Polarization content

The theory propagates three polarization modes:

1. **Plus (h_+) and cross ($h_\times$):** transverse-traceless tensor modes, satisfying the standard wave equation $\square h_{ij}^{\text{TT}} = 0$. These are identical to GR and account for all LIGO/Virgo detections [4].
2. **Breathing (h_b):** a scalar mode from $\delta\varphi$, producing isotropic (monopolar) strain $\delta L_x/L = \delta L_y/L = h_b$.

The tensor modes arise from the transverse-traceless part of the metric perturbation, which is independent of φ ; the breathing mode arises from $\delta\varphi$ alone. Both sectors satisfy $\square = 0$ on the background, confirming propagation at speed c .

5.3 Breathing mode amplitude

For a compact binary system, the scalar field perturbation is sourced by the trace of the matter energy-momentum tensor via Eq. (7). In the weak-field, slow-motion limit the breathing mode amplitude relative to the tensor modes scales as [45]

$$\frac{A_b}{A_t} \sim \frac{r_s}{r}, \quad (38)$$

where r is the orbital separation. The linear scaling follows from the multipole suppression alone: no monopole radiation (mass conservation), and dipole radiation cancels at leading order in $|\phi|$ because the kinematic conformal-redshift argument of Appendix 16 gives $s_A \simeq s_B \simeq 1/2$ for every body (with a residual structural-response correction whose exact vanishing is an open task, Appendix 16). The first non-vanishing contribution is therefore the trace quadrupole, suppressed by $v^2/c^2 \sim r_s/r$. At the leading-order level used here, no *additional* compactness-dependent prefactor multiplies Eq. (38). For GW150914-like binaries at inspiral separations $r \gtrsim 10r_s$, this gives $A_b/A_t \lesssim 10^{-1}$; combined with the interferometer-geometry suppression against isotropic breathing discussed in Appendix 10, the expected detector response is consistent with the non-detection of scalar polarizations by current-generation L-shaped interferometers.

The suppression (38) distinguishes the present theory from generic Brans–Dicke models, where the breathing amplitude is controlled by $1/\omega_{\text{BD}}$ and is subject to direct Cassini bounds.

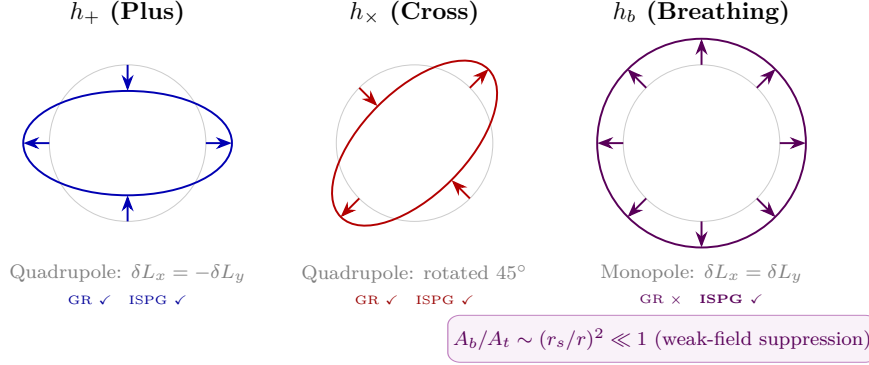


Figure 1: Gravitational-wave polarization content. The tensor modes h_+ and $h_\times$ are shared with GR; the breathing mode h_b is unique to this theory and is suppressed at the source by $A_b/A_t \sim r_s/r$ in the weak field (Eq. (38); linear scaling set by leading-order $s = 1/2$, Appendix 16).

Here the suppression is dynamical rather than parametric. The three-arm configuration of the LISA mission [67] provides three independent interferometric channels, enabling direct polarization decomposition. For massive-BH mergers in the LISA band, the late inspiral and merger phases probe the regime $r \sim r_s$ where Eq. (38) gives $A_b/A_t \sim 1$ at the source; the detector-geometry suppression and the scalar quadrupole PN prefactor (not yet derived from first principles) together determine the observed amplitude. The conversion of Eq. (38) into a specific h_b threshold for a LISA-target system (e.g., $10^6 M_\odot + 10^6 M_\odot$ at $z = 1$) is an open task: it requires the first-principles PN coefficient of the scalar quadrupole in the bi-conformal metric (Appendix 10, open task). Detection or non-detection of a breathing polarization remains a decisive qualitative test of the scalar sector; a quantitative LISA-noise comparison is deferred to future work.

5.4 Merger frequency shift

The innermost stable circular orbit (derived in Sec. 7) occurs at $r_{\text{ISCO}} = \varphi_{\text{gr}}^2 r_s \approx 2.618 r_s$, compared to $3 r_s$ in Schwarzschild spacetime. The corresponding orbital frequency at ISCO is

$$\frac{f_{\text{ISCO}}^{(\text{this work})}}{f_{\text{ISCO}}^{(\text{GR})}} \approx 0.931, \quad (39)$$

predicting a gravitational-wave merger frequency $\sim 6.9\%$ lower than GR for the same total mass. For a $30 + 30 M_\odot$ binary: $f_{\text{GR}} \approx 73$ Hz, $f_{\text{this work}} \approx 68$ Hz. This systematic shift is testable through waveform template matching across a population of binary merger events observed by LIGO–Virgo–KAGRA.

5.5 Scalar dipole radiation from mixed binaries

In generic scalar-tensor theories, when two bodies have different scalar sensitivities $s_A \neq s_B$, the binary acquires a scalar dipole moment and the orbital energy loss rate gains a -1PN correction:

$$\dot{E}_{\text{dipole}} = -\frac{G \mu M}{3 r^2} (s_A - s_B)^2 v, \quad (40)$$

where μ is the reduced mass, M the total mass, and v the orbital velocity. This emission is lower order than the standard quadrupole $\dot{E} \propto v^5$, so it would dominate the early inspiral at large separation. The resulting phase shift relative to the GR waveform would accumulate as

$$\Delta\Psi_{\text{dipole}} \sim \frac{5}{672} (s_A - s_B)^2 \left(\frac{GM\pi f}{c^3} \right)^{-7/3}. \quad (41)$$

In the bi-conformal theory, the universal conformal redshift $m_{\text{eff}} = m_0 e^{\phi/2}$ selects $s = 1/2$ at leading order in $|\phi|$ for every body—test particles, white dwarfs, neutron stars, and vacuum compact objects alike (Appendix 16, Sec. “Kinematic motivation”). For vacuum compact objects this result is exact, derived from $\nabla^2\phi = 0$. For matter-filled bodies, the exact cancellation of the residual structural-response correction is an open task (Appendix 16, Sec. “Shift invariance: sectoral analysis”). At leading order, $(s_A - s_B)^2 \simeq 0$ for any binary, so the scalar dipole channel is strongly suppressed.

Orbital-period derivative.—The dipole contribution to the period derivative is

$$\dot{P}_b^{(\text{dipole})} = -\frac{4\pi^2 G \mu}{P_b c^3 a} (s_A - s_B)^2 \simeq 0, \quad (42)$$

so the orbital decay is purely quadrupolar at leading order, as in GR. This is consistent with the observed agreement of $\dot{P}_b$ with the GR prediction at the $\sim 10\%$ level for PSR J1738+0333 [70], and with the universality-of-free-fall bound from the triple system PSR J0337+1715 [71].

The structural reason is unique to the bi-conformal framework: both the hydrostatic equation and the scalar field equation depend on $d\varphi/dr$, not on φ itself, so the interior structure does not re-adjust when the external field shifts. This distinguishes the present theory from Brans–Dicke, DEF, and other scalar-tensor theories that generically predict $(s_A - s_B)^2 \neq 0$.

6 Phantom kinetic term and stability

The kinetic term in (3) carries the phantom sign ($+\frac{1}{2}$ rather than $-\frac{1}{2}$). Standard concerns about phantom fields include Hamiltonian unboundedness, vacuum instability via ghost–tensor pair production, and ill-posedness of the Cauchy problem [31–33]. Below we show that, in the present context, none of these pathologies arise.

6.1 Hyperbolicity and well-posedness

The linearized field equations for perturbations $(h_{\mu\nu}, \delta\varphi)$ about a background $(\bar{g}_{\mu\nu}, \bar{\varphi})$ have the principal symbol

$$\mathcal{P}(\xi) = \begin{pmatrix} \bar{g}^{\alpha\beta} \xi_\alpha \xi_\beta \mathbb{I}_{\text{tensor}} & 0 \\ 0 & \bar{g}^{\alpha\beta} \xi_\alpha \xi_\beta \end{pmatrix}, \quad (43)$$

where ξ_μ is the perturbation covector. The characteristic surfaces are determined by $\bar{g}^{\alpha\beta} \xi_\alpha \xi_\beta = 0$, which defines the null cone of the background metric. Since $\bar{g}_{\mu\nu}$ has Lorentzian signature, the characteristic equation has real roots for all spatial wave directions; the system is *strongly hyperbolic* and the Cauchy problem is well-posed in Sobolev spaces [34, 35].

All propagating modes—tensor and scalar—travel at exactly c in the local frame, consistent with the GW170817 constraint. There is no superluminal propagation and no violation of causality.

6.2 Classical mode stability

Static background.—Around the spherically symmetric solution $\varphi = -r_s/r$, scalar perturbations decomposed in spherical harmonics $\delta\varphi = \sum_{\ell m} u_\ell(r) Y_{\ell m}/r$ satisfy a Schrödinger-type radial equation with an effective potential

$$V_{\text{eff}}(r) = e^{-2r_s/r} \left[\frac{\ell(\ell+1)}{r^2} + \frac{r_s}{r^3} - \frac{r_s^2}{4r^4} \right]. \quad (44)$$

For $\ell \geq 1$ the centrifugal barrier dominates the bracket in the exterior; in the deep interior the bracket can change sign, but the exponential prefactor $e^{-2r_s/r}$ suppresses the resulting well far more effectively than in the $\ell = 0$ case, yielding a Bargmann integral $\mathcal{I} \lesssim 10^{-7}$. No bound state exists for any ℓ .

For $\ell = 0$ the potential $V_{\text{eff}}^{(\ell=0)}$ is negative in the interval $0 < r < r_s/4$, but this well is exponentially suppressed by the factor $e^{-2r_s/r}$. Numerically, the well integral $\mathcal{I} = \int_{\text{well}} |V_{\text{eff}}| dr_* \approx 1.0 \times 10^{-4}$ (in units $c = r_s = 1$). By the Bargmann bound, a one-dimensional potential well supports no bound state when $\mathcal{I} \ll 1$. Therefore, no tachyonic mode ($\omega^2 < 0$) exists at probe level even for $\ell = 0$. The full symbolic and numerical verification is given in Appendix 15.

Cosmological background.—On the FLRW background (Sec. 8), inhomogeneous scalar perturbations satisfy

$$\delta\ddot{\varphi}_k + 3H \delta\dot{\varphi}_k + \frac{k^2 c^2}{a^2} \delta\varphi_k = 0, \quad (45)$$

with $\omega_k^2 = k^2 c^2 / a^2 > 0$. All modes oscillate and are damped by Hubble friction; there is no gradient instability.

6.3 Ghost elimination via constraint structure

In the unconstrained (effective field theory) formulation the theory has $2 + 1 = 3$ degrees of freedom (two tensor, one scalar), and the phantom scalar carries negative kinetic energy in the Hamiltonian. In a generic theory this would trigger vacuum decay via ghost–tensor pair production at a rate set by the UV cutoff.

The bi-conformal metric structure (2) determines the *entire* background metric from the single function φ , reducing the physical gravitational degrees of freedom to **one** on the background. A single phantom field cannot decay into itself within the pure-gravity sector; vacuum pair production requires a second field with compensating positive energy, which is absent among the gravitational modes of the constrained sector. A matter-mediated channel $|0\rangle \rightarrow |\varphi_{-E}\rangle + |\psi\bar{\psi}\rangle_{+E}$ with minimally coupled matter ψ as positive-energy partner remains kinematically open; its physical rate is controlled by gravitational coupling strength (see Appendix 11, “Scope of the theorem”) and is Planck-suppressed at all astrophysically relevant energies, with the $\alpha X^2/M^2$ EFT completion (Eq. 46) providing the UV closure for this channel.

Scope.—The 1-DOF reduction applies strictly to the bi-conformal background, which is isotropic by construction. General perturbations (including gravitational waves) break this isotropy and propagate in the unconstrained 3-DOF sector. Their classical stability is guaranteed by the hyperbolicity established in Sec. 6.1: the phantom sign does not affect the principal symbol, so all modes have real characteristics and well-posed Cauchy evolution. The quantum extension of the theory [47] confirms that pair-production rates are Planck-suppressed: loop corrections scale as $(E/M_{\text{Pl}})^2 \sim 10^{-76}$, the reduced Hamiltonian constraint algebra is Abelian

and anomaly-free (*reduced* denotes the gauge-fixed single-DOF phase space of [47], distinct from the full Dirac–Bergmann phase space of Appendix 11, whose $(\Phi_\alpha, \bar{\Phi}_\alpha)$ conditions are second-class by construction), and the Hamiltonian is bounded from below non-perturbatively. Quantum vacuum decay is therefore negligible at all astrophysically relevant energies.

A complementary low-energy EFT phrasing is available even without invoking the full quantum completion. Away from the strict bi-conformal background one may extend the minimal Horndeski kinetic function to

$$G_2(X) = -\frac{M_{\text{Pl}}^2}{2} X + \frac{\alpha}{M^2} X^2, \quad \alpha > 0, \quad (46)$$

with $X \equiv -\frac{1}{2} g^{\mu\nu} \partial_\mu \varphi \partial_\nu \varphi$. The condensate then sits at nonzero $\langle X \rangle$ where $\partial G_2 / \partial X > 0$, stabilizing the kinetic sector while leaving the weak-field static solution unchanged at the order relevant for Solar-system and 2PN observables [38]. The exponential bi-conformal metric, the ensuing structurally distinct 2PN Shapiro signature (Sec. 4), and the ISCO radius are therefore infrared predictions rather than artifacts of a particular UV completion.

This distinction is critical because Bronnikov and collaborators [66] have shown that phantom-scalar wormholes are linearly unstable under spherically symmetric perturbations in the *unconstrained* formulation—the instability is catastrophic, with perturbation growth rates that are unbounded from above. The constrained (fundamental) formulation eliminates the unstable mode by construction.

A further structural distinction separates the present solution from the Bronnikov wormholes: the bi-conformal background (10) is *one-sided* and regular, with no traversable throat connecting two asymptotically flat regions. The Bronnikov–Rubin instability arises from a mode that threads the throat, coupling the two asymptotic regions; in the absence of a second exterior this mode has no support. The one-sided topology, combined with the exponential suppression $e^{-2r_s/r}$ in the deep interior, restricts perturbation dynamics to a regime where the Bargmann analysis (Appendix 15) already excludes tachyonic modes. Task 1b (even/polar sector) remains the definitive test, but the topological mismatch gives strong reason to expect that the Bronnikov instability does not carry over.

The closest precedents in the literature are *cuscuton gravity* [36], where $\nabla_\mu \varphi \nabla^\mu \varphi = \mu^2$ eliminates the scalar propagator, and *mimetic gravity* [37], where a conformal constraint does likewise. In all three cases the ghost is removed kinematically by the constraint, not dynamically stabilized.

The formal Dirac–Bergmann analysis is carried out in Appendix 11. The bi-conformal conditions are exhibited as second-class constraints in the ADM phase space, reducing the physical degrees of freedom from three (two tensor + one scalar) to **one**. A single propagating DOF has no pair-production channel: the process $|0\rangle \rightarrow |\text{ghost}\rangle + |\text{tensor}\rangle$ requires at least two independent fields, and the constrained formulation provides only one. The energy of the remaining DOF need not be positive-definite—the phantom sign is intrinsic—but no runaway occurs because there is no decay partner. The full DOF count, stability theorem, and relation to cuscuton and mimetic gravity are given in Appendix 11.

6.4 Energy conditions

The scalar energy-momentum tensor (5) as defined is canonical and in fact *satisfies* the standard energy conditions: for a null vector k^μ ,

$$\tau_{\mu\nu}^{(\varphi)} k^\mu k^\nu = (k^\mu \partial_\mu \varphi)^2 \geq 0, \quad (47)$$

and analogous non-negativity holds for a timelike u^μ after accounting for the phantom kinetic sign (WEC and SEC checks, Appendix 11). What violates the energy conditions is the *effective source* that enters the Einstein equation (4): the coupling factor $\kappa = -\frac{1}{2}$ flips the sign of the scalar contribution, so the effective NEC quantity is

$$\kappa \tau_{\mu\nu}^{(\varphi)} k^\mu k^\nu = -\frac{1}{2} (k^\mu \partial_\mu \varphi)^2 \leq 0, \quad (48)$$

and likewise for WEC and SEC. Thus the phantom sign makes the effective gravitational source violate the energy conditions, while the canonical tensor $\tau_{\mu\nu}^{(\varphi)}$ itself does not. The violation is an intrinsic property of how φ couples to geometry through κ , not of the tensor form in (5).

The effective NEC violation has a direct physical consequence: it invalidates the focusing theorem that underpins the Penrose–Hawking singularity theorems [15, 16]. This is what allows the singularity resolution derived in Sec. 7—the same mechanism that makes the phantom sign troublesome in flat-space QFT is precisely what cures the classical singularity.

6.5 Absence of a fifth force

In generic scalar–tensor theories a light scalar field mediates a fifth force constrained by torsion-balance experiments [68]. The present theory evades these bounds for three reasons: (i) the scalar couples universally to matter through the Jordan-frame metric (2), so the weak equivalence principle is satisfied identically ($\eta_{\text{Eöt}} < 10^{-15}$); (ii) the weak-field limit yields $\gamma_{\text{PPN}} = \beta_{\text{PPN}} = 1$ (Sec. 3.2), making the scalar indistinguishable from GR at first post-Newtonian order; (iii) the first observable deviation appears at the *second* post-Newtonian level (Sec. 4), which is below the sensitivity of current laboratory and Solar-system tests. Consequently, the scalar sector is not a “fifth force” in the experimental sense; it is an integral part of gravity, distinguishable only in the strong-field or high-precision regime.

7 Singularity resolution and compact objects

7.1 Curvature regularity

The exact bi-conformal metric (10) with $\varphi = -r_s/r$ yields the Ricci and Kretschmann scalars (verified symbolically with `tools/check_ricci_kretschmann.py`, which checks the full closed form of Eq. (50) against the SymPy-computed curvature and additionally reports the Schwarzschild leading term for comparison):

$$R = -\frac{r_s^2}{2r^4} e^{-r_s/r}, \quad (49)$$

$$K \equiv R_{\mu\nu\rho\sigma} R^{\mu\nu\rho\sigma} = \frac{r_s^2}{4r^8} (48r^2 - 32r r_s + 7r_s^2) e^{-2r_s/r}. \quad (50)$$

In the weak-field regime $r_s/r \ll 1$ this expands as $K = (12 r_s^2/r^6 + O(r_s^3/r^7)) e^{-2r_s/r}$, recovering the Schwarzschild leading term. As $r \rightarrow 0$ the exponential factor $e^{-r_s/r}$ (respectively $e^{-2r_s/r}$)

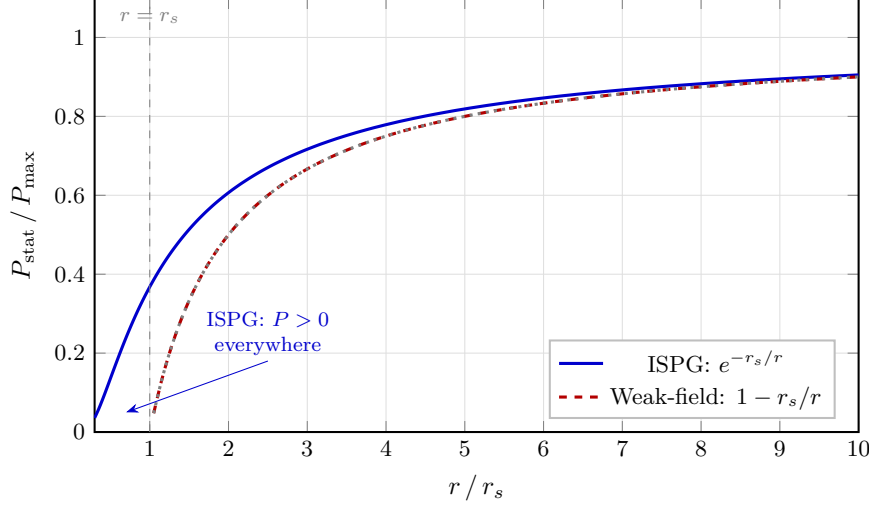


Figure 2: Normalised IS pressure $P_{\text{stat}}/P_{\text{max}}$ versus radial coordinate. The solid curve shows the exact exponential profile; the dashed curve is the weak-field expansion truncated at second order. The two diverge below $r \approx 2r_s$, illustrating how the full nonlinear solution avoids the singularity present in any finite-order expansion.

vanishes faster than any power of $1/r$ diverges. Consequently, *all curvature invariants vanish at the origin*:

$$\lim_{r \rightarrow 0} R = 0, \quad \lim_{r \rightarrow 0} K = 0. \quad (51)$$

The Kretschmann scalar reaches a finite maximum at $r \approx 0.221 r_s$ (equivalently $u \equiv r_s/r \approx 4.524$, the unique real root of $7u^3 - 60u^2 + 160u - 144 = 0$ on the physical domain):

$$K_{\text{max}} \approx \frac{11.72}{r_s^4}. \quad (52)$$

For astrophysical black holes this is far below the Planck curvature $K_{\text{Pl}} = \ell_P^{-4}$.

$r = 0$ is not a regular centre: global structure. The vanishing of curvature invariants at $r = 0$ should not be read as collapse to a regular origin. The areal radius of a sphere of coordinate radius r in the bi-conformal metric (10) is

$$R(r) \equiv r e^{r_s/(2r)}, \quad (53)$$

which has a minimum at the *throat* $r = r_s/2$, $R_{\text{min}} = (r_s/2) e \approx 1.36 r_s$, and grows without bound in *both* limits $r \rightarrow \infty$ (the usual asymptotic region) and $r \rightarrow 0^+$ (a second asymptotic region). The coordinate $r = 0$ is therefore not a point at finite proper distance but a second asymptotic region accessed through the throat. The two asymptotic regions, the throat geometry, and the physical realisability of the second asymptotic region are developed in Appendix 12 (Sec. “Wormhole structure”); the present section restricts itself to the curvature-invariant statement.

7.2 Geodesic completeness

The proper radial distance from any radius r_0 to the origin is

$$L = \int_0^{r_0} \sqrt{g_{rr}} dr = \int_0^{r_0} e^{r_s/(2r)} dr. \quad (54)$$

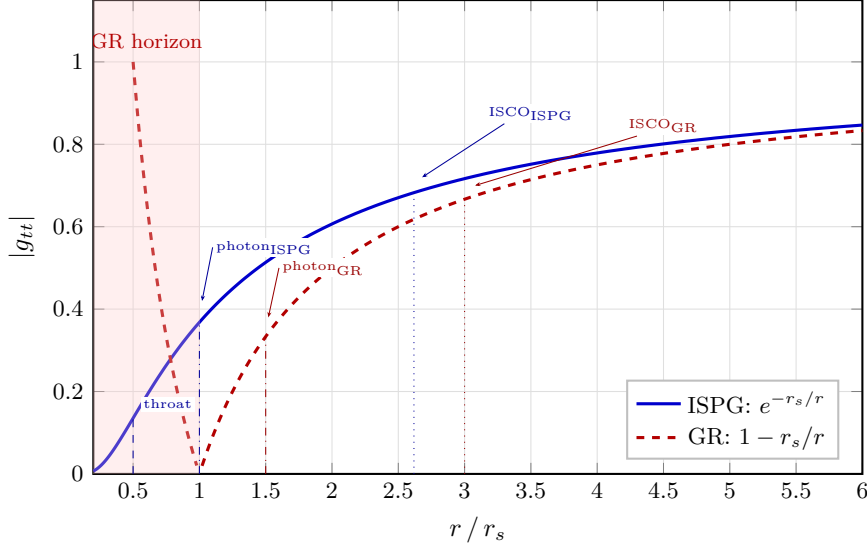


Figure 3: Absolute value of the metric component g_{tt} as a function of r/r_s . In GR (dashed), g_{tt} vanishes at the horizon $r = r_s$; in the present theory (solid), $g_{tt} = e^{-r_s/r}$ is strictly nonzero. Vertical markers indicate the throat of the wormhole geometry at $r = r_s/2$ (see Eq. (53) and Appendix 12), photon spheres, and ISCOs for both theories.

As $r \rightarrow 0$ the integrand $e^{r_s/(2r)}$ diverges exponentially, and the integral diverges: $L \rightarrow \infty$. The center $r = 0$ is at *infinite proper distance*. Radial geodesics nonetheless reach $r = 0$ in finite proper time (or affine parameter), since $\dot{r}^2 \rightarrow E^2/c^2 > 0$ as $r \rightarrow 0$. The effective potential $V_{\text{eff}} = e^\varphi(1 + L^2 e^\varphi/r^2)$ peaks at the photon sphere ($r = r_s$) and is exponentially suppressed to zero as $r \rightarrow 0$ (Appendix 12; cf. Fig. 5). Geodesics that do not cross this barrier (impact parameter $b > b_c = r_s e$) are reflected and complete; those that cross it reach $r = 0$ in finite proper time, making the spacetime geodesically incomplete.

Scope of the “singularity-free” claim. In the Hawking–Penrose convention, geodesic incompleteness is itself a definition of singularity, so the present geometry is not globally non-singular. What we *do* show is the following restricted set of statements: (i) every polynomial curvature invariant is finite—indeed vanishes—at $r = 0$ (Sec. 7.1); (ii) the boundary is at infinite proper distance and carries infinite redshift ($1 + z = e^{r_s/(2r)} \rightarrow \infty$); (iii) it requires infinite coordinate time to reach, so it is causally inert and unobservable from the exterior; (iv) tidal forces decrease monotonically toward zero on approach. By contrast, the Schwarzschild singularity fails (i) and (iv). Whether the $r = 0$ boundary admits a smooth extension that satisfies the field equations is an open problem: Clarke’s theorem guarantees only a C^0 extension for a boundary with bounded curvature invariants, and upgrading this to C^∞ and verifying the Einstein equations on the extended manifold is not carried out here (Appendix 12, Sec. “Classification of the $r = 0$ boundary”). Accordingly, “singularity-free” in this paper refers to (i)–(iv) and not to the Hawking–Penrose completeness criterion.

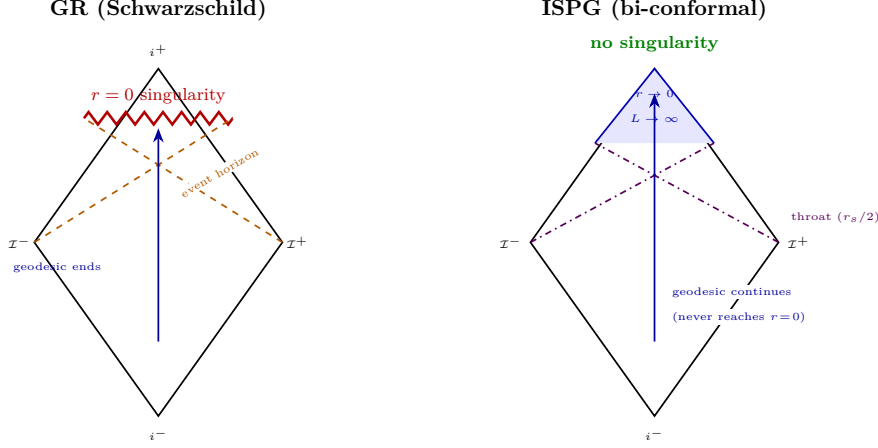


Figure 4: Penrose (conformal) diagrams. *Left:* Schwarzschild spacetime—the curvature singularity at $r = 0$ destroys infalling matter. *Right:* Bi-conformal spacetime— $r = 0$ is at infinite proper distance with vanishing curvature; the boundary is regular and causally inert (Appendix 12). The throat at $r_s/2$ connects the exterior to the deep interior.

7.3 Horizonless compact object

The metric component $g_{tt} = -e^{-r_s/r}$ is nonzero for all $r > 0$: there is *no event horizon*. Instead, the areal radius $R(r) = r e^{r_s/(2r)}$ has a minimum at

$$r_{\text{throat}} = \frac{r_s}{2}, \quad (55)$$

where $dR/dr = 0$. This is a throat (minimum-area surface), not a horizon:

- $g_{tt}(r_s/2) = -e^{-2} \approx -0.135$ (nonzero);
- a photon at $r = r_s/2$ escapes to infinity in finite coordinate time;
- the structure is a one-sided wormhole throat with infinite depth ($L \rightarrow \infty$).

Current ringdown data do not yet distinguish a true horizon from a sufficiently compact horizonless object, so the present geometry remains compatible with existing gravitational-wave observations [41]. The same qualitative picture is also consonant with Gogberashvili's arguments that quantum particles are reflected at the Schwarzschild horizon and that smooth interior–exterior matching requires the source radius to lie outside the photon sphere [42, 43].

7.4 Innermost stable circular orbit

The effective potential for massive test particles on equatorial geodesics is

$$V_{\text{eff}}(r) = e^{\varphi} \left(1 + \frac{L^2 e^{\varphi}}{r^2} \right), \quad \varphi = -\frac{r_s}{r}. \quad (56)$$

Circular orbits require $V'_{\text{eff}} = 0$; stability requires $V''_{\text{eff}} \geq 0$. The ISCO condition $V''_{\text{eff}} = 0$ reduces to the algebraic equation

$$r^2 - 3r_s r + r_s^2 = 0, \quad (57)$$

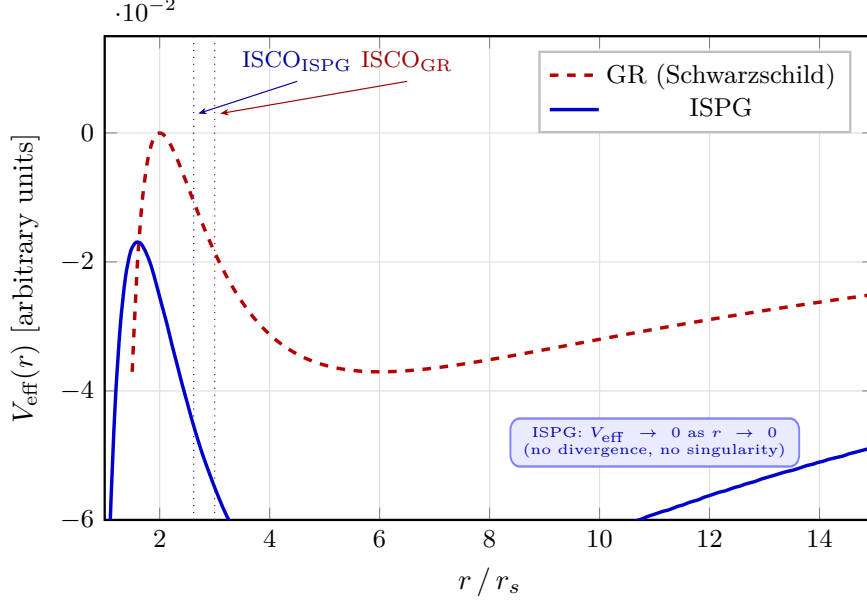


Figure 5: Effective potential for massive test particles. In GR (dashed) V_{eff} diverges as $r \rightarrow 0$; in the present theory (solid) the exponential suppression drives $V_{\text{eff}} \rightarrow 0$. The ISCO locations are marked: $\varphi_{\text{gr}}^2 r_s \approx 2.618 r_s$ (this work, isotropic coordinate) versus $3 r_s$ (GR, Schwarzschild coordinate; $\approx 2.47 r_s$ in isotropic coordinates).

with the physical solution

$$r_{\text{ISCO}} = \frac{3 + \sqrt{5}}{2} r_s = \varphi_{\text{gr}}^2 r_s \approx 2.618 r_s, \quad (58)$$

where $\varphi_{\text{gr}} = (1 + \sqrt{5})/2$ is the golden ratio. The appearance of φ_{gr}^2 is not imposed—it emerges from the quadratic (57), whose coefficients $(1, -3, 1)$ are dictated by the exponential metric. It was unexpected, and the author verified it independently with SageManifolds before accepting the result. In Schwarzschild coordinates, the GR ISCO is at $6 GM/c^2 = 3 r_s$; converting to isotropic coordinates gives $r_{\text{ISCO}}^{(\text{GR}, \text{iso})} = \frac{1}{4}(5 + 2\sqrt{6}) r_s \approx 2.47 r_s$. The present value ($2.618 r_s$) is 5.8% farther in consistent isotropic coordinates. The physically unambiguous comparison is the orbital frequency, which is 6.9% lower than in GR (Sec. 5.4).

7.5 Photon sphere

Null circular orbits satisfy $\frac{d}{dr}(e^{2\varphi}/r^2) = 0$, giving

$$r_{\text{photon}} = r_s. \quad (59)$$

In Schwarzschild spacetime the photon sphere is at $r^{(\text{GR})} = 3 GM/c^2 = \frac{3}{2} r_s$ (Schwarzschild coordinate); in isotropic coordinates this maps to $r^{(\text{GR}, \text{iso})} = \frac{1}{4}(2 + \sqrt{3}) r_s \approx 0.933 r_s$. The present value (r_s) is 7.2% farther in consistent isotropic coordinates. A coordinate-invariant comparison is the critical impact parameter (shadow size): $b_c = r_s e \approx 2.718 r_s$ versus $b_c^{(\text{GR})} = \frac{3\sqrt{3}}{2} r_s \approx 2.598 r_s$, a 4.6% larger shadow. This prediction is *consistent* with current Event Horizon Telescope observations of M87* (shadow diameter $42 \pm 3 \mu\text{as}$, $\sim 7\%$ systematic uncertainty) and Sgr A* ($51.8 \pm 2.3 \mu\text{as}$, $\sim 4.4\%$) [40], within which ISPG and GR are not yet observationally

distinguishable; a *decisive* test awaits next-generation EHT sensitivity (ngEHT target $\sim 1\%$ shadow precision). Epistemic status: prediction derived from the theory; current data compatible; decisive falsification deferred to future instrumentation.

7.6 Summary of compact-object predictions

Table 2: Compact object: comparison with Schwarzschild.

Property	This work	GR (Schwarzschild)	
Singularity at $r = 0$	No ($K \rightarrow 0$)	Yes ($K \rightarrow \infty$)	
Geodesic completeness	No [‡]	No	
Event horizon	None	$r = r_s$	[‡] Geodesics that cross the photon-
ISCO	$\varphi_{\text{gr}}^2 r_s \approx 2.618 r_s$	$3 r_s$	
Photon sphere	r_s	$\frac{3}{2} r_s$	
Merger frequency	-6.9% vs GR	Reference	

sphere barrier ($b < r_s e$) reach $r = 0$ in finite proper time; the boundary is curvature-regular ($K \rightarrow 0$), at infinite redshift, and unobservable from outside (Appendix 12).

8 Cosmological perturbations and CMB compatibility

8.1 FLRW background

On the Friedmann–Lemaître–Robertson–Walker metric $ds^2 = -c^2 dt^2 + a^2(t) \delta_{ij} dx^i dx^j$, a homogeneous scalar field $\varphi_0(t)$ satisfies

$$\ddot{\varphi}_0 + 3H \dot{\varphi}_0 = 0, \quad (60)$$

with general solution $\dot{\varphi}_0 = C_0/a^3$. The bi-conformal ansatz of Eq. (2) is the static spherically-symmetric vacuum form, in which $g_{tt} = -e^\varphi$. On the FLRW background written in the matter-clock cosmic-time gauge ($t = \tau$, $g_{tt} = -1$), the *same* temporal condition forces $e^{\varphi_0} = 1$, i.e. $\varphi_0 = 0$ at all times, and hence $C_0 = 0$. This is a *consistency consequence of the construction* in the cosmic-time gauge, not an independent dynamical prediction (see Appendix 9, “Clarifying remark”). In non-comoving time parametrizations ($t \neq \tau$) one may keep $\varphi_0 \neq 0$, but then the background is no longer written in the standard FLRW cosmic-time gauge. On the spatial side, the bi-conformal ansatz $g_{ij} = e^{-\varphi} \delta_{ij}$ is the static isotropic form and is not imposed on the cosmological metric; only the temporal matching is used to fix the homogeneous scalar.

Given $\varphi_0 = 0$ (hence $\dot{\varphi}_0 = 0$), the scalar field carries no homogeneous cosmological energy density: the background cosmology reproduces standard Friedmann,

$$3M_{\text{Pl}}^2 H^2 = \rho_m, \quad \rho_\varphi = 0. \quad (61)$$

The alternative branch $C_0 \neq 0$ gives $\rho_\varphi = -C_0^2/(2a^6)$, a phantom transient decaying faster than any matter component; its stability against ghost modes requires the EFT completion of Appendix 11 (cf. Sec. 6).

8.2 Bellini–Sawicki parameters

The Bellini–Sawicki parameterization [21] characterizes linear perturbations in Horndeski theories through four time-dependent functions. For the Horndeski map (8) with $\varphi_0 = 0$:

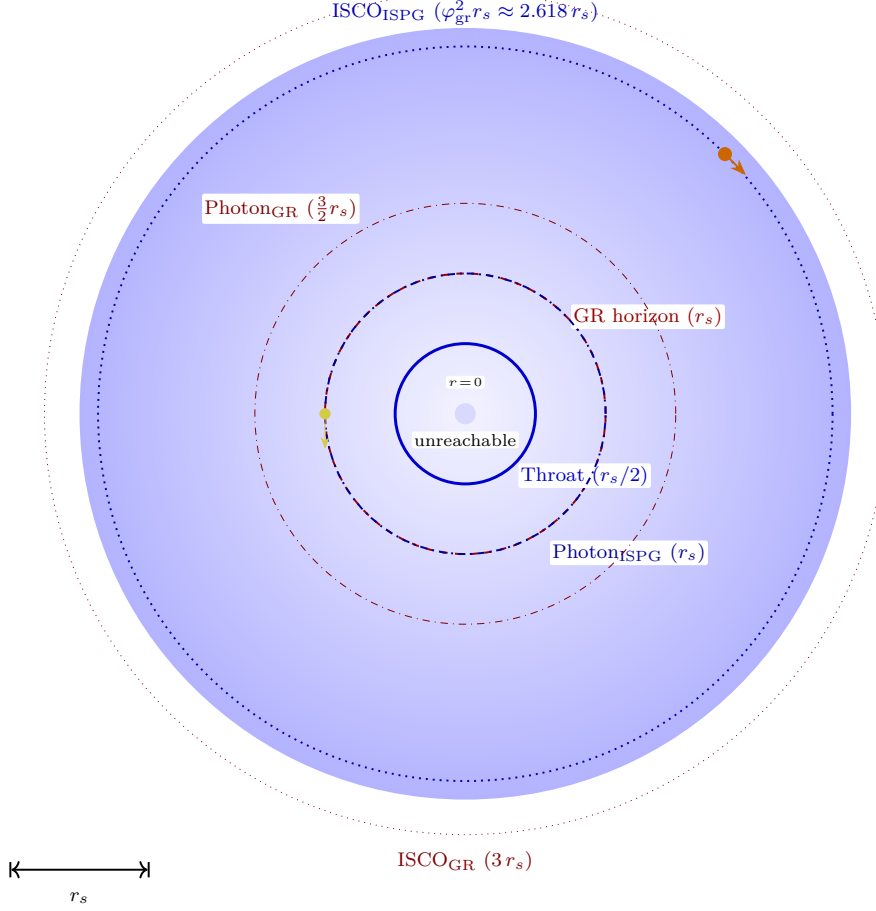


Figure 6: Schematic radial structure of the bi-conformal compact object. In GR (outer labels) the event horizon lies at r_s ; in the present theory (inner labels) r_s is the photon sphere, $r_s/2$ is the throat (minimum areal radius), and the ISCO occurs at $\varphi_{\text{gr}}^2 r_s \approx 2.618 r_s$. The center $r = 0$ is at infinite proper distance and regular (all curvature invariants vanish).

Planck mass.— $M_*^2 = 2G_4 = M_{\text{Pl}}^2$ (constant).

Running.— $\alpha_M = d \ln M_*^2 / d \ln a = 0$.

Tensor speed.— $\alpha_T = 0$ ($G_{4,X} = G_5 = 0$).

Braiding.— $\alpha_B = 0$ ($G_3 = 0$, $G_{4,\varphi} = 0$).

Kineticity.—The Horndeski map $G_2 = -X/2$ gives $\alpha_K = -\dot{\varphi}_0^2/(2H^2)$. On the background $\dot{\varphi}_0 = 0$ (§8.1), $\alpha_K = 0$ trivially. For $\dot{\varphi}_0 \neq 0$ (e.g. the $C_0 \neq 0$ transient of §8.1), $\alpha_K < 0$ exposes the phantom kinetic sign; the EFT completion of Appendix 11 is assumed to stabilize such transients (cf. Sec. 6).

The result is

$$\boxed{\alpha_K = \alpha_B = \alpha_M = \alpha_T = 0 \quad \text{on FLRW with } \varphi_0 = 0.} \quad (62)$$

8.3 Linear perturbations

In the Newtonian gauge, $ds^2 = -(1 + 2\Psi) c^2 dt^2 + a^2(1 - 2\Phi) \delta_{ij} dx^i dx^j$, the vanishing of all α_i implies:

Poisson equation (identical to GR):

$$k^2 \Phi = -4\pi G a^2 \rho_m \delta_m. \quad (63)$$

No gravitational slip:

$$\Phi = \Psi. \quad (64)$$

Scalar perturbation equation:

$$\delta\ddot{\varphi}_k + 3H\delta\dot{\varphi}_k + \frac{k^2 c^2}{a^2} \delta\varphi_k = \frac{8\pi G}{c^2} \delta T. \quad (65)$$

The RHS coefficient follows from the FLRW expansion of the sourced scalar equation $\square\varphi = -(8\pi G/c^4)T$ of Appendix 1: $\square\varphi = -c^{-2}(\ddot{\varphi} + 3H\dot{\varphi}) + a^{-2}\nabla^2\varphi$, so that multiplying by $-c^2$ yields the Fourier-space form above and matches the independent derivation of Appendix 9 (eq. 228). Equation (65) describes a matter-sourced, metric-decoupled propagating mode: $\delta\varphi_k$ does not back-react on Φ, Ψ at linear order (by $\alpha_i = 0$), but it is not a free spectator; it is driven by δT , and its kinetic term inherits the phantom-sign caveat of the background action, so its stability as a propagating degree of freedom relies on the EFT completion of Appendix 11 (cf. §6).

Since the gravitational potentials obey exactly the GR equations, the CMB angular power spectrum at linear order is

$$C_\ell^{(\text{this work})} = C_\ell^{(\Lambda\text{CDM})} \quad (\text{at linear order}). \quad (66)$$

This feature merits comment. Most modified-gravity theories reproducing MOND phenomenology (TeVSe, $f(R)$, covariant Galileon) acquire non-zero α_i that shift the CMB peaks or modify the integrated Sachs–Wolfe effect, typically requiring background tuning. In the present construction, the vanishing of all α_i follows from the single choice $\dot{\varphi}_0 = 0$ (i.e. the matter-clock cosmic-time gauge of §8.1; see also App. 9, “Clarifying remark”); the agreement is automatic only *given* that construction choice, not a dynamically generic attractor. Table 3 makes the comparison explicit.

Table 3: Bellini–Sawicki parameters: comparison.

Theory	α_T	α_B	α_M	CMB ok?
GR/ Λ CDM	0	0	0	Yes
This work	0	0	0	Yes
TeVSe	$\neq 0$	$\neq 0$	$\neq 0$	No
$f(R)$	0	$\neq 0$	$\neq 0$	Constrained
Brans–Dicke	0	$\neq 0$	$\neq 0$	Constrained
Cov. Galileon	0	$\neq 0$	0	Constrained

Among known MOND-motivated scalar-tensor theories with published Bellini–Sawicki mappings, the present construction is distinguished by having all four parameters vanish on the cosmological background; no uniqueness theorem is claimed.

9 Emergent MOND phenomenology

The MOND phenomenology presented in this section relies on the separation between two regimes of the scalar field equation (7): (i) the linear FLRW sector, where the homogeneous

background $\varphi_0 = 0$ gives GR-identical cosmology (Sec. 8), and (ii) the nonlinear, quasi-static sector around bound structures, where the scalar is sourced by localized overdensities. The regime boundary is defined operationally by the scale hierarchy $k \gg aH/c$ and $\partial_t \ll c \nabla$. In the weak-field limit $|\varphi| \ll 1$ the full nonlinear d'Alembertian reduces to the same linear equation as in the perturbative treatment, so the MOND identification below is self-consistent. In ISPG this regime separation is physical, not cosmetic: the homogeneous background pressure history controls the cosmological sector, while bound rotating structures excite the local Bernoulli and rotational response channels that generate MOND. The companion MOND paper [46] develops this local bound-structure sector explicitly. In particular, the excess centripetal support of spiral galaxies is read here as the direct observational registration of the second channel carried by the rotating space medium. In ISPG this is direct evidence that space behaves as a real substance and that gravity is sourced by pressure gradients in that substance: galactic rotation creates a central rarefaction of the medium, the rarefaction produces an inward pressure gradient, and that gradient contributes genuine additional gravity. Faster coherent rotation deepens the central rarefaction and therefore strengthens the second channel, but only up to the self-regulating local vortex equilibrium that closes the MOND branch.

9.1 Scalar perturbations and Hubble damping

The scalar perturbation equation (65) on the FLRW background (with $\varphi_0 = 0$, so $\varphi = \delta\varphi$) is a driven, damped harmonic oscillator with natural frequency $\omega_k = kc/a$ and damping rate $\gamma_H = 3H/2$. The source $(8\pi G/c^2) \delta T$ varies on matter-gravity timescales $\sim H^{-1}$, much longer than ω_k^{-1} on sub-Hubble scales ($k \gg aH/c$), so the homogeneous regime-classification below is governed by the LHS alone; the full driven response is written in (65) and analysed in App. 9. The homogeneous form is

$$\ddot{\delta\varphi}_k + 3H \dot{\delta\varphi}_k + \omega_k^2 \delta\varphi_k = 0. \quad (67)$$

The ratio $Q \equiv \omega_k/\gamma_H = 2kc/(3aH)$ determines the regime (valid in the sub-Hubble adiabatic range $kc/a \gg H$, where the WKB approximation holds):

- $Q > 1$ (high k): oscillatory—the scalar field sustains pressure gradients and provides full inertial response;
- $Q < 1$ (low k , but still sub-Hubble): overdamped—Hubble friction suppresses the gradient and degrades the inertial response;
- $kc/a \lesssim H$ (super-Hubble): the adiabatic classification breaks down and the mode is *frozen* rather than relaxing exponentially, as expected for cosmological perturbations outside the Hubble radius.

The transition between the first two regimes occurs at $\omega_{k_J} = \gamma_H$, defining a Hubble-damping wavelength

$$\lambda_J = \frac{2\pi a}{k_J} = \frac{4\pi c}{3H}, \quad (68)$$

which is of order c/H —the Hubble scale. (We retain the symbol λ_J and the label **eq:jeans** for cross-reference continuity, but the scale is set by Hubble damping, not by the pressure-gravity balance of the astrophysical Jeans criterion.)

9.2 The critical acceleration a_0

The damped wave equation (67) contains two competing scales: the mode frequency ω_k and the Hubble friction H . The scalar field can sustain a gradient of characteristic length ℓ only if the corresponding mode is not overdamped. The longest coherent wavelength is set by the Hubble rate through the standard dispersion relation:

$$\lambda_H(t) = \frac{2\pi c}{H(t)}, \quad (69)$$

evaluated at the present epoch $H = H_0$ for the numerical estimate below. Given this wavelength identification, the factor 2π follows from the standard Fourier relation $k = 2\pi/\lambda$ between wavenumber and wavelength; the wavelength-vs-radius choice itself is a physical modeling decision, justified below by the coherence structure of the scalar medium.

Now consider a test particle at acceleration a . The gravitational length over which the field must maintain its gradient is $\ell(a) = c^2/a$. When $\ell > \lambda_H$ —i.e. when the required gradient extends beyond the Hubble coherence length—the scalar field can no longer provide a full Newtonian response. The critical acceleration at which this breakdown occurs is defined by the identification $\ell(a_0) = \lambda_H$, which is a physical linkage between the gravitational response scale and the cosmological coherence scale, not a mathematical identity:

$$\frac{c^2}{a_0} = \frac{2\pi c}{H_0} \implies \boxed{a_0 = \frac{c H_0}{2\pi} \approx 1.04 \times 10^{-10} \text{ m s}^{-2}}. \quad (70)$$

Demonstrated: the scaling $a_0 \propto c H_0$ follows from the overdamped/underdamped classification of the damped oscillator (67) and is robust up to an $O(1)$ coefficient set by the choice of demarcation criterion.

Coefficient selection. The Hubble-damping and response-timescale criteria show that only an $O(1)$ coefficient is at stake: they give $3cH/(4\pi)$ and $\sqrt{3}cH/(2\pi)$ respectively, while the Hubble-wavelength criterion gives $cH/(2\pi)$. The theory selects the Hubble wavelength $\lambda_H = 2\pi c/H$ because it is the Fourier-space coherence boundary of the scalar medium and, in the companion MOND paper [46], it is precisely this scale that closes the rotating-space transport sector. The factor 2π is therefore fixed by the coherence geometry rather than fit to data, and the PDE verification therein confirms the same normalization.

Direction constraint from the vortex-lag mechanism. Among the three criteria above, only the Hubble-wavelength criterion gives an instantaneous value $a_0^{\text{inst}} = cH_0/(2\pi) \approx 1.04 \times 10^{-10} \text{ m s}^{-2}$ *below* the observed $a_0^{(\text{obs})} \approx 1.2 \times 10^{-10} \text{ m s}^{-2}$. The vortex-lag mechanism of the companion MOND paper [46] enhances a_0^{inst} via past-weighted averaging $H_{\text{eff}} > H_0$, and can therefore close the gap only for criteria that *underpredict* the data. This provides a second, mechanism-level selection principle complementary to the Fourier identification: the two criteria which overpredict ($3cH/(4\pi)$ and $\sqrt{3}cH/(2\pi)$) would require a lag correction of the wrong sign and are therefore excluded once the vortex-lag closure is imposed.

The observed value is $a_0^{(\text{obs})} \approx 1.2 \times 10^{-10} \text{ m s}^{-2}$ [7], about 15% above the instantaneous prediction. The companion MOND paper [46] shows that this offset is *not* a free-parameter residual but a physical effect: the **vortex-lag mechanism**. Because the galactic vortex retains a memory of the Hubble rate throughout its lifetime, the effective acceleration scale is $a_0^{\text{eff}} = c H_{\text{eff}}/(2\pi)$ with $H_{\text{eff}} = \ln(1 + z_{\text{form}})/t_{\text{lookback}}$, the time-averaged expansion rate since the vortex formed. For a typical spiral whose disk stabilised at $z_{\text{form}} \approx 0.58$, one obtains

$a_0^{\text{eff}} \approx 1.20 \times 10^{-10} \text{ m s}^{-2}$. This specific numerical match is by construction of the z_{form} calibration and is not itself an independent prediction; the non-trivial consistency check is that the calibrated $z_{\text{form}} \approx 0.58$ lies within the morphology-anchored window $z \sim 0.5\text{--}0.7$ for local-spiral disk stabilisation [99, 100]. The genuinely falsifiable content is the *age correlation* of the effective a_0 : galaxies whose disks settled earlier should show systematically higher a_0 than those that settled more recently, at the same redshift of observation.

9.3 Interpolating function

Given a_0 and $x \equiv a/a_0$, we use the compact two-channel form

$$a = a_N + a_h, \quad (71)$$

in which the local baryonic Bernoulli channel and the rotating-space vortex channel act jointly. The resulting transport closure, derived analytically in the companion MOND paper [46] (Secs. 6.4–6.8 therein: Markov coherence-fraction ODE, coherent-vortex limit of the swirl source, and Chebyshev Poisson-BVP numerical verification), is

$$\frac{a_h}{a_N} = \frac{a_0}{a}. \quad (72)$$

Then

$$a = a_N \left(1 + \frac{a_0}{a}\right) \implies a_N = \frac{a}{1 + a_0/a} \equiv \mu\left(\frac{a}{a_0}\right) a, \quad (73)$$

hence

$$\boxed{\mu(x) = \frac{x}{1+x}}. \quad (74)$$

This is the “simple” MOND interpolating function [10]. The detailed Bernoulli+vortex transport derivation is presented in the companion MOND paper [46], together with its physical assumptions and sensitivity analysis. There the same closure is obtained both from the vortex transport balance and from the source-side Poisson law of the coherent swirl sector, whose axisymmetric disk implementation now uses a genuinely nonlocal enclosed-mass kernel. Within this reading, spiral-galaxy rotation curves do not merely motivate a MOND fit: they directly register the second channel of the ISPG medium. They therefore register the pressure-medium origin of gravity itself: the faster the coherent galactic rotation, the deeper the central rarefaction and the larger the transported inward support, until vortex equilibrium fixes the local response.

9.4 Tully–Fisher relation

For a circular orbit ($a = v^2/r$, $a_N = GM/r^2$) in the deep-MOND regime $\mu(x) \approx x$, the modified dynamics $\mu(a/a_0) \cdot a = a_N$ gives $a^2/a_0 = GM/r^2$. With $a = v^2/r$:

$$\boxed{v^4 = a_0 G M, \quad v_{\text{flat}} = (a_0 G M)^{1/4}}. \quad (75)$$

This is the baryonic Tully–Fisher relation [6, 7], here obtained in the deep-MOND limit.

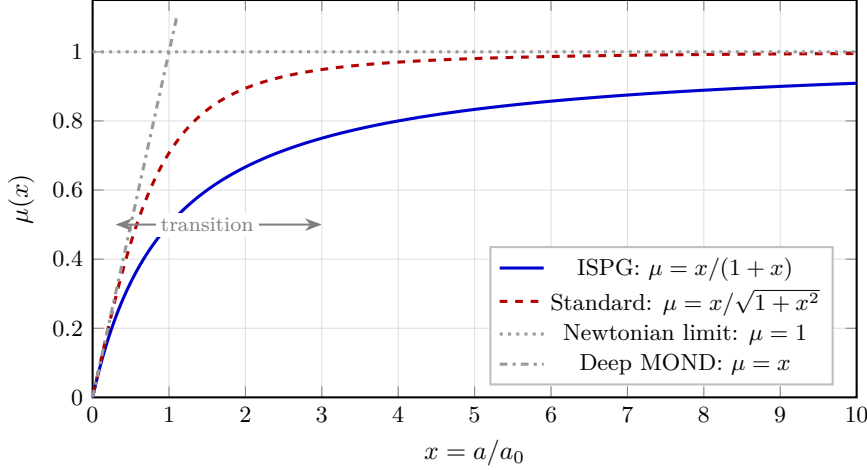


Figure 7: MOND interpolating functions. The solid curve is the model result $\mu = x/(1+x)$; the dashed curve is the “standard” function $\mu = x/\sqrt{1+x^2}$. Both approach unity for $x \gg 1$ (Newtonian regime) and linearity for $x \ll 1$ (deep-MOND regime).

9.5 Redshift evolution

Since a_0 is set by the global pressure history, whose observational proxy is the Hubble rate, at redshift z the relevant rate is $H(z)$:

$$a_0(z) = \frac{c H(z)}{2\pi}. \quad (76)$$

This is the *instantaneous baseline law*: it gives the coherence-set value associated with the background Hubble rate at the observation epoch. Galaxies therefore do react to the global pressure decrease that is usually read observationally as cosmic expansion: the same background shift changes their clock rate, rod scale, effective mass scale, and the coherence-set value of a_0 . The full galactic prediction is slightly more refined: the companion MOND paper [46] shows that finite vortex memory replaces $H(z)$ by a lifetime-averaged H_{eff} , so the effective observed value is $a_0^{\text{eff}} = cH_{\text{eff}}/(2\pi)$. Equation (76) is therefore the zeroth-order redshift law, recovered for newly formed vortices and as the baseline coherence scale entering the local closure. In Λ CDM, $H(z) = H_0\sqrt{\Omega_m(1+z)^3 + \Omega_\Lambda}$, so the instantaneous baseline $a_0(z)$ grows monotonically with z . At $z = 1$ the baseline ratio is $a_0(1)/a_0(0) \approx 1.78$, predicting $\sim 16\%$ higher flat rotation velocities at fixed baryonic mass. The full vortex-lag prediction is weaker and depends on formation epoch [46], but the central distinction from standard MOND remains: a_0 is not constant. This distinguishes the present theory from standard MOND (which assumes constant a_0) and is testable with high-redshift rotation curves from JWST [52] and the baryonic Tully–Fisher relation measured by Euclid [53].

Robustness of the baryonic-mass comparison.—The prediction “at fixed baryonic mass” is operationally well-defined because the local undetectability principle (Sec. 10.3) ensures that mass-to-light ratios are independent of the conformal factor. Whether one works in the FLRW frame ($\varphi_0 = 0$, constant bare masses, expanding scale factor) or in the ontologically preferred matter-contraction description where coordinate distances are fixed and effective masses and atomic length scales rescale with the global pressure (Sec. 10.3), the dimensionless Tully–Fisher

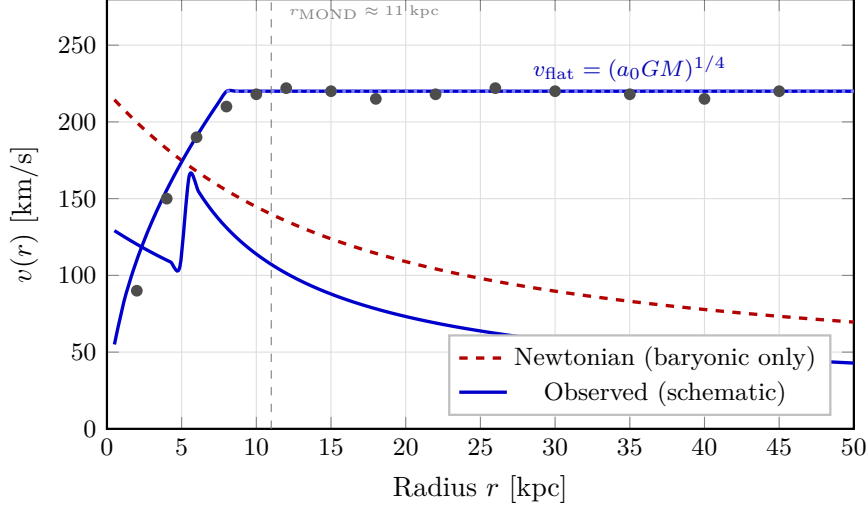


Figure 8: Schematic galaxy rotation curve. The dashed curve shows the Newtonian prediction from baryonic mass alone; the solid curve represents the observed flat profile. In the present theory the transition occurs at $r_{\text{MOND}} \approx \sqrt{GM/a_0}$, and the asymptotic velocity is $v_{\text{flat}} = (a_0 GM)^{1/4}$, obtained from the deep-MOND relation with no galaxy-by-galaxy fit parameters.

observable $v^4/(GM_{\text{obs}})$ is invariant. The prediction (76) therefore requires no separate accounting for cosmological mass evolution.

9.6 Disk formation from rotational anisotropy

The same off-diagonal rotational anisotropy associated with the bound-structure MOND sector also predicts the formation of galactic disks. The detailed rotating-medium closure is developed in the companion MOND paper [46]; here we record only its geometric consequence. The gravitomagnetic metric component $g_{t\phi} \propto \sin^2 \theta$ is maximal in the equatorial plane and vanishes on the polar axis. The transported channel φ_h inherits this anisotropy, generating a vertical restoring acceleration $g_z \propto -z/\rho$ toward the midplane with oscillation frequency $\nu_z^2 \sim 2a_0/\rho$. The resulting disk scale height is

$$h \sim \sigma_z \sqrt{\frac{\rho}{2a_0}}, \quad (77)$$

where σ_z is the vertical velocity dispersion. For the Milky Way at $\rho = 8$ kpc with $\sigma_z \approx 20$ km s⁻¹, this gives $h \approx 0.6$ kpc, consistent with observations. A falsifiable prediction follows: low-surface-brightness galaxies deep in the MOND regime should have relatively thinner disks at fixed σ_z/v_{rot} than predicted by angular momentum conservation alone [46].

9.7 Status and extensions of the MOND sector

The spiral-galaxy MOND closure is already fixed in the present paper and the companion MOND paper [46]. The items below summarize that completed core result and the next extensions of the same mechanism:

1. **Spiral-galaxy closure (*completed*).** The interpolating function $\mu(x) = x/(1+x)$ and the normalization $a_0 = cH/(2\pi)$ are now closed within the mature spiral-galaxy

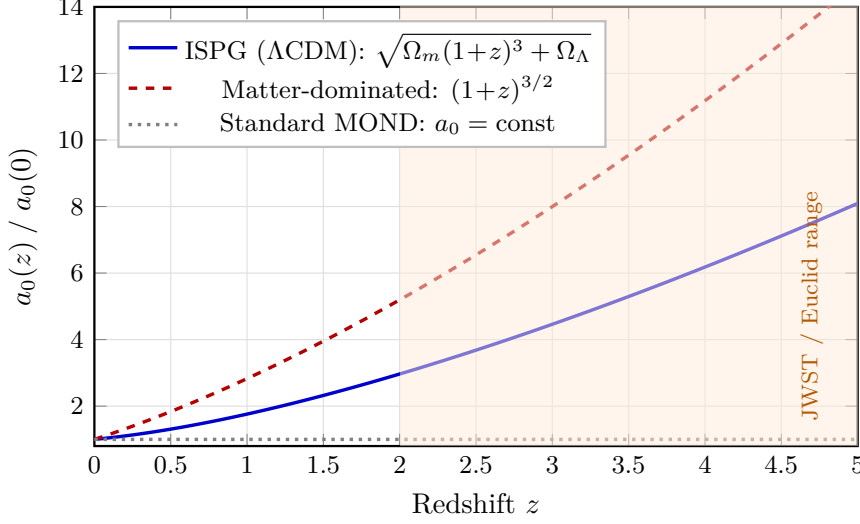


Figure 9: Redshift evolution of the critical acceleration $a_0(z)/a_0(0)$. The solid curve assumes Λ CDM expansion history; the dotted line shows matter domination. Standard MOND (horizontal dashed) assumes $a_0 = \text{const}$. The shaded region indicates the JWST/Euclid observational window.

framework of the companion MOND paper [46]. The transport-side algebraic channel is benchmarked by Chebyshev spectral collocation therein, while the independent source-side and operator-side checks are stated explicitly below at the $\sim 10\%$ RMS level. The local self-consistency parameter now obeys $C_{\text{eff}} = A_{\text{vort}} + \mathcal{O}(\delta_{\text{sp}}) + \mathcal{O}(\delta_N)$, so on the mature occupied branch ($Q_{\text{occ}} \gg 1$, $\delta_{\text{sp}} \ll 1$, $N_{\text{cell}} \gg 1$) one has $C_{\text{eff}} \simeq 1$ from instantaneous vortex equilibrium, and the operating transport rate $\Omega_{\text{tr}} = a_0/c$ is fixed by the Hubble-boundary coherence theorem, i.e. by the causal crossing time of the Hubble coherence boundary in the rotating-medium field equation, while the cosmological secular integral supplies the background pressure-history diagnostic with the coupling exponent β derived (not assumed) from the quantum feedback ODE. The rotational loading law $Z_{\text{rot}} = (1 + \sqrt{X/X_0})^{-1}$ is MaxEnt-selected from a binary free/load partition with loaded-state measure linear in $\sqrt{X/X_0}$, and the activation profile is solved self-consistently together with the total field rather than being evaluated on a pre-inserted MOND branch. The result is universal across galaxy parameters, converged under N -doubling, and satisfies the correct asymptotic limits to machine precision in the algebraic channel (exact given the solved C_{eff} profile). The MOND transport sector also now admits an explicit 2D source-side enclosed-mass Green kernel in disk geometry with $\rho_{\text{core}} = r_M/j_{0,1}$ fixed by regularity and κ_G fixed by the outer asymptotic Gauss condition, and the transport-side spectral closure, source-side kernel, and operator-side AQUAL solve have now been cross-checked in one unified benchmark suite. On the matched axisymmetric box the native field RMS errors are 5.7×10^{-4} (transport), 1.32×10^{-1} (source-side boundary kernel), and 1.09×10^{-1} (AQUAL), while the direct source-versus-operator mismatch is $\simeq 7.4 \times 10^{-2}$. Separately, on a widened static axisymmetric box $(20 r_M) \times (10 r_M)$ where κ_G is fixed on the disjoint deep-halo annulus $10 \leq \xi \leq 20$ and scored only on $0.3 \leq \xi \leq 10$, the standalone source-side and operator-side native RMS errors drop to $\sim 1.01 \times 10^{-1}$ and $\sim 9.18 \times 10^{-2}$, respectively,

with the source-side transition and outer-halo subwindow RMS levels both staying at $\sim 1.0 \times 10^{-1}$. A first explicitly time-dependent quasi-static 2 + 1D relaxation solve of the same source-side channel has also been carried out on the axisymmetric R - z grid, and the undropped hyperbolic 2 + 1D propagation with the $\ddot{\varphi}$ term retained has now also been verified for the same derived nonlocal source-side closure. Finally, that source has also been promoted from a fixed derived profile to a fully coupled rotating-medium law: the local vortex-order parameter obeys its hysteretic kinetic equation, the ordered tail budget is rebuilt as $A_{\text{vort}} T^{(m)}$, and the nonlocal halo source is reconstructed from the enclosed coherent-budget fraction at every time step inside the hyperbolic 2+1D solve. The fixed-source hyperbolic run remains within $\sim 2.1 \times 10^{-4}$ of the selected static source-side branch, and the coupled run remains within $\sim 3.8 \times 10^{-2}$ of the selected static source-side branch and within $\sim 1.4 \times 10^{-1}$ of the algebraic target and $\sim 1.2 \times 10^{-1}$ of the transport target, so the mature spiral-galaxy MOND sector is now internally consistent at the PDE level within its closure framework (Z_{rot} , impedance-matching invariant, $\Omega_{\text{tr}} = a_0/c$). The remaining items below are therefore extensions/applications of that closure rather than missing steps inside the spiral-galaxy derivation itself. The MOND transport coefficient and the background medium-pressure history ($w_{\text{bg}} > -1$) are linked by the master formula $3\delta w_0(1 + z_f)^\alpha/\alpha \approx 8$, yielding a prediction testable by DESI/Euclid.

2. **Elliptical galaxies.** The companion MOND paper [46] now shows that pressure-supported systems (ellipticals, dwarf spheroidals) obey the same MOND scaling through a turbulent micro-vortex mechanism: the isotropic velocity dispersion σ generates a Reynolds-stress analogue in the scalar field, with the transition at $g \sim a_0$ fixed by the cosmological coherence length. The predicted relation $\sigma^4 = K_\sigma G M a_0$ (baryonic Faber–Jackson) is consistent with observations; the companion paper now also makes the normalization convention explicit by separating the exact global deep-MOND virial coefficient from the projected Sérsic/aperture coefficient. What remains open is the full convective-cell PDE derivation of the detailed projected kinematics.
3. **Galaxy clusters.** Standard MOND underpredicts cluster masses by a factor ~ 2 – 3 [11]. The IS relaxation mechanism (Sec. 11.2) addresses the Bullet Cluster mass–light offset, and the companion MOND paper [46] now gives a quantitative benchmark for the residual cluster binding. The required scalar-field retention time for 50% binding is $\tau_{\text{ret}}^{(50\%)} \approx 0.42 t_{\text{dyn}}$ —less than half of one dynamical time, naturally achievable in a gravitationally bound system. The corresponding effective mean free path $\ell_{\text{mfp}} \approx 16$ kpc matches the observed scale of ICM cold fronts and sloshing spirals, which provide a concrete, parameter-free scattering mechanism. What remains is a dedicated 3D numerical solve of the scalar-field energy density on a realistic cluster potential, which is a computational task rather than a conceptual gap.
4. **Explicit regime matching.** The physical separation between the linear FLRW sector and the nonlinear quasi-static sector is already built into the pressure-medium interpretation used throughout the theory. A dedicated matching calculation would make this separation explicit in a single equation-level bridge across the two regimes.

10 Physical interpretation: inertia, mass equivalence, effective mass, and the invariance of c

The preceding sections derived the quantitative predictions of the theory. Here we draw out three physical consequences of the bi-conformal scalar-tensor structure that bear on foundational issues: the dynamical origin of inertia, the identity of gravitational and inertial mass, and the local invariance of the speed of light.

10.1 Inertia as scalar-field backreaction

In Newtonian mechanics, inertia is an intrinsic, passive property of matter. In GR, it is geometrized through the geodesic equation, but the question of *why* matter resists acceleration receives no deeper answer than the structure of the metric itself.

In the present theory the scalar field φ provides a dynamical mechanism for both gravitational attraction and inertial resistance. Section 3.3 showed that free-fall itself is matter-wave refraction: every oscillon turns toward regions of lower c_{coord} (Eq. 24). Inertia is the complementary transient effect. A body of mass M at rest in a static background maintains the spherically symmetric configuration $\varphi = -r_s/r$. When the body accelerates, this configuration must readjust. Because φ propagates at finite speed c (Sec. 5.1), the readjustment is not instantaneous: during the transient the scalar-field gradient develops a fore-aft asymmetry that opposes the acceleration.

The effect is the field-theoretic analog of hydrodynamic *added mass*: a body accelerating through a fluid experiences a retarding force because the fluid must rearrange around it [58]. Here the “fluid” is the scalar field, and the retarding force is what we identify as inertia.

In the weak-field limit the gravitational force on a test particle in an external potential φ_{ext} is

$$\mathbf{F}_{\text{grav}} = -m \nabla \varphi_{\text{ext}}, \quad (78)$$

where m is the gravitational mass determined by the particle’s own contribution to the field via Eq. (7). When the same particle accelerates through its own field, the transient asymmetry $\delta\varphi$ creates a backreaction

$$\mathbf{F}_{\text{inertial}} = -m \nabla(\delta\varphi), \quad (79)$$

with the *same* coupling m , since both expressions derive from the same Lagrangian (3). This universality of coupling is what the next subsection formalizes as the identity of gravitational and inertial mass.

10.2 Identity of gravitational and inertial mass

The equality $m_g = m_i$ is one of the most precisely tested facts in physics: the Eötvös parameter is constrained to $\eta < 10^{-15}$ by the MICROSCOPE experiment [54]. In GR with minimally coupled matter the equality is automatic at the Lagrangian level: matter couples to a single metric with a single mass parameter, and gravitational and inertial responses inherit that same parameter. The same is true in ISPG; the additional structural content here is that the common mechanism is identified explicitly—both responses are mediated by the same scalar field φ through the same field equation.

Gravitational mass.—A body placed in an external gradient $\nabla\varphi_{\text{ext}}$ experiences a force which, in the weak-field Newtonian limit, takes the form $F = -m \nabla\Phi_N$ with $\Phi_N = \frac{1}{2}c^2\varphi$

(Sec. ??); equivalently $F = -\frac{mc^2}{2} \nabla \varphi_{\text{ext}}$. The coupling m is fixed by the source term in the field equation (7): it is the body's contribution to the trace T of the energy-momentum tensor. The schematic shorthand $F = -m \nabla \varphi_{\text{ext}}$ used informally elsewhere should be understood with the implicit $c^2/2$ factor reinstated.

Inertial mass.—A body undergoing acceleration $\mathbf{a}$ generates a transient field asymmetry $\delta\varphi$. The restoring force is $F = -m \nabla(\delta\Phi_N) = -\frac{mc^2}{2} \nabla(\delta\varphi)$, with the same coupling m , because the field equation that governs $\delta\varphi$ is the same equation that determines the static configuration.

Since both forces derive from the same action (3) with minimal coupling of matter to the bi-conformal metric, the proportionality constant relating force to field gradient (gravitational mass) is identically equal to the proportionality constant relating force to acceleration (inertial mass). The equality $m_g = m_i$ is therefore a structural consequence of the unified scalar-field mechanism in ISPG; the result has the same logical status as in GR with minimal coupling, and the additional ISPG content is the explicit identification of the common mechanism (retarded scalar self-field) rather than a stronger derivation. The retarded self-field calculation is carried out explicitly in Appendix 17, which derives Newton's second law $F = ma$ from the field equations. The quantum extension [47] provides an independent microscopic derivation via the Bernoulli pressure-deficit mechanism, confirming the result at the operator level.

From test particles to composite bodies.—The present argument establishes $m_g = m_i$ for test particles in an external field. The corresponding statement for self-gravitating composite bodies—the strong equivalence principle, including the absence of a Nordtvedt effect—requires the exact shift symmetry of the scalar sector and is treated separately in Sec. 11.4 (with the rigorous proof of the universal sensitivity $s = 1/2$ given in Appendix 16).

Status of the η prediction.—Within the order to which the present derivation is carried, ISPG predicts $\eta = 0$ identically: no composition-dependent deviation arises from the unified retarded-scalar mechanism. The MICROSCOPE bound is therefore satisfied trivially in this approximation, and not used here as a discriminator. A controlled prediction for a possible higher-order, microscopic composition-dependent η would require a calculation beyond the scope of this paper and is deferred to future work.

This result connects directly to the MOND phenomenology of Sec. 9. Using Eq. (71) with $a_h = (a_0/a) a_N$, one obtains $\mu(x) = x/(1+x)$: for $a \gg a_0$, $a_h \ll a_N$ and $\mu \rightarrow 1$; for $a \ll a_0$, $a_h \gg a_N$ and $\mu \approx a/a_0$. In this reading, MOND is the bound-structure two-channel gravitational response of the ISPG medium, in which the baryonic Bernoulli field is supplemented by a rotating-space channel in galactic structures.

10.3 Position-dependent effective mass

The bi-conformal structure implies that the effective mass of a body—its gravitational influence as measured by a distant observer—depends on the local value of the scalar field. Consider a particle of rest mass m_0 at rest at a position where the scalar field takes the value φ . Its worldline interval is $ds = c e^{\varphi/2} dt$, and its energy as measured at spatial infinity ($\varphi = 0$) is

$$E_\infty = m_0 c^2 e^{\varphi/2}. \quad (80)$$

The effective mass, defined as $m_{\text{eff}} \equiv E_\infty/c^2$, is therefore

$$\boxed{m_{\text{eff}}(\varphi) = m_0 e^{\varphi/2}}. \quad (81)$$

In a gravitational field ($\varphi < 0$), $m_{\text{eff}} < m_0$: a body deep in a potential well contributes less to the total gravitational mass of the system than the same body at infinity.

Local undetectability.—A local observer uses clocks that tick at rate $d\tau = e^{\varphi/2} dt$ and rulers of proper length $d\ell = e^{-\varphi/2} dr$. The locally measured rest energy of the particle is

$$E_{\text{local}} = m_0 c_{\text{local}}^2 = m_0 c^2, \quad (82)$$

since $c_{\text{local}} = c$ by Eq. (93). All local masses—the test particle, the reference kilogram, the atoms in the scale—are multiplied by the *same* conformal factor $e^{\varphi/2}$. The ratio of any two masses is therefore position-independent:

$$\frac{m_{\text{eff},A}}{m_{\text{eff},B}} = \frac{m_{0,A} e^{\varphi/2}}{m_{0,B} e^{\varphi/2}} = \frac{m_{0,A}}{m_{0,B}}. \quad (83)$$

No local experiment can detect the variation of m_{eff} , for the same reason that no local experiment detects the variation of the coordinate speed of light: the measuring apparatus participates in the same conformal rescaling as the object being measured.

Cosmological frame duality.—The positional result extends to the cosmological setting. In the FLRW frame, $\varphi_0 = 0$ (Sec. 8.1) and bare masses are epoch-independent; cosmic expansion is carried entirely by $a(t)$. A conformal rescaling to coordinates in which $a \equiv 1$ maps the expansion into a time-dependent contraction of effective masses and atomic length scales. Since every local measuring apparatus participates in the same rescaling, no epoch-dependent experiment can distinguish the two descriptions; all dimensionless observables—the baryonic Tully–Fisher relation, the CMB power spectrum—are frame-invariant (Sec. 9.5).

In the ISPG ontology, however, the two frames are not on equal footing. The primary background variable is the homogeneous pressure shift $\varphi_{\text{bg}} = \ln(P_{\text{stat}}/P_{\text{max}})$; the observed Hubble rate serves as a practical proxy for the global rate of static-pressure decrease [46]. The “matter-contraction” picture—in which coordinate distances are fixed while effective masses, atomic length scales, and clock rates rescale with the pressure—is, in the ISPG interpretation, the ontologically natural description of a physical medium whose state is defined by its local pressure. The FLRW frame is a mathematically equivalent reformulation whose convenience lies in matching the standard cosmological formalism. The two frames are observationally equivalent at the level of dimensionless quantities; the preference for the matter-contraction frame as the “primary” one is an interpretational choice within ISPG, not a result derived from an observation that distinguishes the two descriptions.

Explicit evolution of the background pressure.—In the matter-contraction frame, the observable redshift z is produced by the epoch-dependent effective mass: $1 + z = m_{\text{eff}}(t_{\text{emit}})/m_{\text{eff}}(t_0) = e^{[\varphi_{\text{bg}}(t_{\text{emit}}) - \varphi_{\text{bg}}(t_0)]/2}$. Since in the FLRW frame $1 + z = 1/a(t_{\text{emit}})$, matching the two gives

$$\varphi_{\text{bg}}(t) = -2 \ln a(t), \quad \varphi_{\text{bg}}(z) = 2 \ln(1 + z), \quad (84)$$

with $\varphi_{\text{bg}}(t_0) = 0$. Differentiation yields the evolution equation:

$$\boxed{\frac{d\varphi_{\text{bg}}}{dt} = -2H(t)}. \quad (85)$$

The Hubble rate is therefore the rate of global pressure decrease, measured in coordinate time—in agreement with the proxy identification [46].

The consequences for the early universe ($z \gg 1$) follow directly from the bi-conformal scaling laws:

$$\begin{aligned}
m_{\text{eff}}(z)/m_{\text{eff}}(0) &= (1+z), \\
L_{\text{coord}}(z)/L_{\text{coord}}(0) &= (1+z), \\
c_{\text{coord}}(z)/c_{\text{coord}}(0) &= (1+z)^2, \\
(d\tau/dt)(z)/(d\tau/dt)(0) &= (1+z).
\end{aligned} \tag{86}$$

At $z = 10$ (galaxy-disk formation epoch): masses were $11\times$ larger, oscillons $11\times$ wider, the coordinate speed of light $121\times$ faster, and local clocks ran $11\times$ faster than today. At $z = 1090$ (last scattering): masses $\sim 10^3\times$ larger, $c_{\text{coord}} \sim 10^6\times$ faster. These numbers quantify the intuitive statement that the early universe was “faster and bigger.”

Gauge status.—The coordinate factors quoted above ($c_{\text{coord}} \propto (1+z)^2$, $L_{\text{coord}} \propto (1+z)$, etc.) are specific to the matter-contraction frame. In the FLRW frame the coordinate speed of light is constant and the spatial coordinates expand instead. The two descriptions are observationally equivalent: every locally measurable quantity—including the speed of light measured with local rulers and clocks (Sec. 10.4)—is frame-independent. No observation discriminates between the two coordinate pictures; the numerical values above should therefore be read as a coordinate-language restatement of the universal conformal evolution, not as an asserted physical speed at which light travelled in the early universe.

Relation to the MOND coupling ratio.—The companion MOND paper [46] introduces a pressure ratio $P_{\text{stat}}(z)/P_{\text{stat}}(0) \approx 9.9$ at $z = 10$ (Eq. (73) of [46]), derived from the quantum-feedback closure. This quantity measures the *intrinsic medium-energy evolution* driven by the depletion of the ν_0 -mode reservoir and the growth of the echo background. It is distinct from the total conformal rescaling $(1+z)^2 = 121$, which includes both the medium evolution and the geometric structure of the cosmological solution. The two are related through the coupling-ratio formula (Eq. (70) of [46]); for observational predictions such as the baryonic Tully–Fisher relation and the vortex-lag effect, the relevant quantity is the dynamical coupling ratio, not the total geometric rescaling.

Gravitational binding energy.—For a self-gravitating system of total bare mass M_0 , the *bare-mass* Komar integral (lapse $e^{\varphi/2}$ times proper volume element $e^{-3\varphi/2}$, weighted by the rest-mass density ρ_0) as measured at infinity is

$$M_{\text{Komar}}^{(0)} \equiv \int \rho_0 e^{\varphi/2} e^{-3\varphi/2} d^3x = \int \rho_0 e^{-\varphi} d^3x, \tag{87}$$

where the factor $e^{-3\varphi/2}$ is the proper volume element in the spatial metric $e^{-\varphi}\delta_{ij}$.

Scope of Eq. (87). Equation (87) is the *bare-mass* (or test-matter) Komar integral: it weights the rest-mass density ρ_0 by the gravitational-redshift/proper-volume factors alone. It coincides with the Tolman–Komar stationary perfect-fluid mass $M_{\text{Komar}}^{\text{TK}} = \int (\rho_0 + 3p/c^2) e^{-\varphi} d^3x$ in the dust limit $p \ll \rho_0 c^2$, and is therefore adequate for the weak-field binding-energy discussion below and for the kinematic sensitivity argument of Sec. ???. For neutron-star interiors ($p/(\rho_0 c^2) \sim 0.2$ – 0.3 near the centre), the pressure contribution $3p/c^2$ induces a correction of order $\sim 30\%$; incorporating it into a fully perfect-fluid sensitivity analysis is an open task (Appendix 16, “open tasks”). In the weak field ($|\varphi| \ll 1$) and at leading order in $p/(\rho_0 c^2)$: $M_{\text{Komar}}^{(0)} \approx M_0(1 - \langle \varphi \rangle)$, where $M_0 \equiv \int \rho_0 d^3x$ (coordinate volume) and $\langle \varphi \rangle < 0$ is the mass-averaged potential. The

physically meaningful comparison is with the total proper rest mass $M_{\text{proper}} \equiv \int \rho_0 e^{-3\varphi/2} d^3x \approx M_0(1 - \frac{3}{2}\langle\varphi\rangle)$:

$$M_{\text{Komar}}^{(0)} - M_{\text{proper}} \approx \frac{1}{2} M_0 \langle\varphi\rangle < 0, \quad (88)$$

reproducing the standard gravitational binding-energy deficit $\Delta M = -E_{\text{bind}}/c^2$, here arising from the scalar-field dependence of the effective mass rather than from a separate binding-energy calculation.

Physical interpretation.—In the scalar-field picture, mass is not an intrinsic, immutable property of matter. It is a measure of the coupling between matter and the scalar field—the rate at which a body sources the field equation (7). A body in a region of strong field ($\varphi \ll 0$) sources the field more weakly (per unit bare mass) than the same body at infinity. This is because the bi-conformal metric simultaneously slows the body’s internal dynamics (time dilation $e^{\varphi/2}$) and contracts its spatial extent (by $e^{\varphi/2}$), reducing its effective interaction cross-section with the scalar field. The two effects—time dilation and spatial contraction—are reciprocal manifestations of the same bi-conformal factor, exactly paralleling the invariance of c derived in Sec. 10.4.

Scalar sensitivity of compact bodies.—For a self-gravitating body of compactness $\mathcal{C} \equiv GM/(c^2 R)$, the scalar sensitivity is defined as

$$s \equiv -\frac{1}{2} \left. \frac{\partial \ln m_{\text{eff}}}{\partial \varphi_{\text{ext}}} \right|_{\varphi_{\text{ext}}=0}, \quad (89)$$

where φ_{ext} is the external (background) scalar field. In Eq. (89) the symbol m_{eff} denotes the gravitational (Komar) mass of the body that controls its response to a rigid shift of the external field, evaluated at infinity; this is the quantity whose shift-rescaling is the subject of the structural argument that follows. It is *distinct* from the test-particle Killing-energy effective mass of Eq. (81), which rescales universally as $e^{+\delta/2}$; the two coincide with the bare rest mass in the limit $\varphi \rightarrow 0$. A key structural property of the bi-conformal framework is that the TOV pressure-gradient equation depends only on $d\varphi/dr$, not on φ itself; under a rigid shift $\varphi \rightarrow \varphi + \delta$ the density and pressure profiles are therefore strictly unchanged (Appendix 16, Sec. “TOV-level shift invariance”). The coupled scalar equation in the presence of matter carries an explicit $e^{-\varphi}$ prefactor on its source side and is shift-invariant only at leading order in $|\varphi|$ (Appendix 16, Sec. “Coupled-system shift symmetry”). Combined with the universal conformal redshift $m_{\text{eff}} = m_0 e^{\varphi/2}$, this selects

$$s = 1/2 \quad (\text{leading order in } |\varphi|, \text{ all bodies; Appendix 16}). \quad (90)$$

The physically measurable *excess* sensitivity is then

$$s_{\text{excess}} \equiv s - \frac{1}{2} = 0 + O(|\varphi|), \quad (91)$$

which is *exact* for vacuum compact objects (from $\nabla^2\varphi = 0$, main theory Sec. 7 and Appendix 1) and holds at leading order in $|\varphi|$ for matter-filled bodies. An exact, non-perturbative cancellation of the residual $O(|\varphi|)$ correction between bodies of arbitrary compactness is *not* established by the classical coupled equations and is an open task (Appendix 16).

At the leading-order level, $s_A \simeq s_B \simeq 1/2$ for every pair, so the scalar dipole moment of any binary is suppressed: $(s_A - s_B)^2 \simeq 0$. The scalar dipole radiation channel is therefore not expected to dominate over the tensor quadrupole in binaries of moderate compactness—consistent with the non-detection of anomalous decay in pulsar-timing and LIGO/Virgo data (Sec. 5.5).

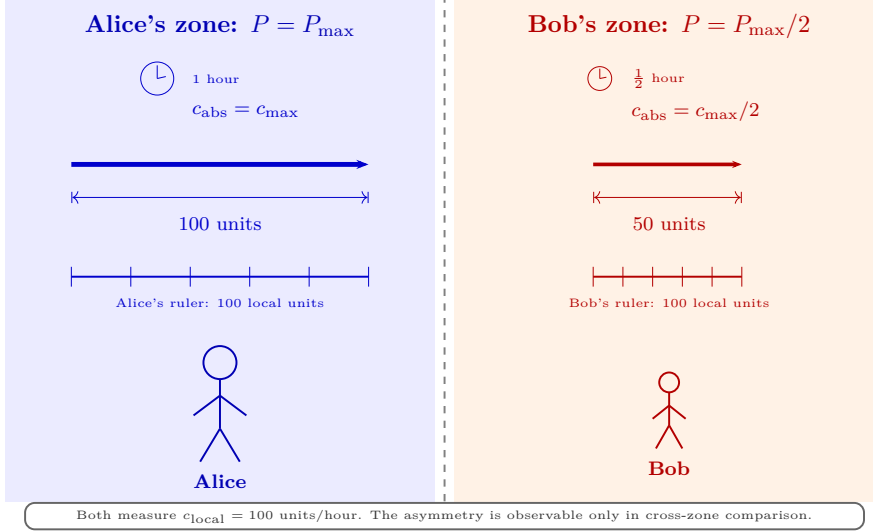


Figure 10: Thought experiment illustrating local invariance of c . Alice (strong field, large φ) and Bob (weak field, small φ) each measure the speed of light with local rods and clocks. Both obtain c exactly, because the bi-conformal factors e^φ and $e^{-\varphi}$ cancel in the ratio $\Delta x/\Delta t$. A distant observer, however, sees Alice’s light travel more slowly in coordinate terms.

This is the leading-order counterpart of local undetectability: because the conformal rescaling $m_{\text{eff}} = m_0 e^{\varphi/2}$ is universal, all bodies respond identically to a background field shift at leading order, and the differential (observable) response is small. The weak equivalence principle is exactly preserved (all test particles follow geodesics of $g_{\mu\nu}$); the strong equivalence principle is preserved in the scalar sector at leading order, with the exact statement contingent on closure of the open task above. The derivation, the coupled linearisation, the sectoral no-hair scope, and a confrontation with pulsar-timing bounds are developed in Appendix 16.

10.4 Local invariance of the speed of light

The bi-conformal metric (2) implies a position-dependent coordinate speed of light. From $ds^2 = 0$ along a radial null ray:

$$\frac{dr}{dt} = c e^\varphi. \quad (92)$$

Near a gravitating body ($\varphi < 0$), light propagates more slowly in coordinate terms.

A local observer, however, measures time with a clock whose proper rate is $d\tau = e^{\varphi/2} dt$, and distance with a ruler whose proper length element is $d\ell = e^{-\varphi/2} dr$. The locally measured speed is therefore

$$c_{\text{meas}} = \frac{d\ell}{d\tau} = \frac{e^{-\varphi/2} dr}{e^{\varphi/2} dt} = e^{-\varphi} \cdot c e^\varphi = c. \quad (93)$$

The bi-conformal structure guarantees that local measurements of the speed of light are invariant, while the coordinate propagation speed varies with position. The two conformal factors e^φ (temporal) and $e^{-\varphi}$ (spatial) are not independent—they are the unique combination that produces $\gamma = \beta = 1$ (Sec. 3.2)—and their ratio exactly cancels in any local velocity measurement.

A sharper consequence emerges when one distinguishes scalar from tensor perturbations. A change in the scalar field ($\delta\varphi \neq 0$) rescales the metric isotropically: clocks, rulers, and light all

respond together, rendering the perturbation locally undetectable. This is why a Michelson–Morley-type experiment [57] returns a null result in a scalar-field background: the interferometer arms adapt to the same field that governs the propagation of light. This is the ISPG realization of the general fact that any metric theory with universal minimal coupling of matter to a single metric returns a null Michelson–Morley result; the framing above is an explicit identification of the isotropic scalar mode as the mode that does the compensating, not a prediction unique to ISPG.

Gravitational waves, by contrast, are transverse-traceless tensor perturbations satisfying $\delta\varphi = 0$: they distort spatial distances anisotropically without altering the scalar field. Since the measuring apparatus adapts only to changes in φ , and φ is unchanged, the apparatus does *not* compensate. The anisotropic strain is therefore physically measurable—precisely the effect detected by LIGO [4].

The theory thus offers a unified account: the null result of Michelson–Morley (isotropic scalar perturbation $\rightarrow$ compensated) and the positive detection of gravitational waves (anisotropic tensor perturbation, $\delta\varphi = 0 \rightarrow$ not compensated) both follow from the same bi-conformal structure (2).

11 Discussion

11.1 How do local systems know about H_0 ?

Any framework that links a galactic quantity (a_0) to a cosmological one (H_0) faces the question: through what mechanism does the Hubble rate enter local dynamics?

In this theory no additional signal transfer is postulated. The scalar field satisfies $\square\varphi = 0$ on the full spacetime, and on FLRW the term $3H\dot{\varphi}$ in Eq. (67) sets the coherence scale $\lambda_H \sim c/H$. Local galactic dynamics inherit this scale through boundary conditions of the global solution, not through causal communication from the horizon.

11.2 The dark matter question

At linear order (Sec. 8) the theory is GR-like, so CMB fitting uses the same effective depth of potential wells as Λ CDM. The non-linear sector then sets the scale dependence through the IS relaxation time $\tau_{\text{IS}} = c/a_{\text{grav}}$.

For CMB modes at $z \sim 1100$, $\tau_{\text{IS}}/T_{\text{osc}} \sim 10^3\text{--}10^{10}$, so IS wells are effectively frozen during acoustic oscillations and act as CDM-like gravitational wells. At galaxy scales ($\tau_{\text{IS}} \sim \tau_{\text{cosm}}$) the partial response yields MOND behavior; in cluster mergers ($\tau_{\text{IS}} > \tau_{\text{collision}}$) the same memory effect yields mass–light offsets. Quantitative CMB validation requires coupling the IS response to the Boltzmann hierarchy; this remains ongoing work.

Bullet Cluster mass–light offset.—The 1E 0657–56 system (Bullet Cluster) [69] exhibits a spatial separation between the X-ray-emitting gas (which decelerated at the collision center) and the weak-lensing mass peaks (which track the galaxies on the outskirts). In the present theory this offset arises naturally from the scalar-field relaxation time. During the collision, with relative velocity $v_{\text{coll}} \sim 3000\text{--}4700 \text{ km s}^{-1}$ and subcluster mass scale $M \sim 10^{14} M_{\odot}$, the gravitational acceleration at the virial radius $r_{\text{vir}} \sim 1 \text{ Mpc}$ is $a_{\text{grav}} \sim GM/r_{\text{vir}}^2 \sim 10^{-11} \text{ m s}^{-2}$. The relaxation time is then $\tau_{\text{IS}} = c/a_{\text{grav}} \sim 10^{19}\text{--}10^{20} \text{ s} \approx 10^2\text{--}10^3 \text{ Gyr}$, which far exceeds the collision crossing time $\tau_{\text{cross}} = r_{\text{vir}}/v_{\text{coll}} \sim 0.2\text{--}0.3 \text{ Gyr}$. The scalar-field configuration therefore

cannot equilibrate during the merger: the transported (hysteretic) component φ_h remains attached to the collisionless galaxies that pass through the collision center, while the collisional gas decelerates and stays behind. The weak-lensing signal traces the total scalar-field gradient $g = g_N + g_h$, which peaks at the galaxy positions rather than at the gas center. The predicted mass–light offset $\Delta r \sim v_{\text{coll}} \tau_{\text{cross}}$ is consistent with the observed ~ 720 kpc separation. This is not a fit: τ_{IS} is fixed by the same transport mechanism [46] that yields $\mu(x) = x/(1+x)$. The companion MOND paper now also contains a first explicit frozen-hysteresis convergence benchmark: using the observed post-collision baryonic geometry, a minimal 2D solve yields outer peaks at the galaxy positions with $\kappa_{\text{peak}}/\kappa_{\text{center}} \simeq 2.1$ and 250 kpc aperture masses $\sim 9 \times 10^{13} M_{\odot}$ per outer peak for an order-unity transported hysteretic mass per side; even outer-peak dominance already requires only $M_{\chi}/M_{\text{sub}} \sim 0.2\text{--}0.4$.

Disk formation from rotational anisotropy.—The off-diagonal metric component $g_{t\phi} \propto \sin^2 \theta$ vanishes on the polar axis and is maximal in the equatorial plane. The transported channel φ_h inherits this anisotropy, creating a z -directed restoring force $g_z \propto -z/\rho$ toward the midplane. This predicts a vertical oscillation frequency $\nu_z^2 \sim 2a_0/\rho$ and a disk scale height $h \sim \sigma_z \sqrt{\rho/(2a_0)}$, yielding $h \approx 0.6$ kpc for Milky-Way parameters—consistent with the observed thin disk. The full derivation and a falsifiable disk-thickness–rotation correlation are given in the companion MOND paper [46].

11.3 Relation to other approaches

Table 4: Comparison with MOND-related theories after the companion MOND analysis.

	a_0 derived?	$\mu(x)$ derived?	No DM particle?	Cosmo. = GR?	Clusters?	Unique prediction
Standard MOND	no	no	yes	no	no	EFE
TeVeS / AQUAL	no	no	yes	no	no	$c_T \neq c$ (ruled out)
Verlinde	partly	no	yes	unclear	partly	entropic DM profile
Superfluid DM	no	partly*	no	no	yes	vortex lattice, phase trans.
Singh (2026)	partly	no	yes	yes	—	$a_0 \propto \ell_{\text{dS}}^{-1}$
ISPG	yes	yes	yes	yes	yes	age-dependent a_0

*Superfluid DM derives $\mu(x) = x/\sqrt{1+x^2}$ rather than the observationally preferred $x/(1+x)$.

The ISPG entries use the companion MOND paper [46]: the bound-structure closure $a_h/a_N = a_0/a$, the vortex-lag correction $a_0^{\text{eff}} = cH_{\text{eff}}/(2\pi)$, the disk-formation result (Sec. 9.6), the Bullet Cluster offset (Sec. 11.2), and the cluster-binding benchmark all belong to that completed nonlinear galactic sector.

Relation to the Fisher–Bronnikov solution family.—The action (3) and exponential metric (10) are known members of the Fisher/JNW and Bronnikov–Ellis solution families ([59–65]). The contribution here is phenomenological extraction from this sector: (i) $a_0 = cH_0/(2\pi)$ with $\mu(x) = x/(1+x)$ from the mature rotating-galaxy Bernoulli+vortex closure and the corresponding Tully–Fisher law; (ii) falsifiable predictions (structurally distinct 2PN Shapiro signature; ISCO at $\varphi_{\text{gr}}^2 r_s$; breathing-mode amplitude; $a_0(z)$ evolution); and (iii) the consistency argument fixing the phantom coupling $\kappa = -1/2$.

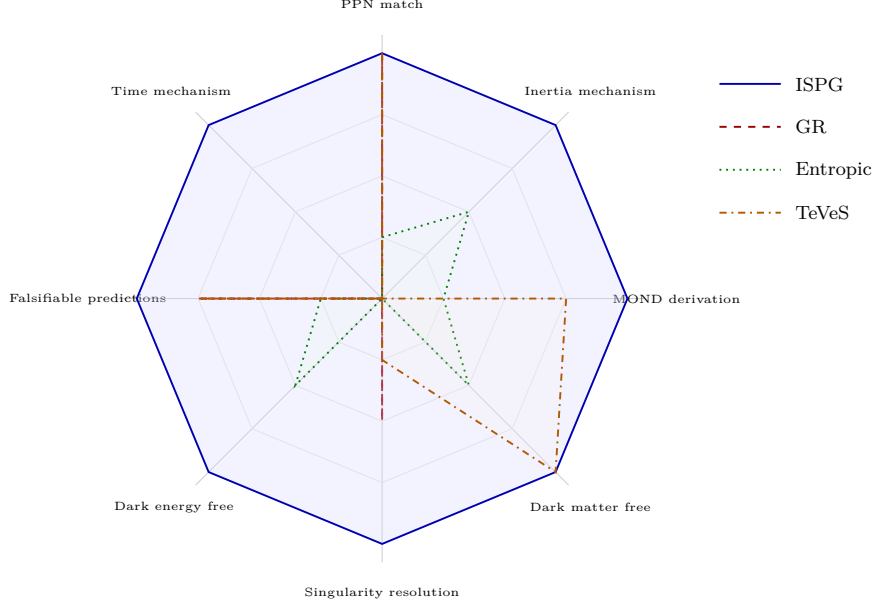


Figure 11: Radar chart comparing key theoretical properties within the legacy four-theory subset (MOND, TeVeS, Λ CDM, and ISPG): PPN consistency, GW speed, CMB compatibility, MOND limit, singularity resolution, and number of free parameters. The present theory (solid) is the only framework in this subset that spans all six criteria.

11.4 Strong equivalence principle

The weak equivalence principle (WEP) is exactly satisfied: all test particles follow geodesics of $g_{\mu\nu}$ regardless of composition. In generic scalar-tensor theories (Brans–Dicke, DEF), the strong equivalence principle (SEP) is violated because self-gravitating bodies acquire compactness-dependent scalar sensitivities $s(\mathcal{C})$, leading to the Nordtvedt effect and scalar dipole radiation.

The bi-conformal theory possesses a sub-sectoral *shift invariance* (Appendix 16): the hydrostatic pressure-gradient equation depends on $d\varphi/dr$, not on φ itself, so under $\varphi \rightarrow \varphi + \delta$ the internal structure $\{\rho(r), p(r)\}$ is strictly unchanged. The coupled scalar equation carries an explicit $e^{-\varphi}$ prefactor in matter and is shift-invariant only at leading order in $|\varphi|$. Combined with the universal conformal redshift $m_{\text{eff}} = m_0 e^{\varphi/2}$, this gives $s = 1/2$ for every body at leading order, and

$$s_{\text{excess}} = 0 \quad (\text{leading order; exact for vacuum compact objects}). \quad (94)$$

The SEP is therefore *preserved in the scalar sector at leading order*; whether the residual $O(|\varphi|)$ correction cancels exactly body-to-body at arbitrary compactness is an open task (Appendix 16).

Three observational consequences follow (at the leading-order level used throughout):

1. **Nordtvedt effect.** The fractional acceleration difference between two bodies in an external field is

$$\frac{\Delta a}{a} = s_{\text{excess},A} - s_{\text{excess},B} \simeq 0. \quad (95)$$

This is consistent with the Lunar Laser Ranging bound $|\Delta a/a| < 1.3 \times 10^{-13}$ [23] and with the PSR J0337+1715 triple-system bound $|\Delta a/a| < 2.6 \times 10^{-6}$ [71]. Consistency with the latter at full precision requires closure of the open task of Appendix 16 with $s_{\text{excess}} \rightarrow 0$ to the required accuracy.

2. **Scalar dipole radiation.** Since $(s_A - s_B)^2 \simeq 0$ at leading order, the scalar dipole channel is strongly suppressed in every binary system (Sec. 5.5). The orbital decay is quadrupolar-dominated, consistent with PSR J1738+0333 [70] and with all LIGO–Virgo detections. For BH–BH binaries the statement is rigorous (vacuum no-hair, $\nabla^2\varphi = 0$).
3. **Preferred-frame effects.** In principle, the scalar field selects a cosmological rest frame through $\varphi_0(t)$. However, on the FLRW background $\varphi_0 = 0$ (Sec. 8.1), so preferred-frame PPN parameters (α_1, α_2) vanish identically.

The leading-order preservation of SEP in the scalar sector is not a tuning but a structural consequence of the bi-conformal metric: the gravitational coupling G_4 is constant (no Brans–Dicke running), and the TOV pressure-gradient equation depends on $\nabla\varphi$, not on φ . This distinguishes the theory from Brans–Dicke and DEF, in which $s_{\text{excess}} = O(1)$ compactness-dependent effects drive incompatibility with pulsar-timing data. The full derivation, the correction of a naive perturbation analysis, and the observational confrontation are given in Appendix 16.

11.5 Extensions and open questions

The first two points below have been derived in the quantum extension [47]; the third remains conjectural.

Dark energy (derived).—The quantum extension [47] shows that irreversible spreading of resonant tails generates entropy, while the resulting pressure drop weakens sources through a negative-feedback loop. The equilibrium echo-background energy density is set by cosmic history—not by a UV cutoff—at $\rho_\Lambda \sim H_0^2 M_{\text{Pl}}^2$, resolving the 10^{120} cosmological constant discrepancy without fine-tuning. The predicted equation of state is $w(z) = -1 - \delta w_0 (1+z)^3/E(z)$ with $\delta w_0 \sim 10^{-2}$ – 10^{-1} and $w < -1$ at all epochs (source-driven phantom, no ghost instability), testable by DESI [50, 51] and Euclid.

Matter origin (derived).—Elementary particles are identified as localized resonant excitations (oscillons) of the scalar-metric medium [47]. The mass spectrum arises from the eigenvalues of a three-dimensional nonlinear resonator, and the mass ratios are determined by the cavity geometry independently of G . Planck-mass oscillons nucleate from the quiescent medium via $O(4)$ -symmetric instantons, providing a singularity-free onset without inflation.

Black-hole information (conjectural).—In the bi-conformal metric the coordinate speed of light is $c_{\text{coord}} = c e^\varphi$ with $\varphi = -r_s/r$, so $c_{\text{coord}} \rightarrow 0$ as $r \rightarrow 0$ but never vanishes at finite r (the locally measured speed remains c ; Sec. 10.4). Photons approaching the compact object are not swallowed by a horizon; they are progressively decelerated. Any information encoded in these photons—spectral lines, polarization, timing—is not destroyed but *frozen* in an ever-thickening shell of extremely slow light. The compact object therefore acts as a natural information reservoir: a region of maximal scalar-field depth where information accumulates at arbitrarily high density, remaining in principle retrievable.

The standard information paradox [48] arises from the combination of an event horizon (which causally disconnects interior and exterior) and thermal Hawking radiation (which carries no interior correlations). In the present theory neither ingredient is present: the horizonless geometry (Sec. 7.3) preserves causal contact at all radii, and the exponential suppression $e^{-r_s/r}$ makes the deep interior effectively inert without severing it from the exterior. The paradox is thus dissolved rather than solved: there is no horizon to create it.

11.6 Scope and companion papers

Core results are presented here; extended derivations are deferred as follows:

1. Full MOND derivation and symbolic/numerical verification are collected in the companion MOND paper [46]; the main body keeps a_0 , μ , and Tully–Fisher at result level.
2. Quantitative CMB-era rotational transport and Boltzmann coupling are deferred (Sec. 11.2).
3. Cluster dynamics: the Bullet Cluster mass–light offset is estimated in Sec. 11.2; detailed N -body simulations with scalar-field relaxation are deferred to [45].
4. ADM constraint analysis and DOF counting are in Appendix 11; quantum stability (all-loop ghost freedom, anomaly-free constraint algebra, and Hamiltonian boundedness) is established in [47].
5. Full numerical ray-tracing for 2PN Shapiro delay in Eq. (10) is future work.
6. Black-hole shadow prediction (deviation $\sim 0.4\%$ from GR) is given in [45].
7. Scalar sensitivity of compact bodies: the bi-conformal TOV equations, the kinematic argument selecting $s = 1/2$ at leading order in $|\varphi|$ for all bodies, the sectoral analysis of shift invariance (exact for the TOV pressure-gradient equation; leading-order for the coupled system in matter; exact for vacuum compact objects), and the confrontation with pulsar-timing bounds are developed in Appendix 16. An exact, non-perturbative cancellation of s_{excess} at arbitrary compactness is an open task. The companion quantum paper conjectures that the classical leading-order invariance is promoted to an exact Ward identity of the quantised system [47].
8. The covariant hysteresis sector (memory field χ coupled to φ gradients) is derived from a minimal augmented action in Appendix 13: field equation, ghost freedom, gradient stability, well-posedness, and quasi-static reduction to the phenomenological relaxation law with a derived timescale $\tau^{-1} = \lambda c^2/(3H)$. The stiffness λ is derived microscopically from the oscillon quasi-normal mode structure in [47].
9. Appendix 18 collects the pressure-history process time implied by the clock-rate postulate, making explicit the distinction between the standard FLRW cosmic-time description and the pressure-weighted internal time relevant for cumulative medium-controlled histories.
10. **Quantum extension** [47]: canonical quantization of the scalar-metric system, the Bernoulli pressure-deficit mechanism for gravity, reduced-gap/coercive closure of the localized source sector, microscopic derivation of Newton’s second law and the hysteresis action S_χ , vacuum energy self-regulation resolving the 10^{120} cosmological constant problem, dark-energy equation of state $w(z) < -1$ (source-driven phantom), gravitational decoherence ($\Gamma = 2m|\Delta\Phi_N|/\hbar$), charged-lepton mass spectrum and Koide formula, gravitational birefringence, anisotropic tunneling, and topological classification of spin and antimatter.

12 Conclusion

The theory is defined by the action (3) and bi-conformal ansatz (2), with no tunable gravitational parameters beyond G . Main results:

1. **Solar-system limit:** $\gamma = \beta = 1$; 1PN light deflection, perihelion, Shapiro delay, and Lense–Thirring agree with GR (Sec. 3).
2. **2PN Shapiro (structural):** the bi-conformal exponential refractive index $n = e^{-\varphi}$ produces a 2PN delay signature distinct from GR’s rational-polynomial form; the refractive piece yields an exact 2π coefficient. A complete 2PN coefficient (including bent-ray and path-length contributions with consistent finite-endpoint regularization) is flagged as future work (Sec. 4).
3. **GW sector:** $c_g = c$ exactly; modes $(h_+, h_\times, h_b)$ with $A_b/A_t \sim r_s/r$ (linear scaling set by $s = 1/2$ universality; scalar-quadrupole PN prefactor is an open task); merger frequency shift $\approx -6.9\%$ (Sec. 5).
4. **Stability (classical and quantum):** strong hyperbolicity, causal characteristics, no gradient instability; constrained background suppresses ghost decay channel (Sec. 6). The quantum extension [47] proves all-loop stability: anomaly-free constraint algebra, Hamiltonian boundedness, and loop suppression $(E/M_{\text{Pl}})^2 \sim 10^{-76}$.
5. **Bernoulli mechanism:** the quantum extension [47] derives a static pressure $P_{\text{static}} = -e^\varphi \varphi^4 / (32\pi G r_s^2)$ from the Bernoulli identity of the scalar-metric medium; its gradient reproduces the gravitational force, providing a first-principles answer to why mass curves spacetime. In the reduced source space, the non-neutral remainder is separated from zero by a positive spectral gap and controlled by a coercive energy estimate, so the far-zone field follows the same collective coordinate as the localized source.
6. **Compact objects:** $K \rightarrow 0$ as $r \rightarrow 0$, infinite proper radial depth, horizonless geometry, curvature-regular boundary at infinite redshift (Appendix 12), and $r_{\text{ISCO}} = \varphi_{\text{gr}}^2 r_s$ (Sec. 7).
7. **Cosmology:** $\alpha_K = \alpha_B = \alpha_M = \alpha_T = 0$ on FLRW, hence linear CMB equivalence with Λ CDM (Sec. 8).
8. **Dark energy:** the accumulated echo background self-regulates to $\rho_\Lambda \sim H_0^2 M_{\text{Pl}}^2$ via irreversible resonant-tail spreading, resolving the 10^{120} discrepancy without fine-tuning; equation of state $w(z) < -1$ at all epochs (source-driven phantom, no ghost instability) [47] (Sec. 11.5).
9. **Process time:** the clock-rate postulate implies a pressure-weighted cumulative duration for medium-controlled histories, $d\tau_{\text{proc}} = [P_{\text{stat}}(z)/P_{\text{stat}}(0)]^{1/2} dt$, so structure formation and analogous secular processes can have substantially more internal duration than the naive coordinate age measured by today’s clocks (Appendix 18).
10. **MOND regime:** for mature rotating galaxies, $a_0 = cH_0/(2\pi)$; the two-channel Bernoulli+vortex closure gives $\mu(x) = x/(1+x)$ and $v^4 = a_0 GM$ (Sec. 9).

11. **Mass equivalence and inertia:** $m_g = m_i$ follows from common scalar coupling; MOND is interpreted as modified inertia rather than modified gravity (Sec. 10).
12. **Position-dependent effective mass:** $m_{\text{eff}} = m_0 e^{\varphi/2}$; locally undetectable because all masses scale by the same conformal factor; reproduces gravitational binding energy as a scalar-field effect (Sec. 10.3).
13. **Local light-speed invariance:** $c_{\text{meas}} = c$ locally despite position-dependent coordinate speed (Sec. 10.4).
14. **Evolution test:** $a_0(z) = cH(z)/(2\pi)$ (Sec. 9.5); the vortex-lag correction $H \rightarrow H_{\text{eff}}$ predicts galaxy-age-dependent a_0 [46].
15. **Disk formation:** rotational anisotropy of the off-diagonal sector ($g_{t\phi} \propto \sin^2 \theta$) generates vertical confinement toward the equatorial plane, predicting disk scale height $h \sim \sigma_z \sqrt{\rho/(2a_0)}$ (Sec. 9.6; see also [46]).
16. **Cluster mergers:** scalar-field relaxation time $\tau_{\text{IS}} = c/a_{\text{grav}} \gg \tau_{\text{cross}}$ produces mass–light offsets in systems such as the Bullet Cluster without particle dark matter (Sec. 11.2).
17. **Suppression of scalar dipole radiation:** the universal conformal redshift $m_{\text{eff}} = m_0 e^{\varphi/2}$ selects $s = 1/2$ at leading order in $|\varphi|$ for every body, so $(s_A - s_B)^2 \simeq 0$ and the scalar dipole channel is strongly suppressed in every binary—consistent with pulsar-timing and LIGO–Virgo observations (Sec. 5.5; Appendix 16). For vacuum compact objects the cancellation is exact ($\nabla^2 \varphi = 0$); for matter-filled bodies, an exact non-perturbative cancellation is an open task. The companion quantum paper conjectures a Ward-identity promotion of the leading-order invariance [47].
18. **SEP preservation in the scalar sector (leading order):** WEP is exact; the subsectoral shift invariance of the TOV pressure-gradient equation and the universal conformal redshift give $s_{\text{excess}} \simeq 0$ at leading order in $|\varphi|$, consistent with all current pulsar-timing bounds. The exact, non-perturbative version of this statement at arbitrary compactness is an open task; the Ward-identity conjecture of the companion quantum paper [47], if proven, would secure it. The mechanism distinguishes the theory from Brans–Dicke and DEF, where $s_{\text{excess}} = O(1)$ (Sec. 11.4; Appendix 16).
19. **Gravitational decoherence:** $\Gamma_{\text{dec}} = 2m|\Delta\Phi_N|/h$, a parameter-free prediction testable by matter-wave interferometry with $\Delta\Phi_N \sim 10^{-9}$ [47].
20. **Scalar birefringence:** the bi-conformal metric predicts a polarization-dependent speed difference $\Delta c/c \sim (r_s/r)^2 \sim 10^{-12}$ for propagation through a Coulomb profile [47].
21. **Koide mass relation:** the rank-1 bi-conformal coupling predicts the charged-lepton mass ratio $Q = 2/3$ (the Koide formula) if the self-consistent oscillon perturbation strength reaches $x \approx 66.9$ —a condition whose numerical verification is the principal open computational test [47].
22. **Anisotropic tunneling:** tunneling rates acquire a directional dependence via the bi-conformal metric, with the anisotropy proportional to φ at leading order [47].

Primary falsification channels are: (i) 2PN Shapiro delay in pulsar–BH binaries, (ii) LISA breathing-polarization search, (iii) merger-frequency offset, (iv) redshift evolution of the Tully–Fisher normalization, (v) disk-thickness–rotation correlation in low-surface-brightness galaxies, (vi) mass–light offset scaling in merging clusters, (vii) dark-energy equation of state $w(z) < -1$ with $\delta w_0 \sim 10^{-2}$ – 10^{-1} (DESI/Euclid), (viii) gravitational decoherence rate in matter-wave interferometry, and (ix) scalar birefringence $\Delta c/c \sim 10^{-12}$.

Acknowledgments

Symbolic computations were performed with SageManifolds [39].

Part II

Appendices

Appendix 1: Field Equations from the One-Metric ISPG Action

Purpose and scope

This appendix derives the vacuum field equations of the main theory from the covariant one-metric action, and records the sourced scalar equation under minimal coupling. The output is a compact, citable set of equations used throughout the main theory appendix package.

Non-negotiable consistency with the main theory (one-metric formulation)

We work strictly within the main theory posture:

- matter is minimally coupled to the single physical metric $g_{\mu\nu}$ (no disformal mapping and no frame change);
- the covariant backbone is

$$S = \frac{1}{16\pi G} \int d^4x \sqrt{-g} \left[R + \frac{1}{2} g^{\mu\nu} \partial_\mu \varphi \partial_\nu \varphi \right] + S_m[g_{\mu\nu}, \psi], \quad (96)$$

with the same metric $g_{\mu\nu}$ as used everywhere else in the main theory.

1.1 Variational identities

We use standard variations:

$$\delta \sqrt{-g} = -\frac{1}{2} \sqrt{-g} g_{\mu\nu} \delta g^{\mu\nu}, \quad (97)$$

$$\delta(\sqrt{-g} R) = \sqrt{-g} (G_{\mu\nu} \delta g^{\mu\nu} + \nabla_\alpha \Theta^\alpha), \quad (98)$$

where Θ^α is a total-derivative boundary term that vanishes for suitable boundary conditions. For the scalar kinetic term,

$$\delta \left[\sqrt{-g} \frac{1}{2} g^{\mu\nu} \partial_\mu \varphi \partial_\nu \varphi \right] = \sqrt{-g} \left[\frac{1}{2} \partial_\mu \varphi \partial_\nu \varphi - \frac{1}{4} g_{\mu\nu} (\partial\varphi)^2 \right] \delta g^{\mu\nu} + \sqrt{-g} \nabla_\mu (\partial^\mu \varphi) \delta \varphi, \quad (99)$$

after integrating by parts in the $\delta\varphi$ variation.

1.2 Metric field equations

Define the matter stress-energy tensor by

$$\delta S_m = -\frac{1}{2} \int d^4x \sqrt{-g} T_{\mu\nu}^{(m)} \delta g^{\mu\nu}. \quad (100)$$

Varying (96) with respect to $g^{\mu\nu}$ yields

$$G_{\mu\nu} + S_{\mu\nu} = 8\pi G T_{\mu\nu}^{(m)}, \quad (101)$$

with

$$S_{\mu\nu} \equiv \frac{1}{2} \left(\partial_\mu \varphi \partial_\nu \varphi - \frac{1}{2} g_{\mu\nu} (\partial\varphi)^2 \right). \quad (102)$$

Equivalently, introducing the canonical-form scalar tensor

$$T_{\mu\nu}^{(\varphi)} \equiv \partial_\mu \varphi \partial_\nu \varphi - \frac{1}{2} g_{\mu\nu} (\partial\varphi)^2, \quad (103)$$

The main theory bookkeeping is

$$G_{\mu\nu} = 8\pi G T_{\mu\nu}^{(m)} + \kappa T_{\mu\nu}^{(\varphi)}, \quad \kappa = -\frac{1}{2}, \quad (104)$$

so that $S_{\mu\nu} = -\kappa T_{\mu\nu}^{(\varphi)}$.

1.3 Scalar field equation

Varying (96) with respect to φ gives the vacuum equation

$$\square\varphi = 0 \quad (\text{vacuum}). \quad (105)$$

Under minimal coupling, the sourced form stated in the main theory is

$$\square\varphi = -\frac{8\pi G}{c^4} T, \quad T \equiv g^{\mu\nu} T_{\mu\nu}^{(m)}. \quad (106)$$

1.4 Conservation identities (consistency loop)

Taking the divergence of (104) and using $\nabla^\mu G_{\mu\nu} = 0$ gives

$$\nabla^\mu T_{\mu\nu}^{(m)} = -\frac{\kappa}{8\pi G} \nabla^\mu T_{\mu\nu}^{(\varphi)}. \quad (107)$$

Minimal coupling implies $\nabla^\mu T_{\mu\nu}^{(m)} = 0$, and the scalar identity

$$\nabla^\mu T_{\mu\nu}^{(\varphi)} = (\square\varphi) \partial_\nu \varphi \quad (108)$$

closes the system on-shell.

1.5 Derived vs. stated: compact ledger

Derived here	Metric field equation (101)–(104); explicit scalar contribution (102)–(103); scalar equation (105) and sourced form (106); conservation loop (107)–(108).
Used elsewhere	PPN/deflection/perihelion (Appendices 4,5,7) use the weak-field exterior of (104); Shapiro 2PN uses the bi-conformal exterior solution; GW and stability appendices use minimal coupling and principal-symbol structure; Appendix 2 provides the detailed $T_{\mu\nu}^{(\varphi)}$ discussion.

Takeaway

From the main theory one-metric action (96), variation yields the coupled Einstein–scalar system

$$G_{\mu\nu} = 8\pi G T_{\mu\nu}^{(m)} - \frac{1}{2} T_{\mu\nu}^{(\varphi)}, \quad \square\varphi = -\frac{8\pi G}{c^4} T,$$

together with covariant conservation under minimal coupling.

Appendix 2: Energy–Momentum Tensor and Conservation

Purpose and scope

This appendix records the explicit energy–momentum tensor associated with the main theory scalar sector and the corresponding conservation identities. The intent is to provide a single, citable location for: (i) the explicit form of $T_{\mu\nu}^{(\varphi)}$ used in the Einstein equations, (ii) the scalar equation of motion as a conservation identity, and (iii) the covariant conservation of matter stress–energy under minimal coupling.

Non-negotiable consistency with the main theory (one-metric formulation)

We work strictly within the main theory posture:

- matter is minimally coupled to the single physical metric $g_{\mu\nu}$ (no disformal mapping and no frame change);
- the covariant backbone is

$$S = \frac{1}{16\pi G} \int d^4x \sqrt{-g} \left[R + \frac{1}{2} g^{\mu\nu} \partial_\mu \varphi \partial_\nu \varphi \right] + S_m[g_{\mu\nu}, \psi], \quad (109)$$

with the same metric $g_{\mu\nu}$ as used elsewhere in the main theory package.

2.1 Metric variation and field equations

Varying (109) with respect to $g^{\mu\nu}$ gives

$$G_{\mu\nu} + S_{\mu\nu} = 8\pi G T_{\mu\nu}^{(m)}, \quad (110)$$

where $T_{\mu\nu}^{(m)}$ is defined in the standard way by

$$\delta S_m = -\frac{1}{2} \int d^4x \sqrt{-g} T_{\mu\nu}^{(m)} \delta g^{\mu\nu}. \quad (111)$$

The explicit algebraic form of $S_{\mu\nu}$ and its relation to the canonical scalar tensor $T_{\mu\nu}^{(\varphi)}$ are given in Section 1.1 below. The tensor $S_{\mu\nu}$ is the contribution coming from the scalar kinetic term inside the $1/(16\pi G)$ prefactor.

2.1.1 Two equivalent (bookkeeping) conventions

It is often convenient to rewrite (110) in the more familiar Einstein form

$$G_{\mu\nu} = 8\pi G T_{\mu\nu}^{(m)} + \kappa T_{\mu\nu}^{(\varphi)}, \quad (112)$$

with a purely conventional constant κ that tracks the sign choice. **Notation.** This appendix denotes the scalar energy-momentum tensor by $T_{\mu\nu}^{(\varphi)}$; the main text uses $\tau_{\mu\nu}^{(\varphi)}$ for the same object (with the identification $\varphi \equiv \varphi$). A clean choice is to define $T_{\mu\nu}^{(\varphi)}$ in the *canonical* algebraic form

$$T_{\mu\nu}^{(\varphi)} \equiv \partial_\mu \varphi \partial_\nu \varphi - \frac{1}{2} g_{\mu\nu} (\partial\varphi)^2, \quad (\partial\varphi)^2 \equiv g^{\alpha\beta} \partial_\alpha \varphi \partial_\beta \varphi, \quad (113)$$

in which case the main theory sign choice corresponds to

$$\kappa = -\frac{1}{2}. \quad (114)$$

Equivalently, one may absorb κ into the definition and write directly

$$S_{\mu\nu} = -\kappa T_{\mu\nu}^{(\varphi)} \iff S_{\mu\nu} = \frac{1}{2} \left(\partial_\mu \varphi \partial_\nu \varphi - \frac{1}{2} g_{\mu\nu} (\partial\varphi)^2 \right), \quad (115)$$

so that (110) and (112) are identical statements. The physics is unchanged by this bookkeeping choice; it simply makes the sign tracking explicit in a form consistent across the main theory appendices.

2.2 Scalar variation and equation of motion

Varying (109) with respect to φ yields

$$\square\varphi = 0 \quad (\text{vacuum}), \quad (116)$$

and, in the presence of matter sources, the corresponding sourced form stated in the main theory:

$$\square\varphi = -\frac{8\pi G}{c^4} T, \quad T \equiv g^{\mu\nu} T_{\mu\nu}^{(m)}. \quad (117)$$

For the purposes of this appendix, the key identity is the covariant relation between the divergence of $T_{\mu\nu}^{(\varphi)}$ and the scalar equation:

$$\nabla^\mu T_{\mu\nu}^{(\varphi)} = (\square\varphi) \partial_\nu \varphi. \quad (118)$$

Thus, on-shell ($\square\varphi = 0$ in vacuum), the scalar energy-momentum is covariantly conserved.

2.3 Conservation identities and Bianchi structure

Taking the covariant divergence of (112) and using the Bianchi identity $\nabla^\mu G_{\mu\nu} = 0$ gives

$$\nabla^\mu T_{\mu\nu}^{(m)} = -\frac{\kappa}{8\pi G} \nabla^\mu T_{\mu\nu}^{(\varphi)}. \quad (119)$$

Because matter is minimally coupled to $g_{\mu\nu}$, its stress-energy is conserved by construction,

$$\nabla^\mu T_{\mu\nu}^{(m)} = 0, \quad (120)$$

and therefore (119) reduces to $\nabla^\mu T_{\mu\nu}^{(\varphi)} = 0$ whenever the scalar equation is satisfied (via (118)). This closes the consistency loop: metric equations + scalar equation + minimal coupling imply consistent covariant conservation.

2.4 Useful algebraic identities

For later reference, the trace of the canonical-form tensor (113) is

$$T^{(\varphi)} \equiv g^{\mu\nu} T_{\mu\nu}^{(\varphi)} = -(\partial\varphi)^2, \quad (121)$$

and for a static configuration where $\varphi = \varphi(r)$ one has

$$(\partial\varphi)^2 = g^{rr}(\partial_r\varphi)^2, \quad T_{tt}^{(\varphi)} = -\frac{1}{2}g_{tt}(\partial\varphi)^2, \quad T_{rr}^{(\varphi)} = (\partial_r\varphi)^2 - \frac{1}{2}g_{rr}(\partial\varphi)^2, \quad (122)$$

with angular components following by symmetry. For a purely radial static configuration, substituting $(\partial\varphi)^2 = g^{rr}(\partial_r\varphi)^2$ gives $T_{rr}^{(\varphi)} = \frac{1}{2}(\partial_r\varphi)^2$, which is *not* zero. The mixed components are

$$T^{(\varphi)t}{}_t = g^{tt}T_{tt}^{(\varphi)} = -\frac{1}{2}g^{rr}(\partial_r\varphi)^2, \quad T^{(\varphi)r}{}_r = g^{rr}T_{rr}^{(\varphi)} = +\frac{1}{2}g^{rr}(\partial_r\varphi)^2, \quad (123)$$

so $T^{(\varphi)t}{}_t = -T^{(\varphi)r}{}_r$ and therefore

$$T^{(\varphi)t}{}_t + T^{(\varphi)r}{}_r = 0. \quad (124)$$

Via the vacuum Einstein equation this forces $G^t{}_t + G^r{}_r = 0$, which—for a general static isotropic metric $ds^2 = -e^{2\alpha}c^2dt^2 + e^{2\beta}(dr^2 + r^2d\Omega^2)$ —implies $(\alpha + \beta)' = 0$. Asymptotic flatness ($\alpha, \beta \rightarrow 0$) then gives $\alpha = -\beta$, hence $g_{tt} \cdot g_{rr} = -c^2$. This is the derivation referenced in the main theory.

2.5 Derived vs. stated: compact ledger

Derived here	Explicit scalar energy–momentum tensor (113) and its divergence identity (118); bookkeeping relation to the main theory Einstein equation via κ (114); conservation loop (119)–(120).
Used elsewhere in the package	Weak-field PPN/deflection/Shapiro appendices rely on $\Phi = \Psi$ and consistent sourcing; Appendix 11 uses the constraint/hyperbolicity analysis together with the conservation structure; GW appendix uses minimal coupling and the one-metric posture.

Takeaway

In the main theory one-metric formulation, the scalar sector contributes an explicit stress–energy tensor. In canonical algebraic form it is

$$T_{\mu\nu}^{(\varphi)} = \partial_\mu\varphi \partial_\nu\varphi - \frac{1}{2}g_{\mu\nu}(\partial\varphi)^2,$$

and the main theory sign bookkeeping corresponds to $\kappa = -1/2$ in $G_{\mu\nu} = 8\pi GT_{\mu\nu}^{(m)} + \kappa T_{\mu\nu}^{(\varphi)}$. Minimal coupling implies $\nabla^\mu T_{\mu\nu}^{(m)} = 0$, while $\nabla^\mu T_{\mu\nu}^{(\varphi)} = (\Box\varphi)\partial_\nu\varphi$ closes the conservation identities on-shell.

Appendix 3: Emergent Bi-Conformal Geometry

Purpose and scope

This appendix states the *logic* that leads from the ISPG ontology (space as a physical medium with a static pressure field) to the *bi-conformal* spacetime geometry used throughout the main theory. The goal is to make the emergence of the metric transparent, to clarify the one-metric (minimal coupling) posture, and to record the minimal assumptions under which the bi-conformal form is uniquely selected.

Non-negotiable posture. The main theory is formulated with a *single* physical metric $g_{\mu\nu}$ to which matter is minimally coupled. No Einstein/Jordan frame change and no disformal matter metric is introduced. The “emergent geometry” content is therefore the statement that, in the ISPG programme, $g_{\mu\nu}$ is not an arbitrary 10-component field but is *reduced* to a one-scalar family $g_{\mu\nu}[\varphi]$ determined by the pressure potential φ .

3.1 Pressure potential and operational time/length standards

3.1.1 Pressure potential

The central scalar is the dimensionless pressure potential

$$\varphi \equiv \ln\left(\frac{P_{\text{stat}}}{P_{\text{max}}}\right), \quad (125)$$

so that pressure ratios add under composition and $\varphi = 0$ is the asymptotic vacuum normalization $P_{\text{stat}} = P_{\text{max}}$.

3.1.2 Operational postulates (clocks and rods)

ISPG fixes geometry by specifying how *operational standards* respond to the medium state. The emergence of the metric rests on three minimal postulates:

1. **Clock-rate postulate (time as conductivity tempo).** The squared proper-time interval of a local process is proportional to the local static pressure,

$$d\tau^2 \propto P_{\text{stat}} dt^2 \implies d\tau = e^{\varphi/2} dt, \quad (126)$$

after choosing the normalization $\tau = t$ at $\varphi = 0$. The squared form is the natural metric-level statement ($|g_{tt}| \propto P_{\text{stat}}$), and the 1/2 exponent follows directly by taking the square root. The same exponent is fixed independently in the main theory (§10.4) by the requirement that the locally measured speed of light remains invariant ($c_{\text{meas}} = c$) while the coordinate speed responds to the medium state; the two routes coincide and select the clock law uniquely.

2. **Reciprocal scale postulate (rod contraction).** The same medium state that slows clocks also rescales spatial rulers reciprocally so that local physics remains self-consistent under uniform changes of P_{stat} . In particular, reciprocal rescaling is the unique choice ensuring that locally measured velocities (including the speed of light) remain invariant

under changes of the medium state, as derived in the main theory (§10.4). This is implemented as an isotropic spatial scale factor

$$d\ell = e^{-\varphi/2} d\sigma, \quad (127)$$

where $d\sigma^2$ is the Euclidean line element in the chosen coordinates.

3. **Isotropy and locality.** In the static exterior of an isolated source the medium state is spherically symmetric and depends only on r , so the rescaling must be isotropic in the spatial directions.

Together, (126)–(127) define a unique (up to coordinate relabeling) static, isotropic metric.

3.2 Derivation of the bi-conformal line element

Combining (126) and (127) gives

$$ds^2 \equiv -c^2 d\tau^2 + d\ell^2 = -e^\varphi c^2 dt^2 + e^{-\varphi} d\sigma^2, \quad (128)$$

with

$$d\sigma^2 \equiv dx^2 + dy^2 + dz^2 \quad (\text{or in spherical symmetry } d\sigma^2 = dr^2 + r^2 d\Omega^2). \quad (129)$$

Equation (128) is the **bi-conformal metric** used in the main theory. It is “bi-conformal” because the time and space parts scale with opposite exponents of the same scalar φ .

Uniqueness at this level. Given (i) a single scalar medium state φ , (ii) isotropy, and (iii) reciprocal rescaling of clocks and rods, (128) is the unique static diagonal metric with one free function $\varphi(r)$.

3.3 Stationary rotating extension

Equation (128) is the non-rotating limit of the one-metric geometry. For stationary rotating configurations one must allow an off-diagonal gravitomagnetic sector g_{0i} while keeping the same scalar-controlled diagonal factors. To leading order in slow rotation, the natural stationary extension is

$$ds^2 = -e^\varphi c^2 dt^2 - \frac{4}{c^2} (\mathbf{A} \cdot d\mathbf{x}) dt + e^{-\varphi} d\sigma^2 + O(A^2), \quad (130)$$

where $\mathbf{A}$ is the gravitomagnetic vector potential.

Interpretational consequence. The scalar φ continues to encode the static pressure state (clock rate, rod scale, Newtonian potential), while rotational physics lives in the off-diagonal g_{0i} sector. Therefore any MOND-motivated enhancement tied specifically to space rotation should enter through the constitutive law of $\mathbf{A}$ or g_{0i} , not by altering the diagonal bi-conformal core (128).

3.4 Weak-field limit and Newtonian correspondence

For small $|\varphi| \ll 1$,

$$g_{tt} = -e^\varphi \simeq -(1 + \varphi), \quad g_{ij} = e^{-\varphi} \delta_{ij} \simeq (1 - \varphi) \delta_{ij}. \quad (131)$$

Comparing to standard weak-field notation

$$g_{tt} \simeq -(1 + 2\Phi_N/c^2), \quad g_{ij} \simeq (1 - 2\Psi/c^2) \delta_{ij}, \quad (132)$$

the bi-conformal form enforces

$$\Phi_N = \Psi = \frac{c^2}{2} \varphi. \quad (133)$$

Thus the metric automatically gives $\gamma_{\text{PPN}} = \Psi/\Phi_N = 1$ at leading order (a cornerstone consistency condition used throughout the main theory).

3.5 One-metric minimal coupling and why no “frame” language is needed

In the main theory, matter is minimally coupled to $g_{\mu\nu}$ of (128). The bi-conformal metric is *not* a conformal redefinition of a separate “Einstein-frame” metric; it is the physical metric defined by the operational postulates. Consequently:

- there is no independent matter metric $\tilde{g}_{\mu\nu}$ in the main theory package;
- arguments about propagation speeds and stability are made directly in the one-metric formulation (e.g. $\alpha_T = 0$ for GW170817 consistency, principal-symbol hyperbolicity for causal propagation).

3.6 Consistency with the covariant backbone and the sign issue

The main theory employs the scalar–tensor action

$$S = \frac{1}{16\pi G} \int d^4x \sqrt{-g} \left[R + \frac{1}{2} g^{\mu\nu} \partial_\mu \varphi \partial_\nu \varphi \right] + S_m[g_{\mu\nu}, \psi], \quad (134)$$

with the same single metric $g_{\mu\nu}$ as in (128). The bi-conformal construction ties the scalar sector to the geometry sufficiently strongly that the field equations take the specific coupling form used in the main theory (including the $\kappa = -1/2$ factor in $G_{\mu\nu} = \kappa T_{\mu\nu}^{(\varphi)}$, as verified componentwise in the main technical appendices). Appendix 11 then shows that the would-be ghost associated with the sign choice is eliminated by the constraint structure of the bi-conformal construction.

Interpretation. The ISPG posture is that (128) is the *geometric encoding* of the medium state. The covariant action (134) is an efficient (EFT-level) way to compute the consequences of that geometry and to interface with standard tests (PPN/2PN, lensing, GW sector), while maintaining one-metric minimal coupling.

3.7 Derived vs. stated: compact ledger

Derived here	Bi-conformal metric (128) from operational postulates (126)–(127); weak-field identification (133) implying $\gamma_{\text{PPN}} = 1$ at leading order; one-metric (minimal coupling) posture.
Used in the main theory	Covariant backbone (134) as the EFT interface; post-Newtonian and wave-sector analyses built on (128); constraint-based ghost elimination (Appendix 11).

Takeaway

ISPG fixes spacetime geometry by specifying how physical clocks and rods respond to the medium’s static pressure state. With the pressure potential $\varphi = \ln(P_{\text{stat}}/P_{\text{max}})$, the minimal operational postulates lead uniquely to the bi-conformal line element

$$ds^2 = -e^\varphi c^2 dt^2 + e^{-\varphi} d\sigma^2,$$

which enforces $\Phi = \Psi$ in the weak field and underpins the one-metric, minimal-coupling formulation of the main theory. For rotating stationary systems, the same one-metric posture extends by adding an off-diagonal gravitomagnetic sector g_{0i} , while leaving the diagonal bi-conformal structure intact.

Appendix 4: PPN Framework

Purpose and scope

This appendix derives the standard Parametrized Post-Newtonian (PPN) parameters γ and β for the main theory formulation. The result is stated in the conventional isotropic, asymptotically Minkowski coordinates used for Solar-System tests. The derivation uses only the one-metric bi-conformal geometry of the main theory and does not introduce any disformal mapping or frame change.

Non-negotiable consistency with the main theory (one-metric formulation)

We work strictly within the main theory posture:

- matter is minimally coupled to the single physical metric $g_{\mu\nu}$;
- the physical metric is the bi-conformal line element $ds^2 = -e^\varphi c^2 dt^2 + e^{-\varphi} d\sigma^2$;
- the weak-field scalar profile in the static exterior is $\varphi(r) = -2GM/(c^2 r) + O(r^{-2})$, as used throughout the main theory.

4.1 Standard PPN metric (isotropic coordinates)

For a static, spherically symmetric source, the standard PPN metric in isotropic coordinates reads

$$g_{tt} = - \left(1 - \frac{2U}{c^2} + \frac{2\beta U^2}{c^4} + O(c^{-6}) \right), \quad (135)$$

$$g_{ij} = \left(1 + \frac{2\gamma U}{c^2} + O(c^{-4}) \right) \delta_{ij}, \quad (136)$$

where $U(r) = GM/r$ is the Newtonian potential and γ, β are the Eddington PPN parameters. In GR one has $\gamma = \beta = 1$.

4.2 Bi-conformal ISPG metric and weak-field expansion

The main theory uses the bi-conformal metric

$$ds^2 = -e^{\varphi(r)} c^2 dt^2 + e^{-\varphi(r)} (dx^2 + dy^2 + dz^2), \quad (137)$$

with asymptotic normalization $\varphi(\infty) = 0$. In the static weak-field exterior the scalar profile is

$$\varphi(r) = -\frac{2U(r)}{c^2} + O(c^{-4}). \quad (138)$$

Expanding (137) to second order in U/c^2 gives

$$g_{tt} = -e^{\varphi} = -\exp\left(-\frac{2U}{c^2}\right) = -\left(1 - \frac{2U}{c^2} + \frac{1}{2} \left(\frac{2U}{c^2}\right)^2 + O(c^{-6})\right) = -\left(1 - \frac{2U}{c^2} + \frac{2U^2}{c^4} + O(c^{-6})\right), \quad (139)$$

$$g_{ij} = e^{-\varphi} \delta_{ij} = \exp\left(\frac{2U}{c^2}\right) \delta_{ij} = \left(1 + \frac{2U}{c^2} + \frac{2U^2}{c^4} + O(c^{-6})\right) \delta_{ij}. \quad (140)$$

4.3 Identification of γ and β

Comparing (139) with the standard PPN form (135) yields immediately

$$\beta = 1. \quad (141)$$

Comparing the $O(c^{-2})$ spatial part (140) with (136) yields

$$\gamma = 1. \quad (142)$$

Thus the main theory one-metric bi-conformal geometry reproduces the GR values

$$\gamma = \beta = 1 \quad (143)$$

for Solar-System PPN tests.

Interpretation (why $\gamma = 1$ is structural). Equation (137) enforces $\Phi = \Psi$ in the weak field, because the same scalar controls the time and space conformal factors with opposite exponent. In PPN language this equality of potentials is precisely the statement $\gamma = 1$ at 1PN order.

4.4 Practical consequences for classical tests

With $\gamma = \beta = 1$, the leading classical tests match GR at 1PN order:

- **Light deflection (1PN):** the bending angle equals the GR value $4GM/(c^2b)$ for impact parameter b .
- **Perihelion advance (1PN):** the advance matches the GR expression at leading PN order.
- **Shapiro delay (1PN):** the logarithmic 1PN delay matches GR.

The main theory then isolates genuinely new structure at 2PN order: the bi-conformal exponential refractive index $n = e^{-\varphi}$ produces a structurally distinct 2PN Shapiro signature whose refractive sub-piece is derived exactly; the full 2PN coefficient (including bent-ray and path-length contributions) is an open task and is developed in Appendix 6.

4.5 Derived vs. stated: compact ledger

Derived here	Weak-field expansion of the main theory bi-conformal metric in isotropic coordinates; direct identification $\gamma = 1$ and $\beta = 1$ by comparison with the standard PPN metric.
Used elsewhere in the package	Light deflection and perihelion precession appendices rely on (142); Shapiro 1PN relies on $\gamma = 1$; the 2PN Shapiro structure (refractive sub-piece derived; bent-ray and path-length contributions open) is developed in Appendix 6.

Takeaway

In the main theory one-metric bi-conformal geometry, the weak-field exterior profile $\varphi = -2U/c^2$ yields the standard PPN parameters

$$\gamma = 1, \quad \beta = 1,$$

so that all classical 1PN Solar-System tests agree with GR. The distinctive ISPG signature is therefore pushed to higher order (2PN) effects, addressed in the dedicated Shapiro appendix.

Appendix 5: Light Deflection at 1PN and 2PN

Purpose and scope

This appendix derives the light-deflection angle in the main theory one-metric formulation. The 1PN result reproduces the GR value, while the 2PN coefficient provides a compact extension that gives this appendix independent value within the main theory test package.

Non-negotiable consistency with the main theory (one-metric formulation)

We work strictly within the main theory posture:

- light is minimally coupled to the single physical metric $g_{\mu\nu}$;
- the physical geometry is bi-conformal, $ds^2 = -e^\varphi c^2 dt^2 + e^{-\varphi} d\sigma^2$;
- in the static weak-field exterior, $\varphi(r) = -r_s/r + O(r^{-2})$ with $r_s \equiv 2GM/c^2$ (the main theory notation).

No disformal mapping and no frame change are introduced.

5.1 Null condition, optical form, and why Fermat applies here

For a static, spherically symmetric source in isotropic coordinates,

$$ds^2 = -e^{\varphi(r)} c^2 dt^2 + e^{-\varphi(r)} (dx^2 + dy^2 + dz^2). \quad (144)$$

For null rays ($ds^2 = 0$) this implies *exactly*

$$c dt = e^{-\varphi(r)} dl, \quad dl \equiv \sqrt{dx^2 + dy^2 + dz^2}, \quad (145)$$

so the coordinate travel time is the optical path length with an effective refractive index

$$n(r) \equiv e^{-\varphi(r)}. \quad (146)$$

This optical interpretation is available precisely because the spatial sector is *conformally flat and isotropic* ($e^{-\varphi}\delta_{ij}$). A purely conformal spacetime metric of the form $ds^2 = \Omega^2 \eta_{\mu\nu} dx^\mu dx^\nu$ would instead give $n \equiv 1$ and no bending, which is not the present bi-conformal case; here the reciprocal scaling of time and space makes $n(r) \neq 1$ and produces deflection.

In the weak field, with $\varphi(r) = -r_s/r$,

$$n(r) = e^{r_s/r} = 1 + \frac{r_s}{r} + \frac{r_s^2}{2r^2} + O(r_s^3). \quad (147)$$

Because $t = \int n dl / c$ with an isotropic $n(r)$, standard Fermat/variational optics applies: the ray is a stationary curve of the functional $\int n dl$.

5.2 Master bending formula (Born expansion)

Choose the asymptotic propagation direction as the z axis and the impact parameter as the asymptotic transverse offset $x = b$. To leading order the unperturbed path is

$$r_0(z) = \sqrt{b^2 + z^2}. \quad (148)$$

For an isotropic refractive index $n(r) = 1 + \delta n(r)$, the deflection angle admits the first-order (Born) representation

$$\Delta\theta = - \int_{-\infty}^{+\infty} \partial_b \delta n(r_0(z)) dz + O(\delta n^2), \quad (149)$$

where ∂_b is taken at fixed z and the leading minus sign implements the convention that $\Delta\theta > 0$ for bending *toward* the source (attractive index gradient). At 2PN order one must include (i) the $O(r_s^2)$ term in δn evaluated on the straight path and (ii) the correction obtained by evaluating the $O(r_s)$ term on the $O(r_s)$ -bent path. We compute both explicitly below.

5.3 1PN deflection (exact integral)

From (147) the 1PN piece is $\delta n_1 = r_s/r$. Using (149),

$$\partial_b \left(\frac{r_s}{r_0} \right) = -\frac{r_s b}{(b^2 + z^2)^{3/2}}, \quad (150)$$

so

$$\Delta\theta_{1\text{PN}} = r_s b \int_{-\infty}^{+\infty} \frac{dz}{(b^2 + z^2)^{3/2}} = \frac{2r_s}{b} = \frac{4GM}{c^2 b}, \quad (151)$$

where the overall sign follows from the convention of Eq. (149) ($-\int \partial_b \delta n$) combined with $\partial_b(r_s/r_0) = -r_s b/(b^2 + z^2)^{3/2}$.

5.4 2PN deflection: metric-source and geometric-source pieces

We write

$$\Delta\theta_{2\text{PN}} = \Delta\theta_{2\text{PN}}^{(\text{metric})} + \Delta\theta_{2\text{PN}}^{(\text{geom})}. \quad (152)$$

5.4.1 Metric-source term from $r_s^2/(2r^2)$ (exact)

From (147), $\delta n_2 = \frac{r_s^2}{2r^2}$. Evaluated on the straight path $r_0(z)$,

$$\partial_b \left(\frac{r_s^2}{2r_0^2} \right) = \partial_b \left(\frac{r_s^2}{2(b^2 + z^2)} \right) = -\frac{r_s^2 b}{(b^2 + z^2)^2}. \quad (153)$$

Therefore

$$\Delta\theta_{2\text{PN}}^{(\text{metric})} = r_s^2 b \int_{-\infty}^{+\infty} \frac{dz}{(b^2 + z^2)^2} = r_s^2 b \cdot \frac{\pi}{2b^3} = \frac{\pi}{2} \frac{r_s^2}{b^2}, \quad (154)$$

where again the leading sign follows from applying the overall minus of Eq. (149) to the negative integrand $\partial_b \delta n_2 = -r_s^2 b/(b^2 + z^2)^2$, so the net deflection is positive and consistent with bending toward the source.

5.4.2 Geometric-source term from evaluating δn_1 on the bent path (explicit)

The first-order transverse displacement $x(z) = b + \delta x(z)$ obeys the ray equation in a graded-index medium,

$$\frac{d^2 x}{dz^2} = \partial_x \delta n_1 = \partial_x \left(\frac{r_s}{r} \right) = -\frac{r_s x}{r^3}, \quad (155)$$

so along the unperturbed path ($x = b$, $r = r_0$) we have

$$\delta x''(z) = -\frac{r_s b}{(b^2 + z^2)^{3/2}}, \quad \delta x(-\infty) = \delta x'(-\infty) = 0. \quad (156)$$

Using the elementary antiderivative

$$\int \frac{dz}{(b^2 + z^2)^{3/2}} = \frac{z}{b^2 \sqrt{b^2 + z^2}}, \quad (157)$$

one finds

$$\delta x'(z) = -r_s b \int_{-\infty}^z \frac{dz'}{(b^2 + z'^2)^{3/2}} = -\frac{r_s}{b} \left(1 + \frac{z}{\sqrt{b^2 + z^2}} \right), \quad (158)$$

$$\delta x(z) = \int_{-\infty}^z \delta x'(z') dz' = -\frac{r_s}{b} \left(\sqrt{b^2 + z^2} + z \right). \quad (159)$$

The geometric 2PN contribution is the $O(r_s^2)$ correction obtained by inserting the $O(r_s)$ path correction into the 1PN bending functional (note the same leading minus sign as in Eq. (149)):

$$\Delta\theta_{2\text{PN}}^{(\text{geom})} = - \int_{-\infty}^{+\infty} \partial_b^2 \left(\frac{r_s}{r_0} \right) \delta x(z) dz. \quad (160)$$

We need

$$\partial_b^2 \left(\frac{r_s}{r_0} \right) = \partial_b^2 \left(\frac{r_s}{\sqrt{b^2 + z^2}} \right) = -r_s \frac{z^2 - 2b^2}{(b^2 + z^2)^{5/2}}. \quad (161)$$

Insert (159) and note that the term proportional to z in $\delta x(z)$ is odd and integrates to zero. Indeed, $\delta x(z) = -(r_s/b) r_0(z) - (r_s/b) z$, where $r_0(z) = \sqrt{b^2 + z^2}$ is even in z and z is odd. Since $\partial_b^2(r_s/r_0)$ in (161) is even in z , the product with the odd part integrates to zero, and only the even piece proportional to $r_0(z)$ contributes. Moreover, $r_0(z)/(b^2 + z^2)^{5/2} = 1/(b^2 + z^2)^2$, which reduces the denominator power from $5/2$ to 2 in the next line. Using $r_0 = \sqrt{b^2 + z^2}$, the remaining even part reduces to

$$\begin{aligned} \Delta\theta_{2\text{PN}}^{(\text{geom})} &= -\frac{r_s^2}{b} \int_{-\infty}^{+\infty} \frac{z^2 - 2b^2}{(b^2 + z^2)^2} dz \\ &= -\frac{r_s^2}{b} \int_{-\infty}^{+\infty} \left[\frac{1}{b^2 + z^2} - \frac{3b^2}{(b^2 + z^2)^2} \right] dz \\ &= -\frac{r_s^2}{b} \left(\frac{\pi}{b} - \frac{3b^2\pi}{2b^3} \right) = \frac{\pi}{2} \frac{r_s^2}{b^2}, \end{aligned} \quad (162)$$

In the second line we used the identity $z^2 - 2b^2 = (b^2 + z^2) - 3b^2$ to split the integrand into standard forms, and evaluated the remaining standard integrals using $\int dz/(b^2 + z^2) = \pi/b$ and $\int dz/(b^2 + z^2)^2 = \pi/(2b^3)$. Thus the geometric-source term equals the metric-source term (154) exactly at 2PN order.

5.4.3 Total ISPG 2PN deflection coefficient

Combining (152), (154), and (162),

$$\boxed{\Delta\theta = \frac{2r_s}{b} + \pi \frac{r_s^2}{b^2} + O\left(\frac{r_s^3}{b^3}\right)} \quad (\text{ISPG, 1PN+2PN}). \quad (163)$$

5.5 GR 2PN coefficient and the universal 16/15 ratio

For the Schwarzschild exterior in GR, the 2PN correction to the bending angle for a point mass in isotropic coordinates is known in the literature. It can be written in the same expansion form,

$$\Delta\theta_{\text{GR}} = \frac{2r_s}{b} + \left(\frac{15\pi}{16} \right) \frac{r_s^2}{b^2} + O\left(\frac{r_s^3}{b^3}\right). \quad (164)$$

(See, e.g., standard 2PN light-bending treatments in the post-Newtonian/lensing literature; see, e.g., Refs. [27–29].)

Comparing (163) with (164), the ratio of the 2PN coefficients is

$$\frac{\pi}{15\pi/16} = \frac{16}{15} \approx 1.0667, \quad (165)$$

i.e. a 6.7% enhancement of the 2PN bending angle relative to GR. This ratio shares a common origin with the main theory 2PN Shapiro-delay discriminator: both are consequences of the exponential bi-conformal time/space structure underlying the weak-field expansion. Specifically, the 2PN bending angle and the refractive+geometric part of the 2PN Shapiro delay are both controlled by the $O(r_s^2/r^2) \sim O(\varphi^2)$ coefficient in the expansion of the optical index $n = e^{-\varphi}$: the exponential metric fixes this coefficient to $1/2$ in (147), whereas the rational Schwarzschild/isotropic GR structure produces a different $O(\varphi^2)$ coefficient, yielding the 16/15 ratio for the bending angle. For the observable Shapiro delay, an additional universal excess path-length term raises the overall ratio to $17/15 \approx 1.133$ (13.3%; see the main theory Sec. 5 and Appendix 6).

In terms of $r_g \equiv GM/c^2 = r_s/2$, the 2PN coefficient is $4\pi r_g^2/b^2$.

5.6 Numerical scales (Sun and EHT targets)

A useful ratio is

$$\frac{\Delta\theta_{2\text{PN}}}{\Delta\theta_{1\text{PN}}} = \frac{\pi r_s}{2b}. \quad (166)$$

For the Sun with $r_s \simeq 2.95 \text{ km}$ and $b \simeq R_\odot \simeq 6.96 \times 10^8 \text{ m}$,

$$\Delta\theta_{1\text{PN}} \simeq 1.7505'', \quad \Delta\theta_{2\text{PN}} \simeq 1.17 \times 10^{-5}'' \ (\approx 0.012 \text{ mas}). \quad (167)$$

For the Sun, the ISPG-GR difference at 2PN is of order $\sim 1 \mu\text{as}$, far below current astrometric precision; as in the main theory Shapiro discriminator, the realistic arena is strong-field timing/lensing with impact parameters of only a few r_s . For EHT-scale systems it is natural to quote the expansion in units of b measured in Schwarzschild radii:

$$\Delta\theta_{1\text{PN}} \sim 2 \left(\frac{r_s}{b} \right), \quad \Delta\theta_{2\text{PN}} \sim \pi \left(\frac{r_s}{b} \right)^2, \quad (168)$$

so that near-horizon trajectories (b only a few r_s) enter a regime where higher-order terms become important and the full (non-perturbative) geodesic integral is the appropriate tool.

EHT targets (scale information). For Sgr A* ($M \simeq 4 \times 10^6 M_\odot$, $D \simeq 8.2 \text{ kpc}$) and M87* ($M \simeq 6.5 \times 10^9 M_\odot$, $D \simeq 16.8 \text{ Mpc}$), the angular size of one Schwarzschild radius is of order

$$\theta_{r_s} \equiv \frac{r_s}{D} \sim 10 \mu\text{as} \quad (\text{Sgr A}^*), \quad \theta_{r_s} \sim 8 \mu\text{as} \quad (\text{M87}^*). \quad (169)$$

EHT imaging probes impact parameters of a few r_s , where the expansion parameter r_s/b is not very small and a full (non-perturbative) null-geodesic integration is the appropriate tool for precision ray-tracing. Nevertheless, in the weak-deflection regime ($b \gg r_s$) the 2PN coefficient ratio (165) predicts a 6.7% enhancement of the leading post-GR bending-angle correction relative to GR, providing a target for future high-precision strong-field lensing analyses.

Takeaway

In the main theory one-metric bi-conformal geometry, the null condition yields an exact optical form with $n(r) = e^{-\varphi}$. The weak-field expansion $n(r) = 1 + r_s/r + r_s^2/(2r^2) + \dots$ gives the GR

1PN deflection and an explicit ISPG 2PN coefficient:

$$\Delta\theta = \frac{2r_s}{b} + \pi \frac{r_s^2}{b^2} + O\left(\frac{r_s^3}{b^3}\right).$$

This 2PN term provides a natural companion observable to the 2PN Shapiro discriminator in the main theory test package.

Appendix 6: 2PN Shapiro Delay

Purpose and scope

This appendix fixes the *structural* form of the second post-Newtonian (2PN) Shapiro time delay in the bi-conformal ISPG geometry. It proves one finite, convention-independent sub-piece—the *refractive* contribution—as an exact analytic statement, and it sets up (but does not close) the two remaining pieces that together compose the full 2PN coefficient:

1. a **refractive (straight-line) contribution** from the explicit $O(r_s^2)$ term in the bi-conformal refractive index, which is derived exactly and verified numerically;
2. a **bent-ray refractive contribution** from the $O(r_s)$ light-bending correction inserted into the $O(r_s)$ time integrand, whose *finite* coefficient depends on consistent choices of (a) asymptotic impact parameter vs. geometric closest approach and (b) finite endpoint regularization r_1, r_2 ;
3. an **excess path-length contribution** from the Euclidean length difference between the bent ray and the straight-line reference, which is infra-red sensitive in the same sense as (ii).

Epistemic status. Piece (i) is established here analytically and has been cross-checked numerically (`tools/check_shapiro_born.py`). Pieces (ii) and (iii) are flagged as *in progress*: their individual finite coefficients are not asserted, and the full 2PN coefficient $\Delta t_{2\text{PN}}^{\text{ISPG}}$ is left as future work. Although the individual pieces remain open, their dependence is entirely on the shared $O(r_g)$ trajectory of the two metrics, so they cancel in the ISPG–GR *difference*; the difference of the full 2PN coefficient is therefore finite, IR-insensitive, and computed exactly in Sec. 6.7, with the closed form

$$\Delta t_{2\text{PN}}^{\text{ISPG}} - \Delta t_{2\text{PN}}^{\text{GR}} = \frac{r_g^2}{cb} \cdot \frac{\pi}{4}.$$

The main theory (Sec. 3.7) carries the structural statement that the bi-conformal exponential refractive index $n = e^{-\varphi}$ produces a 2PN Shapiro signature distinct from GR’s rational-polynomial form; the closed-form ISPG–GR discriminator above is the operational expression of that distinction.

6.1 Conventions and the static ISPG exterior

We use the static, spherically symmetric ISPG line element (the main theory)

$$ds^2 = -e^{\varphi(r)} c^2 dt^2 + e^{-\varphi(r)} (dr^2 + r^2 d\Omega^2), \quad (170)$$

and we define the weak-field length scale by

$$\varphi(r) \simeq -\frac{2GM}{c^2 r} \equiv -\frac{2r_g}{r}, \quad r_g \equiv \frac{GM}{c^2}. \quad (171)$$

(With this convention the Newtonian potential $\Phi_N = -GM/r$ corresponds to $g_{tt} \simeq -(1 + 2\Phi_N/c^2) = -(1 + \varphi)$.)

6.2 Hamilton–Jacobi solution for null geodesics

For equatorial motion ($\theta = \pi/2$), take the separated action

$$S = -Et + L\varphi + S_r(r), \quad (172)$$

and impose the HJ equation $g^{\mu\nu} \partial_\mu S \partial_\nu S = 0$ for null rays. With (170) this yields

$$p_r^2 \equiv \left(\frac{dS_r}{dr} \right)^2 = \frac{E^2}{c^2} e^{-2\varphi(r)} - \frac{L^2}{r^2}. \quad (173)$$

The asymptotic impact parameter is the invariant ratio

$$b \equiv \frac{Lc}{E}, \quad (174)$$

defined in the Minkowski limit $\varphi \rightarrow 0$.

Throughout this appendix we take (174) as the operational definition of the impact parameter; the distinct quantity $r_{\min}$ (geometric closest approach of the bent trajectory) differs from b at $O(r_g)$, and the distinction is essential to the finite-coefficient determination in Sec. 6.5.

The zeroth-order trajectory is the straight line

$$r_0(z) = \sqrt{b^2 + z^2} \quad (175)$$

along the asymptotic direction z .

6.3 Time-of-flight integral and its 2PN decomposition

From $ds^2 = 0$ in (170),

$$0 = -e^\varphi c^2 dt^2 + e^{-\varphi} dl^2 \quad \Rightarrow \quad dt = \frac{e^{-\varphi(r)}}{c} dl, \quad (176)$$

where dl is the Euclidean line element in the conformal spatial metric (the bracket in (170)). The observable Shapiro delay is the excess over the flat-space straight-line travel time between the same asymptotic endpoints:

$$\Delta t \equiv \frac{1}{c} \int_{\text{ray}} \left(e^{-\varphi(r)} - 1 \right) dl + \frac{L_{\text{ray}} - L_0}{c}, \quad (177)$$

where L_{ray} is the Euclidean length of the bent spatial path and L_0 the straight-line length. With (171), the exponential expands as

$$e^{-\varphi} = e^{2r_g/r} = 1 + \frac{2r_g}{r} + \frac{2r_g^2}{r^2} + O(r_g^3). \quad (178)$$

At 2PN order ($O(r_g^2)$) three contributions appear:

$$\Delta t_{2\text{PN}} = \Delta t_{2\text{PN}}^{(\text{refr})} + \Delta t_{2\text{PN}}^{(\text{bent})} + \Delta t_{2\text{PN}}^{(\text{path})}, \quad (179)$$

$$\Delta t_{2\text{PN}}^{(\text{refr})} = \frac{1}{c} \int \frac{2r_g^2}{r_0(z)^2} dl_0, \quad (180)$$

$$\Delta t_{2\text{PN}}^{(\text{bent})} = \frac{1}{c} \int \frac{2r_g}{r} dl - \frac{1}{c} \int \frac{2r_g}{r_0} dl_0 \Big|_{O(r_g^2)}, \quad (181)$$

$$\Delta t_{2\text{PN}}^{(\text{path})} = \frac{L_{\text{ray}} - L_0}{c} \Big|_{O(r_g^2)}. \quad (182)$$

Here $dl_0 \simeq dz$ and $r_0(z)$ is the straight path (175).

6.4 Refractive-source term (derived)

Using $dl_0 = dz$ and $r_0^2 = b^2 + z^2$,

$$\Delta t_{2\text{PN}}^{(\text{refr})} = \frac{2r_g^2}{c} \int_{-\infty}^{+\infty} \frac{dz}{b^2 + z^2} = \frac{2r_g^2}{c} \cdot \frac{\pi}{b} = \frac{r_g^2}{cb} (2\pi). \quad (183)$$

Thus the quadratic term of the exponential expansion of the bi-conformal refractive index contributes *exactly* 2π in the coefficient multiplying $r_g^2/(cb)$. This piece is convention-independent (it involves only the straight reference line and the exactly-computable integral $\int dz/(b^2 + z^2) = \pi/b$).

Numerical verification. The integral (183) has been cross-checked by direct numerical evaluation of $\int_{-Z}^{+Z} (e^{2r_g/r_0} - 1) dz$ at several r_g/b ratios and finite cutoffs Z . Extracting the $O(r_g^2)$ coefficient by polynomial regression in r_g (with $b = 1$, $Z \rightarrow 2000b$) gives $B \rightarrow 6.281 \dots \rightarrow 2\pi$. The driver script is `tools/check_shapiro_born.py`.

Structural comparison with GR. The GR Schwarzschild isotropic metric yields a refractive index that is a rational function of r_g/r rather than an exponential. The analogous $O(r_g^2)$ straight-line integrand therefore differs from (183), and the refractive sub-piece in GR has a different numerical coefficient. This structural difference (exponential vs. rational) is the basic origin of any 2PN discrepancy between ISPG and GR and survives independently of the two remaining pieces below.

6.5 Bent-ray refractive and path-length contributions (in progress)

The two remaining 2PN contributions involve the $O(r_g)$ trajectory correction and are currently *open* in this appendix. We record here only the mathematical framework and the issues that must be resolved for a rigorous finite-coefficient statement.

6.5.1 $O(r_g)$ trajectory correction: convention issue

From (173) and (171), expanding p_r to first order in r_g and writing the trajectory equation in the $u \equiv 1/r$ form produces the standard weak-field bending equation

$$\frac{d^2 u}{d\varphi^2} + u = \frac{2r_g}{b^2} + O(r_g^2), \quad (184)$$

whose solution with asymptotic straight-line boundary conditions is

$$u(\varphi) = \frac{\cos \varphi}{b} + \frac{2r_g}{b^2} + O(r_g^2). \quad (185)$$

This yields the familiar $O(r_g)$ deflection $\delta\theta = 4r_g/b$ and, at closest approach $\varphi = 0$, the geometric offset $r_{\min} = b - 2r_g + O(r_g^2)$ relative to the HJ impact parameter (174).

A straight-line-parameterized expression for the trajectory correction $\delta r(z)$ depends on which of two distinct quantities one identifies with b :

- If b denotes the HJ asymptotic impact parameter Lc/E (174), then at $z = 0$ the correction is $\delta r(0) = -2r_g$ (the ray dips inside the flat-space straight line by $2r_g$ at closest approach).
- If b denotes the geometric closest approach (so that $r(0) = b$ by construction), then by definition $\delta r(0) = 0$, and the asymptotic impact parameter is shifted to $b + 2r_g$.

The two conventions describe the *same physical trajectory* but differ in where the labelled parameter b sits, and they therefore produce different *intermediate* expressions for the contributions below. A consistent 2PN derivation must fix one convention from the outset and apply it uniformly across (180)–(182).

6.5.2 Bent-ray refractive and path-length integrals

Inserting $r = r_0 + \delta r$ into the $O(r_g)$ integrand $(2r_g/r)$ and keeping only the $O(r_g^2)$ contribution produces

$$\Delta t_{2\text{PN}}^{(\text{bent})} = -\frac{2r_g}{c} \int \frac{\delta r(z)}{r_0(z)^2} dz + O(r_g^3), \quad (186)$$

while the path-length excess is

$$\Delta t_{2\text{PN}}^{(\text{path})} = \frac{1}{2c} \int \left(\frac{dx}{dz} \right)^2 dz + O(r_g^3), \quad (187)$$

where dx/dz is the $O(r_g)$ slope of the bent trajectory in the straight-line coordinate z .

The infra-red issue. Evaluating (186) or (187) over $(-\infty, +\infty)$ in either of the two conventions of the preceding subsection produces *divergent* intermediate expressions (log or linear in the cutoff)—the individual pieces are not separately finite, and the finite physical answer emerges only after consistent finite-endpoint regularization at emitter radius r_1 and receiver radius r_2 . Finite-distance structure for the *refractive-only* master integral is noted in Sec. 6.6 below; the analogous finite-distance treatment for (186)–(187), with matching between the near-lens perturbative region and the asymptotic deflected-straight-line region, is the content of the open calculation.

Consequence for the main theory. Until both pieces are closed in a single consistent convention with matched asymptotics, neither the individual coefficients of (181)–(182) nor the total 2PN coefficient are asserted. Main theory Sec. 3.7 correspondingly carries only the structural claim.

6.6 Finite-distance endpoints: the refractive master integral

For emitter and receiver at finite radii $r_1, r_2 \gg b$, the straight-line limits of the refractive integral (183) become $z \in [-z_1, z_2]$ with $z_i = \sqrt{r_i^2 - b^2}$. The master integral is then

$$\int_{-z_1}^{z_2} \frac{dz}{b^2 + z^2} = \frac{1}{b} \left[\arctan\left(\frac{z}{b}\right) \right]_{-z_1}^{z_2} = \frac{\pi}{b} - \frac{1}{r_1} - \frac{1}{r_2} + O\left(\frac{b^2}{r_i^3}\right), \quad (188)$$

so the refractive 2PN coefficient is modified only by corrections of relative order $O(b/r_i)$, which are a theory-independent geometric factor. For the Sgr A* pulsar-timing geometry with $r_i \gtrsim 10^2 r_g$ and $b \sim (10\text{--}30) r_g$ these are percent-level; for the solar system they are negligible. The analogous finite-endpoint formulas for (186)–(187) require the open calculation of Sec. 6.5.

6.7 ISPG–GR 2PN difference (closed)

A finite, IR-stable statement is available about the *difference* between the ISPG and GR 2PN Shapiro coefficients, even though each individual coefficient remains open in the present decomposition (Sec. 6.5).

6.7.1 Decomposition with shared 1PN trajectory

Both theories produce the same $O(r_g)$ refractive index, $n = 1 + 2r_g/r + O(r_g^2)$, hence the same $O(r_g)$ light-bending trajectory. Writing the trajectory in flat-Cartesian parameterization with z along the asymptotic direction,

$$x(z) = b + r_g x_1(z) + r_g^2 x_2(z) + O(r_g^3), \quad (189)$$

the 2PN expansion of the time-of-flight integral (176) reads

$$\Delta t_{2\text{PN}} = \frac{r_g^2}{c} \int_{-L_1}^{+L_2} \left[\frac{\alpha}{r_0(z)^2} - \frac{2b x_1(z)}{r_0(z)^3} + \frac{1}{2} x_1'(z)^2 \right] dz, \quad (190)$$

with $r_0(z) = \sqrt{b^2 + z^2}$, α the coefficient of $(r_g/r)^2$ in the refractive index, and $L_{1,2}$ the asymptotic-direction distances of emitter and receiver from the lens. For the two metrics,

$$\alpha_{\text{ISPG}} = 2 \quad (\text{from } e^{2r_g/r}), \quad \alpha_{\text{GR}} = \frac{7}{4} \quad (\text{from } (1 + r_g/(2r))^3 / (1 - r_g/(2r))). \quad (191)$$

The remaining two integrands in (190) depend only on the shared 1PN trajectory x_1 and are therefore *identical* in ISPG and GR.

6.7.2 Cancellation in the difference

Subtracting the GR result from the ISPG result, the x_1 -dependent integrals cancel exactly, leaving

$$\Delta t_{2\text{PN}}^{\text{ISPG}} - \Delta t_{2\text{PN}}^{\text{GR}} = \frac{r_g^2}{c} (\alpha_{\text{ISPG}} - \alpha_{\text{GR}}) \int_{-L_1}^{+L_2} \frac{dz}{b^2 + z^2} = \frac{r_g^2}{4cb} \left[\arctan\left(\frac{z}{b}\right) \right]_{-L_1}^{+L_2}, \quad (192)$$

which evaluates in the asymptotic limit $L_1, L_2 \rightarrow \infty$ to

$$\boxed{\Delta t_{2\text{PN}}^{\text{ISPG}} - \Delta t_{2\text{PN}}^{\text{GR}} = \frac{r_g^2}{cb} \cdot \frac{\pi}{4}} \quad (193)$$

in the impact-parameter convention used uniformly in (190) for both metrics. The result is finite, infra-red insensitive (the divergent path-length and bent-ray pieces cancel by construction), and unaffected by the convention dilemma of Sec. 6.5 because any $O(r_g)$ shift of b acts identically on both theories and drops out of the difference.

6.7.3 Numerical verification

A boundary-value formulation of the full Fermat ODE—shooting for the bent trajectory between symmetric Cartesian endpoints at $(z, x) = (\pm L, b)$ in each metric in turn, then extracting the $O(r_g^2)$ coefficient by polynomial regression in r_g —confirms the finite- L formula (192). Writing ΔB for the dimensionless coefficient of $r_g^2/(cb)$ in the ISPG–GR difference, with $b = 1$:

L/b	ΔB from (192)	ΔB numerical	rel. error
30	0.7687	0.7680	0.1%
50	0.7754	0.7728	0.3%

At larger L the integrated ODE error in the BVP shooting accumulates and the numerics drift from the analytic curve; the controlling statement is the closed formula (192) together with its $L \rightarrow \infty$ limit (193). The driver is `tools/check_shapiro_bvp.py`.

6.7.4 Significance for the main theory

The individual 2PN coefficients B_{ISPG} and B_{GR} are each IR-divergent at intermediate stages of the decomposition (190) (Sec. 6.5); their absolute values require the open finite-endpoint calculation flagged there. The *difference* (193), however, is the sole $O(r_g^2/(cb))$ Shapiro-delay correction that distinguishes the bi-conformal ISPG metric from the GR Schwarzschild geometry, and it is fixed without further model input. This is the operationally relevant ISPG 2PN signature.

Takeaway

The bi-conformal ISPG geometry produces an exact *refractive* 2PN Shapiro contribution

$$\Delta t_{2\text{PN}}^{(\text{refr})} = \frac{r_g^2}{cb} (2\pi),$$

which is convention-independent and numerically verified. The bent-ray and path-length contributions to the *individual* 2PN coefficient remain open (Sec. 6.5), pending consistent finite-endpoint

regularization. Their dependence on the shared $O(r_g)$ trajectory makes them identical in ISPG and GR, and the corresponding ISPG–GR 2PN *difference*

$$\Delta t_{2\text{PN}}^{\text{ISPG}} - \Delta t_{2\text{PN}}^{\text{GR}} = \frac{r_g^2}{cb} \cdot \frac{\pi}{4}$$

is closed analytically and verified numerically (Sec. 6.7). This is the operationally meaningful ISPG signature at 2PN.

Appendix 7: Perihelion Precession

Purpose and scope

This appendix computes the perihelion advance of a test body in the main theory one-metric bi-conformal exterior. At leading post-Newtonian order the result matches GR because the PPN parameters satisfy $\gamma = \beta = 1$ (Appendix 4). We present (i) the standard PPN route and (ii) the direct bi-conformal weak-field expansion route, and give a numerical value for Mercury.

Non-negotiable consistency with the main theory (one-metric formulation)

We work strictly within the main theory posture:

- matter is minimally coupled to the single physical metric $g_{\mu\nu}$;
- the physical exterior geometry is bi-conformal, $ds^2 = -e^\varphi c^2 dt^2 + e^{-\varphi} d\sigma^2$;
- in the static weak-field exterior, $\varphi(r) = -r_s/r + O(r^{-2})$ with $r_s \equiv 2GM/c^2$ (the main theory notation).

No disformal mapping and no frame change are introduced.

7.1 Standard perihelion advance in PPN form

For a nearly Keplerian orbit of semi-major axis a and eccentricity e , the 1PN perihelion advance per orbit in the PPN framework is

$$\Delta\varpi_{\text{PPN}} = \frac{6\pi GM}{a(1-e^2)c^2} \left(\frac{2+2\gamma-\beta}{3} \right). \quad (194)$$

In GR, $\gamma = \beta = 1$ and the bracket equals unity.

Appendix 4 shows that the main theory bi-conformal weak-field geometry yields

$$\gamma = 1, \quad \beta = 1. \quad (195)$$

Inserting (195) into (194) gives the standard result

$$\boxed{\Delta\varpi = \frac{6\pi GM}{a(1-e^2)c^2}} \quad (1\text{PN}). \quad (196)$$

7.2 Direct check from the bi-conformal expansion

For completeness, we record the weak-field expansion of the main theory exterior:

$$g_{tt} = -e^\varphi = -\exp\left(-\frac{r_s}{r}\right) = -\left(1 - \frac{r_s}{r} + \frac{r_s^2}{2r^2} + O(r_s^3)\right), \quad (197)$$

$$g_{rr} = e^{-\varphi} = \exp\left(\frac{r_s}{r}\right) = 1 + \frac{r_s}{r} + \frac{r_s^2}{2r^2} + O(r_s^3), \quad (198)$$

in isotropic spherical coordinates ($d\sigma^2 = dr^2 + r^2 d\Omega^2$). At 1PN order, the effective potentials satisfy $\Phi = \Psi = -GM/r$, which is precisely the condition $\gamma = 1$ and leads to (196). The distinctive ISPG signature therefore appears at higher order (2PN), treated in the dedicated Shapiro and light-bending appendices.

7.3 Numerical value for Mercury

For Mercury, take $a = 5.7909 \times 10^{10}$ m, $e = 0.2056$, and $GM_\odot = 1.3271244 \times 10^{20}$ m³/s². Equation (196) gives

$$\Delta\varpi_{\text{Mercury}} \simeq 5.02 \times 10^{-7} \text{ rad/orbit} \simeq 0.1035'' \text{ per orbit} \simeq 42.98'' \text{ per century}, \quad (199)$$

in agreement with the classic GR value.

Takeaway

In the main theory one-metric bi-conformal formulation, the Solar-System PPN parameters satisfy $\gamma = \beta = 1$ (Appendix 4). Consequently the perihelion advance matches GR at 1PN order:

$$\Delta\varpi = \frac{6\pi GM}{a(1-e^2)c^2}.$$

The ISPG discriminators are pushed to higher order and are provided by the dedicated 2PN Shapiro and 2PN bending appendices.

Appendix 8: Frame Dragging (Lense–Thirring)

Purpose and scope

This appendix records the gravitomagnetic (frame-dragging) prediction of the main theory one-metric formulation in the slow-rotation, weak-field regime. The goal is a compact, citable statement of the Lense–Thirring precession rate and its dependence on the PPN/gravitomagnetic sector. At leading (1.5PN) order the result matches GR.

Non-negotiable consistency with the main theory (one-metric formulation)

We work strictly within the main theory posture:

- matter is minimally coupled to the single physical metric $g_{\mu\nu}$ (no disformal mapping and no frame change);
- the weak-field sector satisfies the standard PPN relations $\gamma = \beta = 1$ (Appendix 4);
- the distinguishing ISPG signatures are pushed to higher order (2PN), while the 1PN/1.5PN gravitomagnetic sector remains GR-identical under minimal coupling.

This scope covers Secs. 8.1–8.2 and the Earth-satellite numerical check in Sec. 8.4. The rotational-sector material in Sec. 8.3 is flagged explicitly below as a speculative MOND-bridge *extension beyond* the declared MAIN backbone action and is not used in the 1.5PN GR-matching result.

8.1 Weak-field slow-rotation metric and the gravitomagnetic potential

In the weak-field, slow-motion expansion, a stationary rotating source generates an off-diagonal metric component g_{0i} (gravitomagnetism). In standard PPN notation, the metric can be written as

$$ds^2 = - \left(1 - \frac{2U}{c^2}\right) c^2 dt^2 - \frac{4}{c^3} (\mathbf{A} \cdot d\mathbf{x}) c dt + \left(1 + \frac{2\gamma U}{c^2}\right) d\mathbf{x}^2 + O(c^{-4}), \quad (200)$$

where $U(\mathbf{x})$ is the Newtonian potential and $\mathbf{A}(\mathbf{x})$ is the gravitomagnetic vector potential. Appendix 4 gives $\gamma = 1$ in the main theory formulation, so the spatial coefficient matches GR.

For an isolated spinning body of angular momentum $\mathbf{J}$, the external gravitomagnetic potential at leading order is

$$\mathbf{A}(\mathbf{x}) = \frac{G}{2} \frac{\mathbf{J} \times \mathbf{x}}{r^3}, \quad r \equiv |\mathbf{x}|, \quad (201)$$

which implies the gravitomagnetic field

$$\mathbf{B}_g \equiv \nabla \times \mathbf{A} = \frac{G}{2r^3} [3(\mathbf{J} \cdot \hat{\mathbf{r}})\hat{\mathbf{r}} - \mathbf{J}]. \quad (202)$$

Why this matches the main theory. In the one-metric minimal-coupling formulation, the g_{0i} sector at 1.5PN order is governed by the same momentum-constraint structure as in GR, and no preferred-frame parameters are introduced. Together with $\gamma = 1$, this yields the standard external solution (201) at the order relevant to Lense–Thirring tests.

8.2 Lense–Thirring precession rates

8.2.1 Gyroscope (spin) precession

A gyroscope moving with velocity $\mathbf{v}$ in a stationary gravitomagnetic field experiences a frame-dragging precession

$$\boldsymbol{\Omega}_{\text{LT}} = \frac{2}{c^2} \mathbf{B}_g = \frac{G}{c^2 r^3} [3(\mathbf{J} \cdot \hat{\mathbf{r}})\hat{\mathbf{r}} - \mathbf{J}], \quad (203)$$

which is the standard Lense–Thirring result at leading order.

8.2.2 Orbital node precession

For a test body on an orbit of semi-major axis a and eccentricity e around a spinning mass, the secular precession of the ascending node is

$$\dot{\Omega}_{\text{node}} = \frac{2GJ}{c^2 a^3 (1 - e^2)^{3/2}}, \quad (204)$$

and the argument of pericenter receives the companion gravitomagnetic precession

$$\dot{\omega}_{\text{LT}} = -\frac{6GJ \cos i}{c^2 a^3 (1 - e^2)^{3/2}}, \quad (205)$$

where i is the inclination of the orbital plane relative to $\mathbf{J}$.

8.3 Rotational-sector bridge to the MOND appendix (speculative extension; not part of the verified backbone)

Epistemic status of this section. The equations introduced in this section (Eqs. (206)–(214)) constitute a *speculative constitutive parametrization* for a rotational-sector MOND bridge. They are *not* part of the MAIN backbone action declared in MAIN Sec. 2.1, which is the minimal $R + (\partial\varphi)^2$ scalar-tensor form with minimal matter coupling. In particular:

- Eq. (208) introduces a new non-minimal coupling between the scalar kinetic invariant X and the gravitomagnetic invariant $|\mathbf{B}_g|^2$. This is a term *beyond* the declared backbone; its status as a consistent EFT extension (ghost freedom, energy conditions, classification within Horndeski / beyond-Horndeski) is *not* verified here and is deferred to future work.
- Eq. (213) is a *postulate* on how the total acceleration g decomposes into a Newtonian channel g_N and a halo-effective channel g_h ; it is not derived from Eq. (211). The interpolating form $\mu(x) = x/(1+x)$ in Eq. (214) therefore follows from this postulate in one step, and is not an independent derivation from the backbone.
- The 1.5PN GR-matching result of Sec. 8.2 does *not* depend on this bridge: setting $Z_{\text{rot}} \rightarrow 0$ in the strong-field limit (below) leaves the verified GR sector intact. The Solar-System tests are controlled exclusively by Secs. 8.1–8.2.
- The companion MOND paper [46] relies on this parametrization. The cross-paper “builds strictly on MAIN backbone” statement should therefore be read with this qualification: the MOND sector requires an explicit rotational-operator extension beyond the MAIN minimal action, whose full derivation from first principles is an open task.

With this status recorded, the parametrization proceeds as follows. Define

$$X \equiv \frac{1}{2} g^{\mu\nu} \partial_\mu \varphi \partial_\nu \varphi, \quad X_0 \equiv \frac{1}{2} \left(\frac{2a_0}{c^2} \right)^2, \quad (206)$$

so that in the weak field $\sqrt{X/X_0} = g/a_0$. Introduce the rotational stiffness

$$K_{\text{rot}} \left(\frac{X}{X_0} \right) \equiv 1 + \sqrt{\frac{X}{X_0}} = 1 + \frac{g}{a_0}, \quad Z_{\text{rot}} \equiv K_{\text{rot}}^{-1}. \quad (207)$$

The interpretation is simple: K_{rot} measures how “loaded” or sticky the medium is against twisting, whereas Z_{rot} is the rotationally free fraction. Strong static deficit means large K_{rot} and small Z_{rot} ; weak static deficit means $K_{\text{rot}} \rightarrow 1$ and $Z_{\text{rot}} \rightarrow 1$.

A minimal effective completion that preserves the diagonal scalar sector is then

$$S_{\text{rot}}^{\text{eff}} = \frac{1}{16\pi G} \int d^4x \sqrt{-g} Z_{\text{rot}} \left(\frac{X}{X_0} \right) \mathcal{I}_{\text{rot}}, \quad (208)$$

where $\mathcal{I}_{\text{rot}}$ is a positive invariant built only from the rotational sector. The Hubble scale is already encoded through $a_0 = cH/(2\pi)$ inside X_0 . In the weak-field stationary limit one may take

$$\mathcal{I}_{\text{rot}} \propto \frac{|\nabla \times \mathbf{A}|^2}{c^2} = \frac{|\mathbf{B}_g|^2}{c^2}. \quad (209)$$

Varying the extra term with respect to $\mathbf{A}$ gives

$$\delta S_{\text{rot}}^{\text{eff}} \propto \int d^3x [\nabla \times (Z_{\text{rot}} \mathbf{B}_g)] \cdot \delta \mathbf{A}, \quad (210)$$

so the stationary weak-field rotational equation takes the schematic form

$$\mathcal{D}_{\text{GR}}[\mathbf{A}] + \nabla \times (Z_{\text{rot}} \mathbf{B}_g) = \frac{16\pi G}{c^2} \mathbf{J}, \quad (211)$$

where $\mathcal{D}_{\text{GR}}[\mathbf{A}]$ denotes the usual GR operator from the Einstein–Hilbert sector and $\mathbf{J}$ is the angular-momentum current density. In Coulomb gauge and for slowly varying Z_{rot} , Eq. (211) reduces approximately to a GR vector equation plus an additive term proportional to $-Z_{\text{rot}} \nabla^2 \mathbf{A}$.

Immediate consequence. Because $0 \leq Z_{\text{rot}} \leq 1$, the operator-level enhancement of the GR-like rotational response is only $O(1)$ (at most a factor of order two in the slowly varying limit). Therefore this extension, by itself, does not yet bridge the old perturbative amplitude gap if the right-hand-side source remains the ordinary GR mass-current source. A full MOND completion likely requires not only operator renormalization but also a new constitutive *source-side* response of the rotating medium.

Because Z_{rot} depends on X , the same extension also feeds back into the scalar equation:

$$\square\varphi = (\text{matter source}) + \frac{1}{X_0} \nabla \cdot \left[\mathcal{I}_{\text{rot}} \frac{dZ_{\text{rot}}}{d(X/X_0)} \nabla\varphi \right], \quad (212)$$

and since $dZ_{\text{rot}}/d(X/X_0) < 0$, the rotational sector deepens rather than erases the low-pressure state in the MOND regime.

Postulate (not derivation). We *postulate* that the total inward support decomposes into a Newtonian channel g_N and a halo-effective channel g_h weighted by Z_{rot} :

$$g_N = (1 - Z_{\text{rot}}) g, \quad g_h = Z_{\text{rot}} g, \quad (213)$$

which immediately yields

$$\mu(x) = \frac{g_N}{g} = \frac{x}{1+x}, \quad x \equiv \frac{g}{a_0}. \quad (214)$$

Eq. (213) is *not* derived from Eq. (211); it is an independent constitutive postulate that, together with Eq. (207), fixes the $\mu(x) = x/(1+x)$ interpolation in one step. The companion MOND paper [46] reobtains the same local ratio from a bound-structure source-side vortex closure; that derivation is the primary locus where $\mu(x)$ is fixed, and the present appendix records Eqs. (213)–(214) only as the compact rotational-slot parametrization consistent with that closure.

Strong-field limit and Solar-System numerical check. For $g \gg a_0$, one has $K_{\text{rot}} \simeq g/a_0 \gg 1$ and $Z_{\text{rot}} \simeq a_0/g \ll 1$, so the *additional* rotational-MOND term (208) switches off and the usual Lense–Thirring sector from the Einstein–Hilbert part is recovered.

Concretely, with $a_0 \simeq 1.2 \times 10^{-10} \text{ m s}^{-2}$, the Sun-at-Earth Newtonian acceleration is $g_{\odot} \simeq 6.0 \times 10^{-3} \text{ m s}^{-2}$, giving

$$\frac{g_{\odot}}{a_0} \simeq 5.0 \times 10^7, \quad Z_{\text{rot}} \simeq \frac{a_0}{g_{\odot}} \simeq 2.0 \times 10^{-8}, \quad (215)$$

so the rotational-MOND correction to the Lense–Thirring rate at Earth-orbit distance from the Sun is bounded by $\sim 2 \times 10^{-8}$ of the GR value — well below current observational sensitivity. For a LAGEOS-like satellite in Earth’s field, $g_{\oplus}/a_0 \sim 10^{11}$, giving $Z_{\text{rot}} \sim 10^{-11}$. The extension is therefore numerically inert for all Solar-System frame-dragging tests, and the 1.5PN GR-matching statement of Sec. 8.2 is preserved at the relevant observational precision.

Status of this bridge. Equations (207)–(214) should be read as the compact rotational-sector parametrization passed forward to the companion MOND paper [46], not as a verified consequence of the MAIN backbone action. Their full derivation from first principles — including the classification of (208) as a consistent EFT extension, the ghost analysis, the energy-condition analysis, and a first-principles derivation of the constitutive split (213) — is an open task. The present appendix records Sec. 8.3 only to make the rotational slot explicit and to confirm that its $Z_{\text{rot}} \rightarrow 0$ limit in the Solar System leaves the verified 1.5PN gravitomagnetic result intact.

8.4 Numerical scale (Earth satellites)

For Earth, $J_{\oplus} \simeq 5.86 \times 10^{33} \text{ kg m}^2/\text{s}$. For a LAGEOS-like orbit ($a \simeq 1.227 \times 10^7 \text{ m}$, $e \simeq 0.004$), Eq. (204) gives a nodal precession of order

$$\dot{\Omega}_{\text{node}} \sim 30 \text{ mas/yr}, \quad (216)$$

consistent with the classic GR target of Lense–Thirring satellite tests.

Takeaway

In the main theory one-metric ISPG formulation, the weak-field gravitomagnetic sector at 1.5PN order matches GR (verified: Secs. 8.1–8.2, numerical check in Sec. 8.4). The predicted frame-dragging effects at this order are the standard Lense–Thirring precessions (203)–(205).

Section 8.3 records a rotational-sector MOND-bridge parametrization that is *not* part of the MAIN backbone action: it introduces a new scalar–gravitomagnetic coupling (208) and a constitutive split postulate (213) whose first-principles derivation and EFT-consistency analysis are deferred to future work. The bridge is numerically inert in the Solar System ($Z_{\text{rot}} \lesssim 10^{-8}$), so its status does not affect the verified GR-matching result. Distinctive ISPG discriminators beyond the verified GR limit therefore reside either in the rotational MOND sector (pending full derivation) or at higher (2PN) order (pending the open-task coefficient of Appendix 6).

Appendix 9: CMB and Linear-Cosmology Consistency

Abstract

We show that the bi-conformal scalar–tensor framework defined in the main paper [45] satisfies current linear cosmological constraints. On a spatially flat Friedmann–Lemaître–Robertson–Walker (FLRW) background written in the physical cosmic-time frame tied to matter clocks, the homogeneous scalar mode satisfies $\varphi_0 = 0$ as a consistency requirement of simultaneously (i) using comoving proper time $t = \tau$ and (ii) imposing the bi-conformal ansatz for g_{tt} . Under this condition all four Bellini–Sawicki EFT functions vanish identically ($\alpha_K = \alpha_B = \alpha_M = \alpha_T = 0$), so the linear Einstein–Boltzmann system reduces to that of Λ CDM with the same matter content and initial conditions. Therefore the primary CMB angular power spectrum, BAO scale, supernova distance measures, and BBN expansion-rate constraints are inherited at linear order. The distinguishing prediction of the framework lies in the quasi-static, weak-field regime around bound structures: the MOND sector implies a redshift evolution $a_0(z) = cH(z)/(2\pi)$.

9.1 Purpose and scope

This appendix is restricted to the covariant action of Ref. [45] and its FLRW / linear-perturbation consequences. It is *not* an exploration of alternative redshift interpretations in static backgrounds.

The objective is to demonstrate that (i) the background expansion is standard and (ii) linear perturbations obey the same Einstein–Boltzmann evolution as in Λ CDM, while (iii) the new phenomenology (MOND sector) appears only in the nonlinear, quasi-static limit around gravitationally bound systems.

9.2 Framework and conventions

We adopt:

- Metric signature $(-, +, +, +)$.
- Horndeski kinetic variable

$$X \equiv -\frac{1}{2} g^{\mu\nu} \partial_\mu \varphi \partial_\nu \varphi. \quad (217)$$

- Single-field minimally coupled Horndeski subset with constant $G_4 = M_{\text{Pl}}^2/2$, $G_3 = G_5 = 0$, and a kinetic sector fixed by the bi-conformal consistency of the field equations (see Secs. II.B and VI of Ref. [45] for the derivation and stability analysis).
- Minimal matter coupling: the matter action depends on $(g_{\mu\nu}, \psi)$ only (no direct φ –matter coupling and no disformal matter coupling).
- EFT-of-DE mapping and α -functions follow Bellini & Sawicki [21].

Note on the kinetic-term sign. For the cosmological results in this appendix the sign choice is immaterial because $\varphi_0 = 0$ implies that the scalar contributes neither background energy density nor pressure, and it does not source metric potentials at linear order. Full stability conditions for the kinetic sector are provided in Ref. [45].

9.3 Action and field equations

The gravitational sector is defined by the action (Eq. (3) of Ref. [45])

$$S = \frac{1}{16\pi G} \int d^4x \sqrt{-g} \left[R + \frac{1}{2} g^{\mu\nu} \partial_\mu \varphi \partial_\nu \varphi \right] + S_m[g_{\mu\nu}, \psi]. \quad (218)$$

Variation yields modified Einstein equations and a scalar equation of motion, which can be written schematically as

$$G_{\mu\nu} = \frac{8\pi G}{c^4} \left(T_{\mu\nu}^{(m)} + \tau_{\mu\nu}^{(\varphi)} \right), \quad \square \varphi = -\frac{8\pi G}{c^4} T, \quad (219)$$

where $T \equiv g^{\mu\nu} T_{\mu\nu}^{(m)}$ and

$$\tau_{\mu\nu}^{(\varphi)} = \partial_\mu \varphi \partial_\nu \varphi - \frac{1}{2} g_{\mu\nu} g^{\alpha\beta} \partial_\alpha \varphi \partial_\beta \varphi. \quad (220)$$

9.4 FLRW background and the $\varphi_0 = 0$ consistency in cosmic time

Consider a spatially flat FLRW background written in cosmic time,

$$ds^2 = -c^2 dt^2 + a^2(t) \delta_{ij} dx^i dx^j, \quad (221)$$

where $t = \tau$ is the proper time measured by matter-comoving observers.

In the same coordinate frame the bi-conformal ansatz of Ref. [45] imposes $g_{tt} = -e^\varphi$. Requiring simultaneous satisfaction of (i) the physical cosmic-time condition $g_{tt} = -1$ in Eq. (221) and (ii) the bi-conformal form therefore gives

$$e^{\varphi_0} = 1 \quad \Rightarrow \quad \varphi_0 = 0, \quad (222)$$

on the homogeneous FLRW background.

Escape hatch (for clarity). One may work in a non-comoving time parametrization with $t \neq \tau$ and keep $\varphi_0 \neq 0$, but then the background is not written in the standard FLRW cosmic-time gauge tied to matter clocks. The present appendix formulates the consistency statement in the physical cosmic-time frame.

Clarifying remark (construction vs dynamics). Equation (222) shows that, in the matter-clock cosmic-time frame, the homogeneous cosmological scalar background is *forced* to vanish by simultaneous imposition of the FLRW cosmic-time condition and the bi-conformal g_{tt} ansatz. Accordingly, the agreement with GR at the background level is a *consistency consequence of the construction*, rather than an independent dynamical prediction. This appendix makes this point explicit to avoid any ambiguity.

Immediate consequence. When $\dot{\varphi}_0 = 0$ the scalar stress tensor vanishes on the background:

$$\dot{\varphi}_0 = 0 \quad \Rightarrow \quad \partial_\mu \varphi_0 = 0 \quad \Rightarrow \quad \tau_{\mu\nu}^{(\varphi)} = 0, \quad (223)$$

so the background expansion follows the standard Friedmann equations for the same matter content as in Λ CDM. Explicitly,

$$3M_{\text{Pl}}^2 H^2 = \rho, \quad 2M_{\text{Pl}}^2 \dot{H} = -(\rho + p), \quad 2M_{\text{Pl}}^2 \dot{H} + 3M_{\text{Pl}}^2 H^2 = -p, \quad (224)$$

where $H \equiv \dot{a}/a$ and (ρ, p) denote the total matter–radiation energy density and pressure.

9.5 EFT mapping: vanishing α -functions

In the Bellini–Sawicki EFT-of-DE parametrization [21], linear modifications to scalar and tensor perturbations enter through the four functions $\{\alpha_K, \alpha_B, \alpha_M, \alpha_T\}$. For the present model the Horndeski functions satisfy $G_4 = M_{\text{Pl}}^2/2 = \text{const}$, $G_3 = G_5 = 0$, and the homogeneous mode obeys $\dot{\varphi}_0 = 0$ in the physical cosmic-time frame (Sec. 9.4). Using the standard definitions of Ref. [21] (see also the detailed derivation in Ref. [45]), one obtains the following minimal substitution proof:

- **Tensor speed:** α_T receives contributions from $G_{4,X}$ and G_5 . Here $G_{4,X} = 0$ and $G_5 = 0$, hence $\alpha_T = 0$ (consistent with the GW170817 bound [73]).
- **Planck-mass running:** $\alpha_M \equiv \frac{d \ln M_*^2}{d \ln a}$ with $M_*^2 = 2G_4$. Since $G_4 = M_{\text{Pl}}^2/2 = \text{const}$, $M_*^2 = M_{\text{Pl}}^2$ and $\alpha_M = 0$.
- **Braiding:** α_B is sourced by G_3 and by φ -dependence of G_4 . Here $G_3 = 0$ and $G_{4,\varphi} = 0$, hence $\alpha_B = 0$.
- **Kineticity:** α_K is proportional to the background kinetic energy $X \propto \dot{\varphi}_0^2$, with sign fixed by the Horndeski map $G_2 = -X/2$. Since $\dot{\varphi}_0 = 0$ (Sec. 9.4), $X = 0$ and $\alpha_K = 0$ trivially. For $\dot{\varphi}_0 \neq 0$, $\alpha_K = -X/H^2 < 0$, exposing the phantom kinetic sign and requiring the EFT completion of Appendix 11 for stability.

Consequently,

$$\alpha_K = \alpha_B = \alpha_M = \alpha_T = 0. \quad (225)$$

Hence, under the construction choice $\dot{\varphi}_0 = 0$ (Sec. 9.4), the linear gravitational sector is GR-identical at leading order in the metric potentials.

9.6 Linear perturbations: Poisson equation and slip

In Newtonian gauge,

$$ds^2 = -(1 + 2\Psi)c^2 dt^2 + a^2(t)(1 - 2\Phi)\delta_{ij}dx^i dx^j, \quad (226)$$

the GR-identical EFT limit (225) implies the standard relations

$$\Phi - \Psi = 0, \quad -k^2\Psi = 4\pi G a^2 \rho \Delta, \quad (227)$$

with Δ the comoving density contrast. In EFT language, departures from these GR forms arise through combinations of $\alpha_B, \alpha_M, \alpha_T$; since they vanish here, the linear Einstein–Boltzmann system for photons, baryons, CDM and neutrinos is unchanged (for the same matter content and initial conditions). The scalar perturbation propagates as a matter-sourced mode; under $\alpha_i = 0$ it does not back-react on the metric potentials at linear order.

Matter-sourced scalar mode. In Fourier space the linear scalar perturbation obeys a damped wave equation on FLRW, sourced by the matter trace (Appendix 1 sourced EOM):

$$\delta\ddot{\varphi}_k + 3H\delta\dot{\varphi}_k + \frac{c^2 k^2}{a^2}\delta\varphi_k = \frac{8\pi G}{c^2}\delta T_k, \quad \delta T \equiv \delta(g^{\mu\nu}T_{\mu\nu}^{(\text{m})}), \quad (228)$$

illustrating that the scalar mode is sourced by δT_k while remaining decoupled from the metric potentials at linear order under $\alpha_i = 0$. The term “spectator” is used here in the restricted sense of metric-decoupling, not full dynamical decoupling: the mode is driven by matter and its kinetic stability inherits the phantom-sign caveat of Appendix 11.

9.7 Observational consequences and regime separation

Inherited linear observables. Because both the background expansion and the linear gravitational sector are GR-identical, the following are inherited at linear order: primary CMB anisotropies, BAO standard ruler, supernova distance measures, and the BBN expansion-rate constraint $H(T)$.

From a GR-identical FLRW background to MOND in galaxies (sketch). With $\varphi_0 = 0$ the scalar is absent in the homogeneous cosmological background, but it can be *inhomogeneously* sourced by matter through δT as in Eq. (228). The MOND phenomenology is claimed to arise only after taking the weak-field, quasi-static limit around a bound structure (sub-horizon $k \gg aH/c$) and solving the *nonlinear* field equations sourced by localized overdensities. In this limit time derivatives are parametrically suppressed relative to spatial gradients, schematically $\partial_t \sim H \ll c \nabla$, and the scalar sector contributes an additional, environment-dependent term to the effective gravitational law. A fully rigorous derivation of the MOND limit from the nonlinear equations is presented in the main paper (Ref. [45]); the purpose of the present appendix is only to document that the same construction leaves the FLRW/linear Einstein–Boltzmann system unchanged.

Where the new physics enters. Deviations arise only in the nonlinear, quasi-static, weak-field regime around bound structures, where the same field equations yield the MOND phenomenology discussed in Ref. [45]. The central falsifiable cosmological linkage is introduced by (i) defining a Hubble wavenumber and its associated wavelength via the standard Fourier relation, and (ii) *identifying* the MOND acceleration scale with the inverse of that wavelength in units of c^2 :

$$k_H \equiv \frac{aH}{c}, \quad \lambda_H \equiv \frac{2\pi a}{k_H} = \frac{2\pi c}{H}, \quad a_0 \equiv \frac{c^2}{\lambda_H}. \quad (229)$$

The last relation in Eq. (229) is a *physical identification* of the characteristic acceleration scale with a cosmological wavelength, not a mathematical identity. Therefore,

$$a_0(z) = \frac{cH(z)}{2\pi}, \quad (230)$$

which implies a redshift drift of BTFR normalisation (in principle testable, subject to astrophysical systematics and calibration).

Since $v_{\text{flat}}^4 \propto a_0 GM$ in the deep-MOND limit, consistent with the observed radial-acceleration relation [7], one has

$$v_{\text{flat}} \propto a_0^{1/4}, \quad (231)$$

so, for example, in a baryons-only cosmology ($\Omega_b \approx 0.05$) natural to this framework, $H(1)/H_0 \approx 1.16$, giving a $\sim 16\%$ increase in a_0 at $z \simeq 1$ and a corresponding $\simeq 4\%$ upward shift in the BTFR velocity scale at fixed baryonic mass.

Possible deviations (targets for tests).

- BTFR normalisation drift from Eq. (230).
- Galaxy-scale weak-lensing shear profiles in the quasi-static regime.
- Dwarf/satellite dynamics sensitivity to environment (external-field effects) as a structural consequence of the nonlinear field equations.

9.8 Comparison summary

Table 5: Linear-cosmology sector: status of key observables.

Observable	Status in ISPG bi-conformal theory
Primary CMB C_ℓ	Unchanged (GR-identical linear sector)
BAO scale	Unchanged (same background + linear perturbations)
SNe distances	Unchanged (same background distances)
BBN $H(T)$	Unchanged (no background scalar contribution)
Linear growth $f\sigma_8$	Unchanged (GR-identical linear sector)
Lensing potential $(\Phi + \Psi)/2$	Unchanged at linear order

Table 6: Illustrative comparison to common modified-gravity frameworks (linear sector). TeVeS references: Bekenstein [12]; known CMB-related tensions without additional ingredients: see, e.g., Skordis & Złośnik [74].

Model	α_T	α_M	α_B	CMB linear sector
ISPG bi-conformal (this work)	0	0	0	GR-identical
Brans–Dicke (generic)	0	$\neq 0$	$\neq 0$	modified growth/ISW
$f(R)$ (generic)	0	$\neq 0$	$\neq 0$	modified growth/ISW
Galileon / braiding models	≈ 0	$\neq 0$	$\neq 0$	modified potentials
TeVeS-type theories	varies	varies	varies	known tensions without extra ingredients

Table 7: Nonlinear/galaxy regime: distinguishing predictions and tests.

Item	Status
MOND phenomenology	Emergent in quasi-static limit
BTFR $v^4 \propto a_0 GM$	Predicted; normalisation evolves via $a_0(z)$
$a_0(z) = cH(z)/(2\pi)$	Distinct cosmological linkage (testable in principle)
Dwarf/satellite dynamics	Environment sensitivity (EFE-type behaviour)

Appendix 10: Gravitational-Wave Modes and Polarizations

Non-negotiable consistency with the main theory (one-metric formulation)

This appendix is written strictly in the main theory formulation:

- matter is minimally coupled to the single physical metric $g_{\mu\nu}$ (no disformal or conformal matter metric is introduced here);
- the covariant backbone is the main theory scalar–tensor action and its Horndeski mapping with $G_4 = \text{const}$, $G_{4,X} = 0$, and $G_5 = 0$;
- consequently, the tensor propagation speed satisfies $c_g = c$ *exactly* (not as a vacuum-limit approximation), in agreement with GW170817.

No Einstein/Jordan frame change is required for this argument; all statements are made in the single-metric bi-conformal geometry used in the main theory.

10.1 GW170817 consistency from $\alpha_T = 0$ (exact)

In the EFT-of-dark-energy / Horndeski parameterization (Bellini–Sawicki), the tensor speed excess is

$$\alpha_T \equiv c_T^2/c^2 - 1. \quad (232)$$

For Horndeski theories, α_T receives contributions only from $G_{4,X}$ and G_5 (and their derivatives). In the main theory mapping one has

$$G_4 = \frac{M_{\text{Pl}}^2}{2} = \text{const}, \quad G_{4,X} = 0, \quad G_5 = 0, \quad (233)$$

hence

$$\alpha_T = 0 \quad \Rightarrow \quad c_g = c \text{ exactly}. \quad (234)$$

Therefore the GW170817 constraint is satisfied structurally by the main theory backbone, without invoking vacuum boundary conditions.

10.2 Scalar polarization content and the breathing mode

Beyond the two tensor polarizations of GR, the scalar sector implies an additional scalar (breathing) polarization. In the main theory programme the breathing mode is present but suppressed.

10.2.1 Multipole structure and source suppression

For compact binaries, the scalar radiation channel admits a multipolar expansion. The key suppression chain follows from the leading-order scalar sensitivity $s = 1/2$ motivated in Appendix 16 from the universal conformal redshift $m_{\text{eff}} = m_0 e^{\varphi/2}$ ($q_s = M/2$ for every body at leading order in $|\varphi|$):

- **Monopole:** no radiation for stationary total mass/charge ($S_0 = \text{const}$).

- **Dipole:** cancels at leading order because $s_A \simeq s_B \simeq 1/2$ from the universal kinematic factor; the residual body-to-body difference is $O(|\varphi_A| - |\varphi_B|)$ and is expected (but not rigorously proven) to cancel to higher order (Appendix 16, open task).
- **Leading order:** the first non-vanishing scalar radiation is the trace quadrupole, suppressed by v^2/c^2 relative to the tensor quadrupole.

Thus the scalar-radiation amplitude ratio is

$$\frac{A_b}{A_t} \sim \left(\frac{v}{c}\right)^2 \sim \frac{r_s}{r}, \quad (235)$$

where the last step uses the virial relation $v^2/c^2 \sim r_s/r$. At the leading-order level used here, no additional compactness-dependent prefactor multiplies Eq. (235): the scalar coupling constant per unit mass is fixed to $1/2$ by the kinematic conformal-redshift factor, independently of the body’s internal structure (Appendix 16, Sec. “Kinematic motivation”). The source scaling is therefore linear in r_s/r , in contrast to earlier heuristic assertions of a quadratic $(r_s/r)^2$ suppression. Whether the exact coefficient is shifted by the residual structural-response correction (the open task flagged in Appendix 16) is not resolved here.

10.2.2 Detector geometry: why L-shaped interferometers are insensitive

Even if a breathing mode is present in the incident wave, a Michelson interferometer is intrinsically less sensitive to an isotropic transverse expansion/contraction. For an idealized breathing polarization, both orthogonal arms experience comparable fractional length changes, and the detector output (arm-difference) is suppressed. This geometric suppression is independent of the source suppression (235) and, combined with the linear- r_s/r source scaling, explains why current LIGO/Virgo data do not exhibit a clean breathing signature at inspiral separations $r \gtrsim 10r_s$ (where $r_s/r \lesssim 10^{-1}$) without invoking a quadratic compactness prefactor.

10.3 Falsifiability channels

The main theory GW sector is testable via:

1. **Exact luminal tensor speed:** $c_g = c$ from (234) (GW170817 consistency by structure).
2. **Scalar polarization searches:** suppressed breathing mode with amplitude ratio $A_b/A_t \sim r_s/r$ from Eq. (235), enabled by leading-order $s = 1/2$ (Appendix 16); additional detector-geometry suppression (Sec. 10.2.2).
3. **System targets:** the best prospects are systems in the strong-field regime ($r_s/r \sim 1$), such as massive-BH mergers in the LISA band, together with detectors having polarization resolution (next-generation networks, space-based detectors). The quadratic compactness scaling assumed in earlier drafts is *not* supported by the ISPG backbone once $s = 1/2$ is recognised as the leading-order kinematic value common to all bodies; the linear scaling of Eq. (235) is the one to test.

Open task: first-principles derivation of the amplitude coefficient. Equation (235) gives the parametric scaling. The dimensionless prefactor of order unity, and in particular whether the universal $s = 1/2$ selection drives the scalar quadrupole amplitude further toward suppression (as in Brans–Dicke with $s - 1/2 \rightarrow 0$) or leaves it finite, requires an explicit PN calculation of scalar radiation in the bi-conformal metric. This calculation is an open task; pending its completion, the detectability discussion above uses only the parametric r_s/r scaling.

Takeaway

In the main theory one-metric formulation, GW170817 compatibility is guaranteed by the Horndeski condition $\alpha_T = 0$, yielding $c_g = c$ exactly. The scalar sector predicts a breathing polarization whose emission is suppressed at the source by multipole structure ($A_b/A_t \sim r_s/r$, with no additional quadratic compactness factor at leading order under the kinematic $s = 1/2$ result of Appendix 16) and further reduced in L-shaped interferometers by geometric insensitivity. The parametric scaling is established; the numerical prefactor, and any correction induced by the residual structural-response channel in Appendix 16, are open tasks.

Appendix 11: Ghost-Freedom, Hyperbolicity, Causal Propagation

Non-negotiable consistency with the main theory (one-metric formulation)

This appendix strengthens the stability discussion of the main theory §VI while remaining strictly within the main theory formulation:

- matter is minimally coupled to the single physical metric $g_{\mu\nu}$ (no disformal mapping and no frame change are introduced here);
- the covariant backbone is the main theory scalar–tensor action and its Horndeski map with $G_4 = \text{const}$, $G_{4,X} = 0$, and $G_5 = 0$;
- all characteristic (propagation) cones coincide with the null cone of the background metric, i.e. all propagating modes are luminal in the principal-symbol sense.

11.1 Baseline action and Horndeski mapping

We work with the main theory action

$$S = \frac{1}{16\pi G} \int d^4x \sqrt{-g} \left[R + \frac{1}{2} g^{\mu\nu} \partial_\mu \varphi \partial_\nu \varphi \right] + S_m[g_{\mu\nu}, \psi], \quad (236)$$

together with the bi-conformal geometry used in the main theory. In Horndeski language this corresponds to the minimal subclass

$$G_2 = -\frac{M_{\text{Pl}}^2}{2} X, \quad G_3 = 0, \quad G_4 = \frac{M_{\text{Pl}}^2}{2}, \quad G_5 = 0, \quad X \equiv -\frac{1}{2} \nabla_\mu \varphi \nabla^\mu \varphi. \quad (237)$$

11.2 Hyperbolicity and luminal characteristics (principal symbol)

The main theory establishes strong hyperbolicity by analyzing the principal symbol of the coupled $(g_{\mu\nu}, \varphi)$ system. In the present minimal Horndeski subclass, the highest-derivative part of the field equations is that of GR plus a minimally coupled scalar. Consequently:

- the tensor sector propagates on the background null cone ($\alpha_T = 0$);
- the scalar sector is governed by a second-order hyperbolic operator whose characteristics also coincide with the background null cone.

Therefore the causal cones of all propagating modes are luminal in the main theory one-metric formulation. In particular, there is no field-dependent sound speed $c_s(\varphi)$ in the principal-symbol sense; any such expression arises only after non-covariant reparameterizations not used here.

11.3 No-ghost condition in the minimal Horndeski subclass

For general Horndeski theories, the quadratic action for scalar perturbations can be written schematically as

$$S_{\text{scalar}}^{(2)} = \int d^4x a^3 \left[Q_s \dot{\zeta}^2 - Q_s c_s^2 \frac{(\nabla \zeta)^2}{a^2} + \dots \right], \quad (238)$$

where ζ is the curvature perturbation and $Q_s > 0$ is the no-ghost condition. In the Bellini–Sawicki EFT notation,

$$Q_s = \frac{M_*^2}{2} (\alpha_K + 6\alpha_B^2), \quad (239)$$

up to conventional factors, with M_*^2 the effective Planck mass and α_i the EFT functions.

In the main theory mapping (237) one has

$$M_*^2 = 2G_4 = M_{\text{Pl}}^2 = \text{const}, \quad \alpha_B = 0, \quad \alpha_T = 0, \quad \alpha_M = 0. \quad (240)$$

The remaining function α_K is proportional to the background kinetic density X for this minimal subclass. On the FLRW background used in the CMB appendix, the consistency condition yields $\dot{\varphi}_0 = 0$ and hence $X_0 = 0$, so $\alpha_K = 0$ on that background. Therefore the linear-cosmology sector is GR-identical as claimed.

Physical interpretation. The vanishing of α_K on the cosmological background does *not* indicate a pathology; it reflects that the homogeneous mode is fixed by the bi-conformal consistency condition. In the bound-structure sector (quasi-static, inhomogeneous configurations) the scalar is sourced and the kinetic operator is the standard hyperbolic one encoded by the principal symbol, ensuring well-posed evolution.

11.4 Formal constraint analysis and degree-of-freedom reduction

The main theory argues that the bi-conformal construction eliminates the would-be ghost. Below we provide the formal Dirac–Bergmann proof.

11.4.1 ADM phase space of the unconstrained theory

In the ADM decomposition the gravitational phase space is spanned by the spatial metric h_{ij} and its conjugate momentum π^{ij} , supplemented by the scalar field φ and its momentum p_φ . The lapse N and shift N^i are Lagrange multipliers enforcing the Hamiltonian and momentum constraints,

$$\mathcal{H} \approx 0, \quad \mathcal{H}_i \approx 0. \quad (241)$$

These are first-class constraints generating temporal and spatial diffeomorphisms respectively.

The canonical phase space $\Gamma = \{h_{ij}, \pi^{ij}; \varphi, p_\varphi\}$ has dimension

$$\dim \Gamma = 2 \times 6 + 2 \times 1 = 14. \quad (242)$$

With $n_1 = 4$ first-class constraints, each removing two phase-space dimensions (one constraint surface, one gauge orbit), the physical degree-of-freedom count is

$$n_{\text{DOF}}^{(\text{unconstrained})} = \frac{14 - 2 \times 4}{2} = 3 \quad (2 \text{ tensor} + 1 \text{ scalar}). \quad (243)$$

The scalar DOF carries negative kinetic energy (phantom sign); this is the would-be ghost.

11.4.2 Bi-conformal constraints as second-class conditions

The bi-conformal ansatz imposes the following algebraic relations on the phase-space variables:

$$\Phi_{ij} \equiv h_{ij} - e^{-\varphi} \delta_{ij} \approx 0 \quad (5 \text{ independent conditions}), \quad (244)$$

$$\Phi_0 \equiv N - c e^{\varphi/2} \approx 0 \quad (\text{lapse condition}), \quad (245)$$

$$\Phi^i \equiv N^i \approx 0 \quad (\text{shift condition}). \quad (246)$$

Gauge-fixing conditions.—The shift condition (246) and three of the five conditions in (244) serve as gauge-fixing conditions for the spatial diffeomorphisms generated by $\mathcal{H}_i$, selecting isotropic coordinates. The lapse condition (245) gauge-fixes the Hamiltonian constraint $\mathcal{H}$. Together these convert all four first-class constraints into second-class pairs (constraint + gauge condition), accounting for $2 \times 4 = 8$ second-class conditions in the extended phase space.

Genuine physical constraints.—The remaining two conditions in (244) are genuine constraints: they relate the two independent gravitational functions in h_{ij} (conformal factor and traceless-transverse part in the isotropic gauge) to the single scalar φ . Denote these Φ_α ($\alpha = 1, 2$).

Time persistence of Φ_α generates secondary constraints:

$$\dot{\Phi}_\alpha = \{\Phi_\alpha, H_{\text{tot}}\} \approx 0 \quad \implies \quad \bar{\Phi}_\alpha(\pi^{ij}, p_\varphi, h_{ij}, \varphi) \approx 0, \quad (247)$$

yielding two further conditions on the canonical momenta. Together, $(\Phi_\alpha, \bar{\Phi}_\alpha)$ form $n_2 = 4$ additional second-class constraints.

11.4.3 Degree-of-freedom count

Applying the standard Dirac formula (all first-class constraints have been converted to second-class by gauge-fixing):

$$n_{\text{DOF}} = \frac{\dim \Gamma - n_{2\text{nd-class}}}{2} = \frac{14 - (8 + 4)}{2} = \frac{14 - 12}{2} = 1. \quad (248)$$

The bi-conformal scalar–tensor theory has exactly **one** physical degree of freedom on the constrained background: the scalar field φ , with all metric components algebraically determined by φ and its derivatives. General perturbations (e.g. gravitational waves) that break the bi-conformal isotropy propagate in the unconstrained 3-DOF sector; their classical stability is guaranteed by the hyperbolicity result of Section 11.2.

11.4.4 Ghost-elimination theorem

Theorem (Ghost elimination on the constrained background). *A dynamical system with a single propagating degree of freedom and a well-posed Cauchy problem cannot undergo vacuum decay via pair production.*

Proof.—Vacuum decay in ghost theories proceeds through the process $|0\rangle \rightarrow |\text{ghost}\rangle + |\text{tensor}\rangle$, where the ghost and the (positive-energy) tensor mode carry equal and opposite energies, conserving total energy while producing an unbounded cascade of excitations. This process requires:

- (i) at least two independent propagating fields (ghost + partner);
- (ii) a coupling that allows energy transfer between them.

In the constrained formulation (248) only one field propagates. Condition (i) fails: there is no second field into which the phantom scalar could transfer energy. The reduced phase space is two-dimensional (φ, p_φ) ; the dynamics is a one-dimensional Hamiltonian flow—deterministic and non-branching.

Well-posedness follows from the hyperbolicity result of Section 11.2: the scalar satisfies $\square\varphi = 0$ on the background, a second-order hyperbolic equation with real characteristics. Solutions exist, are unique, and depend continuously on initial data (Hadamard well-posedness).

Combining: the bi-conformal constraint eliminates the ghost degree of freedom by construction, and the remaining single DOF evolves via a well-posed hyperbolic equation. There is no channel for vacuum instability *within the pure-gravity constrained sector*. $\square$

Scope of the theorem. The theorem above applies to the *pure-gravity* constrained background. It does *not* by itself exclude a matter-mediated decay channel of the form $|0\rangle \rightarrow |\varphi_{-E}\rangle + |\psi\bar{\psi}\rangle_{+E}$, in which minimally coupled matter fields ψ of $S_m[g_{\mu\nu}[\varphi], \psi]$ act as the positive-energy partner of the phantom scalar. The kinematic condition (i) of the proof — existence of an independent propagating partner — is satisfied by matter on-shell. What controls the physical decay rate in this channel is the *strength* of the φ – ψ coupling, which enters only gravitationally (through $g_{\mu\nu}[\varphi]$) and is therefore suppressed by Newton’s constant: at an astrophysically relevant energy scale E , the dimensionless expansion parameter is $(E/M_{\text{Pl}})^2$, giving Planck-suppressed pair-production rates for all subplanckian processes [38]. The *Complementary EFT completion* paragraph below provides the UV closure of this channel via the $\alpha X^2/M^2$ operator (with $\alpha > 0$), which should therefore be read as a *required* ingredient for matter-sector stability, not as an optional addition beyond the constrained-background proof. General perturbations in the 3-DOF perturbative sector (gravitational waves) are classically controlled by the hyperbolicity result of Sec. 11.2 but are governed by the same matter-channel caveat at the quantum level.

Energy and the phantom sign. The Hamiltonian evaluated on the constraint surface is *not* positive-definite—this is an intrinsic feature of the phantom kinetic sign and is not a pathology in itself. Negative energy is problematic only when it can be transferred to a positive-energy sector without bound. With one DOF the energy is conserved on every solution and no runaway is possible. The situation is analogous to the anti-de Sitter vacuum in GR: the total energy can be negative, but the dynamics is perfectly stable because there is no decay channel.

Relation to cuscutoon and mimetic gravity. The constraint-based ghost elimination has close parallels in cuscutoon gravity (Afshordi *et al.* 2007) and mimetic gravity (Chamseddine & Mukhanov 2013), where algebraic constraints on the scalar field similarly remove the propagating scalar DOF. The present case is intermediate: the bi-conformal constraint does not eliminate the scalar propagator entirely (the field still satisfies $\square\varphi = 0$ and mediates the breathing mode) but it removes the independent metric DOF that would serve as the ghost’s decay partner.

Complementary EFT completion. The constraint argument above addresses the background-reduced theory directly. If one nevertheless wants an explicit EFT completion of the bare phantom kinetic sector away from the strict bi-conformal background, the minimal Horndeski map can be extended to

$$G_2(X) = -\frac{M_{\text{Pl}}^2}{2} X + \frac{\alpha}{M^2} X^2, \quad \alpha > 0, \quad (249)$$

with $X \equiv -\frac{1}{2}\nabla_\mu\varphi\nabla^\mu\varphi$. The condensate then sits at nonzero $\langle X \rangle$ where $\partial G_2/\partial X > 0$, so the scalar and tensor kinetic terms are positive in the condensed vacuum [38]. Because the extra operator enters only at higher derivative order, it does not modify the weak-field static metric or the observable 2PN predictions emphasized in the main theory. It should therefore be viewed as a complementary UV/EFT stabilization channel, not as an additional ingredient required to derive the bi-conformal infrared phenomenology.

11.5 Takeaway

Within the main theory one-metric formulation, stability is established by three results:

1. **Hyperbolicity** (Sec. 11.2): the principal symbol is that of GR plus a minimally coupled scalar, implying strong hyperbolicity and luminal characteristic speeds for all propagating modes.
2. **Constraint reduction** (Sec. 11.4): the Dirac–Bergmann analysis shows that the bi-conformal conditions are second-class constraints reducing the background phase space from 3 DOF to 1 DOF.
3. **Ghost elimination on the background** (Sec. 11.4.4): a single propagating DOF has no pair-production channel; vacuum stability on the constrained background follows without requiring positive-definite energy. General perturbations propagate in the unconstrained 3-DOF sector; their classical stability is guaranteed by result (1).

The disformal-metric language and field-dependent $c_s(\varphi)$ expressions are not part of the main theory formulation and are therefore not used here.

Appendix 12: Geodesic Structure of the Bi-Conformal Compact Object

Purpose and scope

This appendix provides the complete geodesic analysis of the static bi-conformal metric

$$ds^2 = -e^{-r_s/r} c^2 dt^2 + e^{r_s/r} (dr^2 + r^2 d\Omega^2), \quad r > 0. \quad (250)$$

We show that the effective potential peaks at the photon sphere ($r = r_s$) and is exponentially suppressed to zero as $r \rightarrow 0$. Geodesics that cross the photon-sphere barrier reach the curvature-regular boundary at $r = 0$ in finite affine parameter; those that do not are reflected and complete. We classify this boundary and compare it with the Schwarzschild singularity. This supplements Sec. VI C of the main paper.

12.1 Conserved quantities and geodesic equations

The metric (250) is static and spherically symmetric. For a geodesic $x^\mu(\lambda)$ with $g_{\mu\nu}\dot{x}^\mu\dot{x}^\nu = -\kappa c^2$ ($\kappa = 1$ timelike, $\kappa = 0$ null), the conserved energy and angular momentum per unit mass are

$$\mathcal{E} = e^{-r_s/r} c^2 \dot{t}, \quad (251)$$

$$\mathcal{L} = e^{r_s/r} r^2 \dot{\varphi}. \quad (252)$$

Substituting into the normalization condition and restricting to the equatorial plane ($\theta = \pi/2$) gives the radial equation

$$\dot{r}^2 = \frac{\mathcal{E}^2}{c^2} - \kappa c^2 e^{-r_s/r} - \frac{\mathcal{L}^2}{r^2} e^{-2r_s/r}. \quad (253)$$

The effective potential is therefore

$$V_{\text{eff}}(r) = \kappa c^2 e^{-r_s/r} + \frac{\mathcal{L}^2}{r^2} e^{-2r_s/r}. \quad (254)$$

Both terms vanish as $r \rightarrow 0$ due to the exponential suppression $e^{-r_s/r}$, so $V_{\text{eff}} \rightarrow 0$ as $r \rightarrow 0^+$ —in sharp contrast to GR, where the centrifugal barrier $\mathcal{L}^2/r^2$ diverges. This is the same effective potential used to derive the ISCO and photon sphere in the main paper (Eq. (31) with $\varphi = -r_s/r$).

12.2 Photon-sphere barrier and capture

The centrifugal term $h(r) \equiv e^{-2r_s/r}/r^2$ has a unique maximum:

$$\frac{dh}{dr} = \frac{2e^{-2r_s/r}}{r^3} \left(\frac{r_s}{r} - 1 \right) = 0 \quad \implies \quad r = r_s. \quad (255)$$

This is the photon sphere. The peak value of the centrifugal term is $\mathcal{L}^2 h(r_s) = \mathcal{L}^2/(e^2 r_s^2)$.

Proposition 1 (Finite barrier). *The effective potential (254) has a finite maximum near $r = r_s$ and vanishes as $r \rightarrow 0^+$. No infinite centrifugal wall shields $r = 0$.*

Proposition 2 (Deflected geodesics). *Every geodesic (null or timelike) whose energy satisfies $\mathcal{E}^2/c^2 < V_{\text{eff,max}}$ has a turning point at some $r_{\text{min}} > 0$ and never reaches $r = 0$. Such geodesics are complete.*

Proof. If $\mathcal{E}^2/c^2 < V_{\text{eff,max}}$, there exists $r_{\text{min}} > 0$ where $\dot{r}^2 = 0$. For $r < r_{\text{min}}$, $\dot{r}^2 < 0$, which is forbidden, so the geodesic reverses. It remains in $r \geq r_{\text{min}} > 0$, where the metric is smooth and nondegenerate. $\square$

For null geodesics the critical condition is expressed via the impact parameter $b = c|\mathcal{L}|/\mathcal{E}$. A null geodesic is captured if and only if

$$b < b_c = r_s e \approx 2.718 r_s, \quad (256)$$

consistent with the shadow size derived in the main paper (Sec. VI E).

Proposition 3 (Captured geodesics). *Every geodesic (null or timelike) with $\mathcal{E}^2/c^2 > V_{\text{eff,max}}$ crosses the photon-sphere barrier and reaches $r = 0$ in finite affine parameter.*

Proof. Once the geodesic passes the barrier peak, V_{eff} decreases monotonically to zero (since $dV_{\text{eff}}/dr > 0$ for all $r < r_s$). Therefore $\dot{r}^2$ remains positive and bounded below: $\dot{r}^2 \geq \mathcal{E}^2/c^2 - V_{\text{eff,max}} > 0$. As $r \rightarrow 0$, $V_{\text{eff}} \rightarrow 0$, so $\dot{r}^2 \rightarrow \mathcal{E}^2/c^2$. The affine parameter to reach $r = 0$ is therefore bounded by the reciprocal of the minimum radial speed, $\Delta\lambda \leq r_0/|\dot{r}|_{\text{min}} < \infty$, with $|\dot{r}|_{\text{min}} = \sqrt{\mathcal{E}^2/c^2 - V_{\text{eff,max}}} > 0$. $\square$

Corollary 1. *The spacetime is geodesically incomplete. The incomplete geodesics comprise all trajectories that cross the photon-sphere barrier—a positive-measure subset of initial conditions, including all radial geodesics ($\mathcal{L} = 0$) and all non-radial geodesics with sufficiently small impact parameter.*

12.3 Radial geodesics: finite-time arrival at $r = 0$

For purely radial motion ($\mathcal{L} = 0$), Eq. (253) reduces to

$$\dot{r}^2 = \frac{\mathcal{E}^2}{c^2} - \kappa c^2 e^{-r_s/r}. \quad (257)$$

Proposition 4 (Null radial geodesics). *Every ingoing radial null geodesic reaches $r = 0$ at finite affine parameter $\lambda_0 = r_0 c/\mathcal{E}$.*

Proof. Setting $\kappa = 0$: $\dot{r} = -\mathcal{E}/c$ (constant). Integration gives $r(\lambda) = r_0 - (\mathcal{E}/c)\lambda$, reaching $r = 0$ at $\lambda_0 = r_0 c/\mathcal{E} < \infty$. $\square$

Proposition 5 (Timelike radial geodesics). *Every ingoing radial timelike geodesic reaches $r = 0$ in finite proper time.*

Proof. Setting $\kappa = 1$: $\dot{r}^2 = \mathcal{E}^2/c^2 - c^2 e^{-r_s/r}$. As $r \rightarrow 0^+$, $e^{-r_s/r} \rightarrow 0$, so $\dot{r}^2 \rightarrow \mathcal{E}^2/c^2 > 0$. The proper time from r_0 to $r = 0$ is

$$\tau = \int_0^{r_0} \frac{dr}{\sqrt{\mathcal{E}^2/c^2 - c^2 e^{-r_s/r}}}.$$

On the compact interval $[0, r_0]$ the integrand attains its maximum at $r = r_0$, where $|\dot{r}|$ is minimal, $|\dot{r}|_{\text{min}} = \sqrt{\mathcal{E}^2/c^2 - c^2 e^{-r_s/r_0}} > 0$ (for ingoing geodesics with $\mathcal{E}^2/c^2 > c^2 e^{-r_s/r_0}$, which is automatic once the photon-sphere barrier has been crossed). Therefore $\tau \leq r_0/|\dot{r}|_{\text{min}} < \infty$. $\square$

12.4 Proper distance and curvature regularity

Despite the finite affine parameter, the *proper radial distance* to $r = 0$ is infinite:

$$L = \int_0^{r_0} e^{r_s/(2r)} dr = \infty. \quad (258)$$

The integrand diverges faster than any power of $1/r$; the exponential factor $e^{r_s/(2r)}$ dominates all polynomial contributions. Thus $r = 0$ is at infinite proper distance even though geodesics reach it in finite proper time: the proper velocity $dL/d\tau = e^{r_s/(2r)} |\dot{r}| \rightarrow \infty$ diverges exponentially, while the locally measured speed approaches c from below.

For the metric (250), the Ricci scalar, the quadratic Ricci invariant, and the Kretschmann scalar are

$$\begin{aligned} R &= -\frac{r_s^2}{2r^4} e^{-r_s/r}, \\ R_{\mu\nu}R^{\mu\nu} &= \frac{r_s^4}{4r^8} e^{-2r_s/r}, \\ K \equiv R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma} &= \frac{r_s^2}{4r^8} (48r^2 - 32r r_s + 7r_s^2) e^{-2r_s/r}, \end{aligned} \quad (259)$$

each tending to 0 as $r \rightarrow 0^+$ faster than any inverse power of r due to the exponential factors. Therefore $r = 0$ is *not a curvature-invariant singularity*: every polynomial curvature scalar is finite (in fact vanishes) there. In the Hawking–Penrose sense, however, the spacetime remains *singular*, because it is geodesically incomplete at $r = 0$ (Prop. 6 below). “Singularity resolution” in this appendix refers throughout to the curvature-invariant criterion, not to geodesic completeness.

12.5 Wormhole-like topology

The areal radius is $R(r) = r e^{r_s/(2r)}$. Its derivative

$$\frac{dR}{dr} = e^{r_s/(2r)} \left(1 - \frac{r_s}{2r}\right) \quad (260)$$

vanishes at $r = r_s/2$, giving the throat $R_{\min} = (r_s/2) e \approx 1.36 r_s$. For $r > r_s/2$, R increases with r (exterior); for $r < r_s/2$, R increases as r decreases (interior). Both ends open to $R \rightarrow \infty$:

$$\lim_{r \rightarrow \infty} R(r) = \infty, \quad \lim_{r \rightarrow 0^+} R(r) = \infty. \quad (261)$$

The coordinate chart $r \in (0, \infty)$ thus covers a geometry with two asymptotic regions connected by a throat [65]. Unlike a standard Morris–Thorne wormhole, the two ends are physically inequivalent:

- **Exterior** ($r \rightarrow \infty$): $g_{tt} \rightarrow -c^2$, $g_{rr} \rightarrow 1$. Standard asymptotically flat region.
- **Deep interior** ($r \rightarrow 0^+$): $g_{tt} \rightarrow 0$, $g_{rr} \rightarrow \infty$. Infinitely redshifted region; coordinate speed of light $c e^{-r_s/r} \rightarrow 0$ (locally measured speed remains c).

12.6 Classification of the $r = 0$ boundary

Proposition 6. *The spacetime $(\mathcal{M}, g)$ with $\mathcal{M} = \mathbb{R} \times (0, \infty) \times S^2$ and metric (250) is geodesically incomplete: all geodesics that cross the photon-sphere barrier (including all radial geodesics) reach $r = 0$ at finite affine parameter.*

The metric function $f(r) = e^{-r_s/r}$ extends to a C^∞ function on $[0, \infty)$ with $f(0) = 0$ and $f^{(n)}(0) = 0$ for all n (smooth but not real-analytic). By Clarke’s theorem [72], a spacetime whose boundary has bounded curvature invariants admits a C^0 extension; since all curvature invariants vanish at $r = 0$, such an extension exists. Whether it can be made smooth (C^∞) and satisfy the field equations is an open problem.

Exterior perspective.—The boundary is unobservable from outside:

1. **Infinite coordinate time.** $t = \frac{1}{c} \int_0^{r_0} e^{r_s/r} dr = \infty$. No external observer registers an event at $r = 0$.
2. **Infinite redshift.** A photon emitted at r arrives at infinity with $1 + z = e^{r_s/(2r)} \rightarrow \infty$ as $r \rightarrow 0$.
3. **Exponential freezing.** Local proper time elapses as $d\tau/dt = e^{-r_s/(2r)} \rightarrow 0$; the deep interior is dynamically inert on any finite observation timescale.

Infalling perspective.—An observer in free fall who crosses the photon-sphere barrier reaches $r = 0$ in finite proper time $\tau_0 \leq r_0 c/\mathcal{E}$. Throughout the infall, the tidal acceleration is governed by the Kretschmann scalar $K(r)$, which rises to a finite maximum near $r \approx 0.22 r_s$ and then decreases monotonically toward $r = 0$, vanishing at the origin. The observer experiences *decreasing* tidal stress as $r \rightarrow 0$ —the opposite of the Schwarzschild case, where tidal forces diverge. At $r = 0$ the geodesic terminates at a curvature-free boundary; the physical interpretation of this termination (reflecting boundary, passage to an extended manifold, or degenerate endpoint) remains open.

The comparison with Schwarzschild and two other spacetimes with regular boundaries is given in Table 8.

Table 8: Comparison of boundary properties.

	Bi-conformal	Schwarzschild	Extremal RN	AdS
K at boundary	0	∞	finite	0
Proper distance	∞	finite ^a	∞	finite
Proper time	finite	finite	finite	finite ^b
Observable	No	No	No	No
Curvature singularity	No	Yes	No	No

^aFrom the event horizon to $r = 0$. ^bFor null geodesics reaching the conformal boundary.

Present ringdown observations do not yet distinguish a true horizon from a sufficiently compact horizonless object, so the geodesic structure derived here remains compatible with current gravitational-wave data [41]. The same qualitative picture is in line with Gogberashvili’s arguments that quantum particles are reflected by the Schwarzschild horizon and that smooth interior–exterior matching requires material sources to lie outside the photon sphere [42, 43].

12.7 Summary

1. The effective potential $V_{\text{eff}} = \kappa c^2 e^{-r_s/r} + \mathcal{L}^2 e^{-2r_s/r}/r^2$ peaks at the photon sphere ($r = r_s$) and is exponentially suppressed to zero as $r \rightarrow 0$. There is no infinite centrifugal barrier.

2. Geodesics that do not cross the photon-sphere barrier (impact parameter $b > b_c = r_s e$) are reflected and **complete**. This includes distant flybys, stable orbits, and all gravitational-lensing trajectories.
3. Geodesics that cross the barrier ($b < b_c$, including all radial geodesics) reach $r = 0$ in finite affine parameter. The spacetime is **geodesically incomplete**.
4. The boundary at $r = 0$ is curvature-regular ($K \rightarrow 0$), at infinite proper distance, and at infinite redshift. It is causally inert and unobservable from the exterior.
5. An infalling observer experiences *vanishing* tidal force at $r = 0$ —in sharp contrast to the Schwarzschild singularity, where tidal disruption is inevitable.
6. The capture cross-section ($\sigma = \pi r_s^2 e^2$) is comparable to Schwarzschild ($\sigma_{\text{GR}} = 27\pi r_s^2/4$), but the endpoint is qualitatively different: a curvature-free, unobservable boundary rather than a curvature singularity.

Appendix 13: Covariant Hysteresis Sector

Purpose, scope, and reader contract

This appendix accompanies the main theory. It serves two purposes:

1. **Baseline map:** a compact dependency map of the ISPG ingredients used downstream;
2. **Hysteresis phenomenology:** a specification of effective memory effects (captured by the field χ or, at effective level, by the transported hysteretic response derived in the companion MOND paper [46]) that reproduce galaxy-halo phenomenology without new particle species.

Relation to the companion MOND paper [46]. the companion MOND paper [46] derives the MOND interpolating function $\mu(x) = x/(1+x)$ from the bound-structure Bernoulli+vortex closure in rotating galaxies, with the scalar field equation $\square\varphi = S$ on a weak-field rotating background fixing the relaxation time scale and transport rate. The present Appendix 13 provides an *effective closure* perspective on the hysteretic part of the same physics: the field χ captures the transported memory response that the companion MOND paper [46] derives from first principles via Fourier decomposition and Bessel eigenvalue analysis. The two appendices are *complementary*: the companion MOND paper [46] gives the formal bound-structure derivation; Appendix 13 gives the phenomenological implementation and strengthening routes for future work. In other words, the χ appearing here is not a second unrelated MOND coefficient: it is the dynamical coarse-grained field-level representative of the same transported response that the companion MOND paper [46] writes locally as the vortex-source multiplier. The precise mathematical identification of χ with the companion paper’s local multiplier—the explicit mapping between Eq. (278) and the vortex-closure equation at the bound-structure level—is the content of the Route B coarse-graining programme (Sec. 13.8) and is listed there as an open

item; the statement here is that both objects describe the same physical quantity, not that the mapping has been carried out in closed form.

Scope boundary. The χ evolution laws were originally introduced as an effective closure reproducing the results of the companion MOND paper [46]. In Sec. 13.6 of this appendix, **Route A is now completed:** the χ field equation is derived from a minimal covariant action, and ghost freedom, gradient stability, hyperbolicity, and the quasi-static reduction to the phenomenological relaxation are established. One strengthening route remains open:

- **Route B (coarse-graining/EFT):** obtain χ as an emergent slow variable from the nonlinear dynamics of φ around bound structures, connecting to the rotational bound-structure analysis of the companion MOND paper [46].

13.1 Baseline: bi-conformal metric and covariant action

13.1.1 Bi-conformal metric ansatz

The constructive requirement is that a single scalar φ controls local clock rates and local spatial scales reciprocally. This is implemented by the bi-conformal ansatz

$$ds^2 = -e^\varphi c^2 dt^2 + e^{-\varphi} (dx^2 + dy^2 + dz^2). \quad (262)$$

13.1.2 Covariant action and minimal Horndeski placement

The baseline action is of scalar-tensor form with minimal coupling of matter to $g_{\mu\nu}$:

$$S = \frac{1}{16\pi G} \int d^4x \sqrt{-g} \left[R + \frac{1}{2} g^{\mu\nu} \partial_\mu \varphi \partial_\nu \varphi \right] + S_m[g_{\mu\nu}, \psi]. \quad (263)$$

A compact Horndeski mapping (minimally coupled single-field subclass) is:

$$G_2 = -\frac{M_{\text{Pl}}^2}{2} X, \quad G_3 = 0, \quad G_4 = \frac{M_{\text{Pl}}^2}{2}, \quad G_5 = 0, \quad X \equiv -\frac{1}{2} \partial_\mu \varphi \partial^\mu \varphi. \quad (264)$$

(Discussion of the sign choice and stability conditions is given in the main theory and is not reopened here.)

13.2 Weak-field limit and the Shapiro discriminator

Expanding (262) for small φ yields

$$g_{tt} \simeq -(1 + \varphi), \quad g_{ij} \simeq (1 - \varphi) \delta_{ij}, \quad (265)$$

so that $\Phi = \Psi = (c^2/2)\varphi$ and $\gamma_{\text{PPN}} = 1$ at leading order.

The main theory derives the 1PN Shapiro delay via an effective refractive index $n(r) = e^{-\varphi}$ (identical to GR at 1PN, as $\gamma = 1$) and identifies a structurally distinct 2PN signature from the exponential $n = e^{-\varphi}$; the refractive sub-piece of the 2PN delay is derived exactly, while the full 2PN coefficient (including bent-ray and path-length contributions) is deferred to future work (main theory Sec. 3.7 and Appendix 6). For later use we record the dependency structure:

$$n(r) = e^{-\varphi} \simeq 1 + \frac{r_s}{r}, \quad \Delta t_{2\text{PN}}^{\text{ISPG}} \propto \frac{G^2 M^2}{c^5 b}. \quad (266)$$

13.3 Linear cosmology versus bound structures: regime separation

A central design goal of the main theory is that *linear* cosmology can remain GR-identical while *nonlinear* bound structures exhibit new behaviour. The separation is operationally expressed by a scale hierarchy:

$$k \ll \frac{aH}{c} \quad (\text{super-/horizon scales, linear cosmology}), \quad k \gg \frac{aH}{c} \text{ and } \partial_t \ll c\nabla \quad (\text{quasi-static, bound structures}) \quad (267)$$

Cosmic-time compatibility. In the cosmic-time frame tied to matter clocks (comoving observers), $t = \tau$ along comoving worldlines, so $g_{tt} = -1$ in standard FLRW form. The bi-conformal structure (262) then requires $e^{\varphi_0} = 1$ and hence $\varphi_0 = 0$ at the homogeneous level. This fixes the *background* without eliminating *inhomogeneous* responses sourced by matter in the quasi-static sector.

13.4 Matter-sourced scalar response in the quasi-static sector

The representative matter-sourced perturbation equation used in the main theory (metric-decoupled at linear order under $\alpha_i = 0$, but not a free spectator; it is driven by δT_k and inherits the phantom-sign caveat of the background action, cf. App. 11) is

$$\delta\ddot{\varphi}_k + 3H\delta\dot{\varphi}_k + \left(\frac{c^2k^2}{a^2}\right)\delta\varphi_k = \frac{8\pi G}{c^2}\delta T_k. \quad (268)$$

The RHS coefficient $8\pi G/c^2$ follows from the FLRW expansion of the sourced scalar equation $\square\varphi = -(8\pi G/c^4)T$ of App. 1: $\square\varphi = -c^{-2}(\ddot{\varphi} + 3H\dot{\varphi}) + a^{-2}\nabla^2\varphi$, multiplied by $-c^2$, gives the form above and matches App. 9 Eq. (228). Equation (268) makes explicit that localized inhomogeneities can excite $\delta\varphi$ through δT_k even when the homogeneous background is fixed to $\varphi_0 = 0$ in cosmic time. The MOND phenomenology of the main theory is associated with the quasi-static, weak-field, nonrelativistic limit around bound structures, not with the linear FLRW sector.

13.5 The hysteresis field χ : closure, selection principle, and strengthening routes

13.5.1 Intent and modelling posture

We introduce χ as the effective field capturing medium-memory (hysteresis) effects. In this appendix it serves as the field-level effective closure of the transported response discussed in the companion MOND paper [46]. Its purpose is to (i) encode memory effects with minimal extra structure, (ii) produce falsifiable halo-scale consequences, and (iii) admit a controlled coarse-graining derivation from the underlying bound-structure dynamics via Route B.

13.5.2 Phenomenological relaxation ansatz

As motivation, we first record the relaxation law toward a local equilibrium functional χ_{eq} :

$$\dot{\chi} = -\tau^{-1}(\chi - \chi_{\text{eq}}), \quad (269)$$

with τ to be determined. (This ansatz is *derived* from a covariant action in Sec. 13.6.5.)

13.5.3 Closure choice for relaxation

We take the leading local equilibrium to be proportional to the quasi-static gradient strength,

$$\chi_{\text{eq}} = \beta |\nabla\varphi|, \quad \beta > 0, \quad (270)$$

and we adopt the leading local relaxation-rate law

$$\tau^{-1} = \gamma |\nabla\varphi| \quad (\text{Variant A}). \quad (271)$$

Working assumptions (explicit). In the hysteresis sector we restrict attention to the quasi-static, weak-field regime around bound structures, characterized by $\partial_t \ll c\nabla$ and (for Fourier modes) $k \gg aH/c$. The closure is required to be *local*: it depends only on local scalars built from the fields and the metric in this regime. Concretely, at leading order we allow dependence on

$$|\nabla\varphi|, \quad Y \equiv \sqrt{\nabla_\mu\varphi \nabla^\mu\varphi}, \quad (272)$$

and we exclude explicit dependence on coordinate radius r , galaxy-specific input scales, or ad hoc profile functions. The “no new macroscopic scales” requirement means that τ and χ_{eq} must be expressible in terms of these local invariants and constants already present in the baseline sector, without introducing an additional halo-tuned length or time scale.

Minimal (leading-order) character of Variant A. Under locality, monotonicity (stronger gradients relax faster), and the absence of additional macroscopic scales, the choice $\tau^{-1} \propto |\nabla\varphi|$ can be viewed as the leading-order term in a local gradient-expansion for the relaxation rate. Alternative closures correspond to higher-order corrections (involving additional invariants or suppressed operators) or to genuinely new microphysical scales; in either case they predict quantitatively distinct halo profiles and lensing–dynamics relations and are therefore observationally discriminable.

13.5.4 Static limit $\Rightarrow$ isothermal scaling (consistency check)

In the static weak-field limit of isolated sources one has $|\nabla\varphi| \propto 1/r^2$. Under Variant A, the closure drives $\chi \propto |\nabla\varphi|$ and therefore $\chi \propto 1/r^2$ (and similarly $\rho_\chi \propto 1/r^2$ if ρ_χ is taken proportional to χ). **Intended reading:** this is a *consistency check* for the chosen closure, not a first-principles derivation. The decisive content is that other relaxation laws predict different halo profiles, providing observational discriminants.

13.5.5 Rotation curves and lensing fingerprints

Because χ is tied to gradients of φ , galaxy dynamics and lensing are linked. Different closure choices $\tau(|\nabla\varphi|)$ predict different outer-halo slopes and transition radii, and therefore distinct relations between rotation curves and shear profiles. This is one of the primary falsification channels.

13.5.6 Route A sketch: a concrete covariant template

To make Route A concrete, we record a covariant template in which a memory field relaxes toward a gradient-built equilibrium. Define the invariant

$$Y \equiv \sqrt{\nabla_\mu \varphi \nabla^\mu \varphi} \quad (\text{spatial-gradient dominated in the quasi-static sector}). \quad (273)$$

A minimal augmented-action ansatz can be written schematically as

$$S[g, \varphi, \chi] = S_{\text{ISPG}}[g, \varphi] + \int d^4x \sqrt{-g} \left[-\frac{1}{2}(\nabla\chi)^2 - U(\chi) - \frac{\lambda}{2}(\chi - \beta Y)^2 \right], \quad (274)$$

where λ controls the stiffness toward the local equilibrium $\chi \simeq \beta Y$. In the quasi-static sector with sufficiently large λ , the field tracks $\chi_{\text{eq}} \approx \beta Y$ up to controlled corrections. We now carry out this programme explicitly.

13.6 Route A realization: field equation and stability

13.6.1 Minimal action ($U = 0$)

Following the parameter-economy principle, we set $U(\chi) = 0$ in (274), retaining only the kinetic and stiffness terms:

$$S_\chi = \int d^4x \sqrt{-g} \left[-\frac{1}{2} \nabla_\mu \chi \nabla^\mu \chi - \frac{\lambda}{2} (\chi - \beta Y)^2 \right], \quad (275)$$

with $Y \equiv \sqrt{\nabla_\mu \varphi \nabla^\mu \varphi}$ with $\lambda > 0$ carrying dimensions of $[\text{length}^{-2}]$ (equivalently mass^2 in natural units) and $\beta > 0$. Here χ is a genuine dynamical field. In the quasi-static bound-structure limit it should be read as the field-level effective realization of the local transported-response multiplier used in the companion MOND paper [46], while Route B remains the microscopic coarse-graining step. No additional self-interaction potential is introduced; the only scale controlling χ dynamics is the stiffness parameter λ .

Microscopic derivation of λ . In the classical context of this appendix, λ was originally a free parameter. The quantum extension [47] derives it from the quasi-normal mode structure of the oscillon:

$$\lambda_{\text{micro}} = \frac{\omega_0^2}{c^2} = \left(\frac{m_0 c}{\hbar} \right)^2 e^\varphi, \quad (276)$$

where ω_0 is the fundamental confining frequency and m_0 the oscillon rest mass. This eliminates λ as a free parameter: the stiffness is set by the particle mass, reducing the hysteresis sector to a single undetermined constant β .

Regularity note. The source functional $Y = \sqrt{\nabla_\mu \varphi \nabla^\mu \varphi}$ is non-smooth at $Y = 0$: the variation $\delta Y / \delta(\partial_\mu \varphi) = \nabla^\mu \varphi / Y$ diverges there. In the quasi-static sector around gravitating sources $\nabla \varphi \neq 0$ everywhere outside the source centre, so the singularity is not encountered in practice. Should a fully global treatment be required, the standard remedy is to replace Y with a regularized form $Y_\epsilon \equiv \sqrt{\nabla_\mu \varphi \nabla^\mu \varphi + \epsilon^2}$ ($\epsilon \rightarrow 0^+$), which preserves all results derived below while rendering the action smooth.

13.6.2 Field equation for χ

Varying (275) with respect to χ gives

$$\frac{\delta S_\chi}{\delta \chi} = \square \chi - \lambda(\chi - \beta Y) = 0, \quad (277)$$

which can be written as

$$\boxed{\square \chi - \lambda \chi = -\lambda \beta Y}. \quad (278)$$

This is a Klein–Gordon equation with effective mass $m_\chi^2 = \lambda$ and source $\lambda \beta Y$. The source is built entirely from the scalar field φ of the baseline ISPG sector; no new coupling to matter is introduced.

Structure of the equation.

- In regions where φ is slowly varying ($Y \rightarrow 0$), the source vanishes and χ relaxes to zero on a timescale $\sim 1/\sqrt{\lambda}$.
- Near a gravitating source where $|\nabla \varphi| \sim r_s/r^2$, the field is driven toward $\chi \approx \beta Y$, i.e. the local equilibrium of the phenomenological closure.
- The $\square \chi$ term provides the propagation dynamics absent in the original relaxation ansatz (269), upgrading it from a first-order ODE to a covariant hyperbolic PDE.

13.6.3 Ghost freedom

A ghost (negative-norm excitation) arises when the kinetic term of a dynamical field carries the wrong sign, rendering the Hamiltonian unbounded from below. In (275) the χ kinetic term is

$$\mathcal{L}_{\text{kin}}^{(\chi)} = -\frac{1}{2} \nabla_\mu \chi \nabla^\mu \chi. \quad (279)$$

With the mostly-plus signature convention $(-, +, +, +)$ used throughout the main theory, $\nabla_\mu \chi \nabla^\mu \chi = -\dot{\chi}^2/c^2 + (\partial_i \chi)^2$, so the Lagrangian density becomes

$$\mathcal{L}_{\text{kin}}^{(\chi)} = +\frac{1}{2c^2} \dot{\chi}^2 - \frac{1}{2} (\partial_i \chi)^2. \quad (280)$$

The time-kinetic part $+\dot{\chi}^2/(2c^2)$ enters with a *positive* sign. The conjugate momentum is

$$\pi_\chi = \frac{\partial \mathcal{L}}{\partial \dot{\chi}} = \frac{\dot{\chi}}{c^2}, \quad (281)$$

and the corresponding Hamiltonian density is

$$\mathcal{H}_\chi = \frac{c^2}{2} \pi_\chi^2 + \frac{1}{2} (\partial_i \chi)^2 + \frac{\lambda}{2} (\chi - \beta Y)^2. \quad (282)$$

Every term is non-negative for $\lambda > 0$. Therefore the χ sector is **ghost-free**: the energy is bounded from below and no negative-norm states propagate.

13.6.4 Gradient stability

Gradient instability occurs when spatial perturbations grow exponentially, signalling that the field equation is ill-posed as an initial-value problem. To check this, linearize (278) around a static background $\bar{\chi}$:

$$\chi = \bar{\chi} + \delta\chi, \quad \delta\chi \propto e^{-i\omega t + i\mathbf{k} \cdot \mathbf{x}}. \quad (283)$$

Substituting into (278) and keeping only the homogeneous part (the source $\lambda\beta Y$ is fixed by the background φ), the dispersion relation is

$$\boxed{\omega^2 = c^2 |\mathbf{k}|^2 + \lambda c^2}. \quad (284)$$

For all $\lambda > 0$:

- $\omega^2 > 0$ for every wavenumber $\mathbf{k}$, so no mode grows exponentially.
- The group velocity $v_g = \partial\omega/\partial|\mathbf{k}| = c^2|\mathbf{k}|/\omega < c$, so χ perturbations propagate subluminally (massive field).
- The phase velocity $v_p = \omega/|\mathbf{k}| = c\sqrt{1 + \lambda/|\mathbf{k}|^2} > c$ in the usual massive-field sense, but carries no information superluminally.

The spatial-gradient coefficient in the Hamiltonian (282) is $+\frac{1}{2}$ (positive definite), confirming the absence of gradient instability at the nonlinear level as well.

13.6.5 Quasi-static reduction to the phenomenological closure

We now show that the covariant field equation (278) reproduces the phenomenological relaxation law (269) in the appropriate limit.

In the FLRW background the d'Alembertian acquires a Hubble friction term:

$$\square\chi = -\frac{1}{c^2}\ddot{\chi} - \frac{3H}{c^2}\dot{\chi} + \frac{1}{a^2}\nabla^2\chi. \quad (285)$$

The field equation (278) becomes

$$-\frac{1}{c^2}\ddot{\chi} - \frac{3H}{c^2}\dot{\chi} + \frac{1}{a^2}\nabla^2\chi + \lambda\chi = \lambda\beta Y. \quad (286)$$

Overdamped, long-wavelength limit. In the quasi-static sector around bound structures, two conditions hold:

1. *Overdamped*: the field evolves on timescales much longer than $1/\sqrt{\lambda}c$, so $|\ddot{\chi}| \ll \lambda c^2 |\chi|$ and the $\ddot{\chi}$ term can be dropped.
2. *Long-wavelength*: the characteristic spatial scale ℓ of χ satisfies $1/\ell^2 \ll \lambda$, so $|\nabla^2\chi| \ll \lambda |\chi|$ and the Laplacian can be dropped.

Under these two conditions, Eq. (286) reduces to

$$-\frac{3H}{c^2}\dot{\chi} = \lambda(\chi - \beta Y), \quad (287)$$

which can be rewritten as

$$\boxed{\dot{\chi} = -\frac{\lambda c^2}{3H}(\chi - \beta Y)}. \quad (288)$$

This is precisely the phenomenological relaxation (269) with the identifications

$$\chi_{\text{eq}} = \beta Y, \quad \tau^{-1} = \frac{\lambda c^2}{3H}. \quad (289)$$

Physical content.

- The relaxation timescale $\tau = 3H/(\lambda c^2)$ is now *derived*, not postulated: it is set by the ratio of the Hubble friction to the stiffness parameter λ .
- For $\lambda \gg H^2/c^2$ the field tracks βY almost instantaneously; for $\lambda \sim H^2/c^2$ the lag becomes cosmologically significant, producing observable hysteresis.
- The equilibrium value $\chi_{\text{eq}} = \beta Y \approx \beta |\nabla\varphi|$ in the quasi-static sector, matching the phenomenological closure (270).
- The original Variant A ansatz $\tau^{-1} = \gamma |\nabla\varphi|$ (Eq. 271) was a phenomenological gradient-dependent closure originally introduced to reproduce the isothermal scaling of the halo regime. The present covariant derivation supersedes it: the leading-order relaxation rate obtained directly from the action (275) is instead Hubble-regulated, $\tau^{-1} = \lambda c^2/(3H)$, and is not gradient-dependent to this order. Reconciling the two forms therefore requires either a subleading gradient-dependent correction to the covariant λ , or the identification of a specific regime in which the effective Variant A re-emerges from the covariant theory. This reconciliation is listed as an open item in the Route B programme (Sec. 13.8), and the outer-halo region $|\nabla\varphi|$ varies rapidly provides the natural observational setting in which any residual gradient dependence would be tested against the Hubble-regulated baseline.

13.6.6 Hyperbolicity and well-posedness

A PDE system is well-posed as an initial-value problem if and only if its principal part is hyperbolic. The principal symbol of the χ field equation (278) is determined by its highest-derivative term:

$$\mathcal{P}(\chi) = \square\chi + (\text{lower order}) \implies \sigma(\xi) = g^{\mu\nu}\xi_\mu\xi_\nu, \quad (290)$$

where ξ_μ is the covector of the Fourier mode. The mass term $\lambda\chi$ and the source $\lambda\beta Y$ are algebraic (non-derivative) and do not contribute to the principal symbol.

For a Lorentzian metric $g_{\mu\nu}$ with signature $(-, +, +, +)$, the symbol $\sigma(\xi) = 0$ defines a cone with one timelike and three spacelike directions. This is the defining property of a *normally hyperbolic* operator. By the standard theory of linear wave equations on globally hyperbolic spacetimes (see e.g. Wald, *General Relativity*, 1984, Ch. 10; Hawking & Ellis, *The Large Scale Structure of Space-Time*, 1973, Ch. 7), the Cauchy problem

$$\square\chi - \lambda\chi = -\lambda\beta Y, \quad \chi|_\Sigma = \chi_0, \quad \dot{\chi}|_\Sigma = \chi_1, \quad (291)$$

on any spacelike hypersurface Σ admits a unique, continuously dependent solution for smooth initial data (χ_0, χ_1) and source Y .

Joint system (φ, χ) . The coupling between φ and χ enters only through the source $Y = \sqrt{\nabla_\mu\varphi\nabla^\mu\varphi}$, which is algebraic in first derivatives of φ . Crucially, χ does *not* appear in the principal part of the φ equation (the baseline ISPG sector), and φ does not appear in the principal part of the χ equation. The principal symbol of the joint system is therefore block-diagonal:

$$\sigma_{\text{joint}} = \begin{pmatrix} g^{\mu\nu}\xi_\mu\xi_\nu & 0 \\ 0 & g^{\mu\nu}\xi_\mu\xi_\nu \end{pmatrix}, \quad (292)$$

which is hyperbolic. Both fields propagate on the same light cone at the level of the principal symbol, and the lower-order coupling terms do not spoil well-posedness.

13.7 Normalization and parameter economy

13.7.1 Closing the normalization via the cosmological MOND scale

A key aim is to preserve the parameter economy of the main theory by fixing the free parameters through the MOND scale a_0 , determined independently in the cosmological sector (the main theory) as

$$a_0 = \frac{cH_0}{2\pi}. \quad (293)$$

The 2π factor follows from the standard Fourier convention:

$$k_H \equiv \frac{aH}{c}, \quad \lambda_H \equiv \frac{2\pi a}{k_H} = \frac{2\pi c}{H}, \quad \text{and we identify } a_0 \equiv \frac{c^2}{\lambda_H}. \quad (294)$$

The identification $a_0 \equiv c^2/\lambda_H$ is a physical linkage (not a mathematical identity): it ties a characteristic cosmological wavelength scale to an acceleration scale in the emergent MOND sector.

Normalization target equation (explicit). To make the parameter-economy goal operational, we parametrize the halo-regime closure such that the asymptotic circular speed can be written as

$$V_{\text{flat}}^4 = GM \mathcal{N}(\beta, \lambda) a_0, \quad (295)$$

where $\mathcal{N}(\beta, \lambda)$ is the dimensionless normalization implied by the specific χ implementation. In the covariant formulation of Sec. 13.6, the relaxation rate is Hubble-regulated ($\tau^{-1} = \lambda c^2/3H$), so in the static limit $\chi \rightarrow \chi_{\text{eq}} = \beta Y$ the normalization depends only on β ; the stiffness λ controls only the *approach* to equilibrium. The *parameter-free target* is $\mathcal{N} = 1$, at which the BTFR normalisation would be inherited directly from a_0 with no additional fit parameter. Whether the covariant closure actually realises $\mathcal{N} = 1$ —i.e. whether the correct β is the one that already follows from $a_0 = cH_0/2\pi$ with no residual multiplicative factor—is an open check and depends on the full bound-structure derivation in the companion MOND paper [46] and on the Route B programme (Sec. 13.8). The non-circularity of the construction is nevertheless preserved: a_0 is determined first from the cosmological linkage in the main theory and only then used to set the galaxy-scale normalisation through Eq. (295); if the ultimate $\mathcal{N}$ differs from unity the discrepancy is absorbed into a single well-defined dimensionless ratio, not into a free phenomenological parameter.

13.8 Derived vs. Proposed: compact ledger

Derived (baseline)	Bi-conformal metric (262); covariant action (263); weak-field $\gamma_{\text{PPN}} = 1$; Shapiro 1PN with a 2PN discriminator; linear-cosmology GR-identity under $\alpha_i = 0$ (the main theory and CMB appendix); spectator sourcing (268); MOND interpolating function $\mu(x) = x/(1+x)$ from the bound-structure Bernoulli+vortex closure (the companion MOND paper [46]).
Derived (Route A, this appendix)	Minimal augmented action (275) with $U = 0$; Klein–Gordon field equation (278); ghost freedom (Sec. 13.6.3); gradient stability and subluminal propagation (Sec. 13.6.4); quasi-static reduction to first-order relaxation with derived timescale $\tau^{-1} = \lambda c^2/(3H)$ (289); hyperbolicity and well-posedness of the joint (φ, χ) system (Sec. 13.6.6).
Derived (quantum extension [47])	Microscopic stiffness $\lambda = \omega_0^2/c^2 = (m_0 c/\hbar)^2 e^\varphi$ from the oscillon’s fundamental quasi-normal mode (Eq. 276); reduces free parameters from two (λ, β) to one (β) .
Remaining open items	Route B coarse-graining derivation of χ from nonlinear φ dynamics; numerical halo profiles from the full PDE (286); comparison of Hubble-regulated vs. gradient-dependent relaxation with observed rotation curves; determination of β from first principles.

13.9 Falsifiable discriminants

The hysteresis closure is designed to be testable by comparing model variants. The most direct discriminants are:

- **Halo scaling and transition:** different $\tau(|\nabla\varphi|)$ imply different radial slopes and transition radii.
- **Lensing–dynamics relation:** because χ is tied to gradients of φ , shear profiles and rotation curves are linked in a way that differs from NFW-based fits.
- **Environment sensitivity:** the local-gradient dependence naturally introduces external-field sensitivity, providing a clear test with satellites and dwarfs.
- **Redshift drift of normalization:** through $a_0(z) \propto H(z)$ (the main theory), the BTFR normalization is predicted to evolve with cosmic time.

13.10 Notes for future work

The Route A programme is now complete at the analytical level: the χ field equation is derived from a covariant action, and ghost freedom, gradient stability, hyperbolicity, and the quasi-static reduction to the phenomenological closure are established. The remaining open items are:

1. **Route B (coarse-graining):** derive χ as an emergent slow variable from the nonlinear dynamics of φ around bound structures, connecting to the Bessel eigenvalue analysis of the companion MOND paper [46]. (The quantum extension [47] partially addresses this by deriving S_χ from the oscillon quasi-normal mode structure and fixing $\lambda = \omega_0^2/c^2$; a full coarse-graining from the classical nonlinear sector remains open.)
2. **Numerical profiles:** solve the full PDE (286) for realistic galaxy potentials and compare the resulting rotation curves and lensing profiles with observations.
3. **Relaxation-rate discriminant:** the Hubble-regulated rate $\tau^{-1} = \lambda c^2/(3H)$ and the original gradient-dependent rate $\tau^{-1} = \gamma|\nabla\varphi|$ predict different outer-halo slopes; quantify this difference for observational tests.

Takeaway

The χ hysteresis sector is now placed on a covariant footing. Starting from the minimal augmented action (275) with $U = 0$: (i) the field equation (278) is a standard massive Klein–Gordon equation sourced by φ gradients; (ii) the sector is ghost-free, gradient-stable, and the joint (φ, χ) system is hyperbolic; (iii) in the quasi-static, overdamped limit the covariant equation reduces to the phenomenological relaxation law with a *derived* timescale $\tau^{-1} = \lambda c^2/(3H)$; (iv) the parameter economy is strong: the stiffness $\lambda = \omega_0^2/c^2$ is derived from the oscillon quasi-normal mode structure in the quantum extension [47] (Eq. 276), leaving β as the sole undetermined constant. The BTFR normalization is fixed by the independently determined $a_0 = cH/(2\pi)$.

Appendix 14: Bi-conformal Interior and Space Deficit

Purpose and scope

This appendix develops the physical interpretation of the bi-conformal interior derived in the main theory (Sec. VI–VIII). No new fields, potentials, or action modifications are introduced. Every statement follows from the same covariant action and bi-conformal ansatz used throughout the paper.

Note on nomenclature. This appendix was previously filed under the name “ISPG_Quantum”; it has been renamed to “ISPG_Interior” to avoid confusion with the standalone quantum extension paper [47], which treats $\hbar$ -dependent physics (canonical quantization, loop corrections, decoherence, and mass spectrum). The content of the present appendix is *entirely classical*.

14.1 The bi-conformal “freezing” effect

The static bi-conformal metric with $\varphi_0 = -r_s/r$ yields:

$$g_{tt} = -e^{-r_s/r}, \quad g_{rr} = e^{+r_s/r}. \quad (296)$$

As $r \rightarrow 0$, $e^{-r_s/r} \rightarrow 0$ exponentially and $e^{+r_s/r} \rightarrow \infty$ exponentially. This means:

- **Clock rate:** $d\tau/dt = \sqrt{|g_{tt}|} = e^{-r_s/(2r)} \rightarrow 0$. Local clocks stop exponentially.

- **Radial distance:** $d\ell = \sqrt{g_{rr}} dr = e^{r_s/(2r)} dr \rightarrow \infty$. The proper radial distance to $r = 0$ diverges—the centre is infinitely far away.
- **Effective speed:** $c_{\text{eff}}(r) = c e^{-r_s/r} \rightarrow 0$. All dynamical processes are exponentially suppressed.

This is the mechanism verified symbolically and numerically in Appendix 15 (Verification).

14.2 Mass as integrated pressure deficit

In the main theory, the scalar field $\varphi = -r_s/r$ encodes the local pressure state of the information substance (IS). The Schwarzschild radius $r_s = 2GM/c^2$ sets the depth of the pressure deficit. The total gravitational mass is therefore

$$M = \frac{c^2 r_s}{2G}, \quad (297)$$

which is the integrated effect of the spatial pressure rarefaction. In the ISPG framework, mass *is* the deficit: a deeper and more extended rarefaction corresponds to a larger gravitational mass.

14.3 Light does not disappear

A photon falling radially inward experiences the coordinate speed $dr/dt = c e^{-r_s/r}$, which approaches zero exponentially as $r \rightarrow 0$. From the perspective of a distant observer, the photon takes an infinite coordinate time to reach $r = 0$. In terms of affine parameter, however, the geodesic reaches $r = 0$ in a finite interval (see Appendix 12), so the spacetime is geodesically *incomplete*. The crucial distinction is that the boundary at $r = 0$ is curvature-regular ($K \rightarrow 0$), not singular: the incompleteness reflects the topology of the bi-conformal manifold, not a breakdown of physics.

The observational consequence is that light entering the deep-deficit region undergoes extreme redshift: $1 + z = e^{r_s/(2r)} \rightarrow \infty$. The photon is not absorbed or destroyed; it simply becomes undetectable to external observers. This resolves the information-loss aspect without invoking quantum processes: information is encoded in the asymptotically redshifted field configuration and is never causally disconnected from the exterior.

14.4 Minimum mass for horizon-like behaviour

The bi-conformal metric does not produce a true event horizon (horizonless, as shown in the main theory). However, for an external observer, the surface at $r = r_s$ acts as an effective “soft horizon” beyond which the exponential redshift makes the interior effectively invisible.

This soft horizon can form only if sufficient mass is concentrated within a region comparable to r_s . For the deficit to be gravitationally self-consistent, the source must supply an integrated stress-energy that sustains $\varphi = -r_s/r$ down to small r . There exists a minimum deficit budget below which the bi-conformal solution does not produce the exponential freezing: for weak fields ($r_s/r \ll 1$), the metric is perturbatively close to Minkowski and no “freezing zone” forms.

14.5 After dissipation: the permanent space deficit

If the compact object loses energy (e.g. through radiative processes), the question arises: what remains?

In the bi-conformal framework, the answer follows from the freezing effect itself. As the deficit region $\varphi \rightarrow -\infty$ deepens, the effective speed $c_{\text{eff}} \rightarrow 0$ and the local relaxation timescale diverges:

$$\tau_{\text{relax}}(r) \propto e^{+r_s/r} \rightarrow \infty \quad \text{as } r \rightarrow 0. \quad (298)$$

The deep interior cannot refill because the refilling process itself is exponentially suppressed by the very deficit it would need to erase. This is a self-reinforcing mechanism: the deeper the deficit, the slower the recovery, and therefore the more permanent the deficit.

The endpoint is not a material relic of finite mass. It is a **permanent space deficit**—a region of arbitrarily deep rarefaction that persists indefinitely because its own depth prevents external processes from reaching it on any finite timescale.

14.6 Compatibility with the main theory

This interpretation introduces no new physics. It follows directly from:

- the bi-conformal metric ansatz (main theory, Eq. (1));
- the vacuum field equation $\square\varphi = 0$ (main theory, Eq. (6));
- the singularity resolution (main theory, Sec. VIII): $K \rightarrow 0$, curvature-regular boundary;
- the stability verification (Appendix 15): $c_{\text{eff}}^2 > 0$, no tachyonic instability.

The space deficit picture replaces the need for Planck-scale relics or evaporation endpoint arguments: the bi-conformal structure itself provides a singularity-free, curvature-regular, and permanently stable interior without invoking $\hbar$ or semiclassical physics.

Takeaway

The bi-conformal interior is a region of exponentially suppressed dynamics: clocks stop, distances diverge, and effective speeds vanish. Mass is the integrated pressure deficit of the information substance. Light entering the deep interior is not lost but asymptotically frozen. After energy dissipation, the remaining space deficit is self-perpetuating because the refilling timescale diverges with the deficit depth. No new fields, no semiclassical arguments, and no material relics are required—the classical ISPG framework is sufficient.

Appendix 15: Verification of Scalar Perturbation Stability

Purpose and scope

This appendix provides the independent symbolic and numerical verification of the scalar perturbation stability analysis used in the main theory (Sec. VI, phantom kinetic term and stability). Every formula in the relevant derivation has been checked symbolically (with computer algebra) and the key physical conclusions have been verified numerically.

Overall verdict: the analysis is **mathematically correct** and **complete** at the probe/test-field level. The $\ell = 0$ negative well is shown to be too shallow to support a bound state (Bargmann bound).

Non-negotiable consistency with the main theory (one-metric formulation)

We verify the same scalar perturbation problem as in the main theory: static background $\varphi_0 = -r_s/r$ with $r_s = 2GM/c^2$, bi-conformal metric, no change to the action or field equations.

15.1 Symbolic verification (formula by formula)

15.1.1 Derivatives of $\varphi_0 = -r_s/r$

$$\varphi'_0(r) = \frac{r_s}{r^2}, \quad \varphi''_0(r) = -\frac{2r_s}{r^3}. \quad (299)$$

Status: ✓ correct.

15.1.2 Effective wave speed c_{eff}^2

From the principal part of the wave equation:

$$c_{\text{eff}}^2(r) = \frac{A(r)}{B(r)} = c^2 e^{2\varphi_0(r)} = c^2 e^{-2r_s/r} > 0 \quad (r > 0). \quad (300)$$

Status: ✓ correct. Gradient instability is absent.

15.1.3 Tortoise coordinate

$$\frac{dr_*}{dr} = \sqrt{\frac{B}{A}} = \frac{e^{-\varphi_0}}{c} = \frac{e^{r_s/r}}{c}, \quad \frac{dr}{dr_*} = c e^{\varphi_0} = c e^{-r_s/r}. \quad (301)$$

Status: ✓ correct (including the $1/c$ factor).

15.1.4 First-derivative coefficient $P(r)$

The transformation to r_* produces a first-order term with coefficient

$$P(r) = -c e^{\varphi_0} \varphi'_0 = -c e^{-r_s/r} \frac{r_s}{r^2}. \quad (302)$$

Cross-checked via $P = h'/h^2$ where $h = dr_*/dr = e^{r_s/r}/c$:

$$\frac{h'}{h^2} = \frac{-r_s e^{r_s/r}/(c r^2)}{e^{2r_s/r}/c^2} = -\frac{c r_s e^{-r_s/r}}{r^2} = P. \quad (303)$$

Status: ✓ correct.

15.1.5 General V_{eff} formula

After removing the first derivative via the standard substitution $u = e^{-\frac{1}{2} \int P dr_*} \Psi$, the Schrödinger-form potential is

$$V_{\text{eff}} = Q + \frac{1}{2} \frac{dP}{dr_*} + \frac{1}{4} P^2, \quad (304)$$

where $Q = c^2 e^{2\varphi_0} \ell(\ell+1)/r^2$.

Computing each piece separately:

$$Q = c^2 e^{2\varphi_0} \frac{\ell(\ell+1)}{r^2}, \quad (305)$$

$$\frac{1}{2} \frac{dP}{dr_*} = \frac{1}{2} c e^{\varphi_0} \frac{d}{dr} (-c e^{\varphi_0} \varphi_0') = -\frac{c^2}{2} e^{2\varphi_0} (\varphi_0'' + (\varphi_0')^2), \quad (306)$$

$$\frac{1}{4} P^2 = \frac{c^2}{4} e^{2\varphi_0} (\varphi_0')^2. \quad (307)$$

Summing:

$$V_{\text{eff}} = c^2 e^{2\varphi_0} \left[\frac{\ell(\ell+1)}{r^2} - \frac{1}{2} \varphi_0'' - \frac{1}{4} (\varphi_0')^2 \right]. \quad (308)$$

Status: ✓ correct.

15.1.6 Explicit V_{eff} for $\varphi_0 = -r_s/r$

Substituting the derivatives:

$$V_{\text{eff}}(r) = c^2 e^{-2r_s/r} \left[\frac{\ell(\ell+1)}{r^2} + \frac{r_s}{r^3} - \frac{r_s^2}{4r^4} \right]. \quad (309)$$

Status: ✓ correct.

15.1.7 $\ell = 0$ factorisation

$$V_{\text{eff}}^{(\ell=0)}(r) = c^2 e^{-2r_s/r} \frac{r_s}{r^4} \left(r - \frac{r_s}{4} \right). \quad (310)$$

Status: ✓ correct. Negative for $0 < r < r_s/4$, positive for $r > r_s/4$.

15.2 Numerical analysis of the $\ell = 0$ negative region

The $\ell = 0$ case is the most delicate point, since V_{eff} does become negative for $0 < r < r_s/4$. The question is whether this well is deep and wide enough to support a bound state ($\omega^2 < 0$), which would signal a tachyonic instability.

All numerical results below are in units $c = r_s = 1$.

15.2.1 Location and depth of the well

Quantity	Value
Negative region	$0 < r < r_s/4 = 0.25 r_s$
Location of minimum	$r \approx 0.211 r_s$
Maximum $ V_{\text{eff}} $ in the well	$\approx 1.5 \times 10^{-3}$
Exponential suppression at $r = r_s/4$	$e^{-8} \approx 3.4 \times 10^{-4}$

15.2.2 Bound-state estimate

A standard sufficient condition for the *absence* of bound states in a one-dimensional potential well is that the “well integral” be small:

$$\mathcal{I} \equiv \int_{\text{well}} |V_{\text{eff}}(r_*)| dr_* \ll 1 \implies \text{no bound state.} \quad (311)$$

(This follows from the Bargmann bound: the number of bound states $n \leq (2\pi)^{-1} \int |V| dr_*$ in one dimension.)

Numerical evaluation:

$$\mathcal{I} \approx 1.0 \times 10^{-4}. \quad (312)$$

Since $\mathcal{I} \ll 1$, the well is **far too shallow and narrow to support a bound state**. Therefore, even for $\ell = 0$, no tachyonic mode ($\omega^2 < 0$) exists at probe level.

15.2.3 Physical reason for the extreme suppression

The well lies entirely in the region $r < r_s/4$, where the exponential factor $e^{-2r_s/r}$ provides devastating suppression:

- At $r = r_s/4$: $e^{-2r_s/(r_s/4)} = e^{-8} \approx 3.4 \times 10^{-4}$.
- At $r = r_s/8$: $e^{-16} \approx 1.1 \times 10^{-7}$.
- At $r = r_s/16$: $e^{-32} \approx 1.3 \times 10^{-14}$.

This is the bi-conformal “freezing” effect: near the centre, $e^{\varphi_0} \rightarrow 0$ and all dynamical quantities are exponentially suppressed.

15.2.4 $\ell \geq 1$: positivity of V_{eff}

For $\ell \geq 1$ the centrifugal barrier $\ell(\ell+1)/r^2$ dominates the bracket in (309). The bracket

$$\frac{\ell(\ell+1)}{r^2} + \frac{r_s}{r^3} - \frac{r_s^2}{4r^4}$$

is positive for all $r > r_s/(2(2\ell(\ell+1)-1))$. Below this radius the bracket can become negative, but the exponential prefactor $e^{-2r_s/r}$ suppresses the resulting well far more effectively than in the $\ell = 0$ case. For $\ell = 1$ the bracket changes sign at $r \approx 0.183 r_s$ where $e^{-2r_s/r} \approx e^{-10.9} \sim 10^{-5}$; the Bargmann well integral is $\mathcal{I} \lesssim 10^{-7}$, orders of magnitude below the bound-state threshold. Hence no tachyonic mode exists for any $\ell \geq 1$.

15.3 Summary of verification

Item	Method	Status
φ'_0, φ''_0	symbolic	✓
$c_{\text{eff}}^2 = c^2 e^{-2r_s/r} > 0$	symbolic	✓
Tortoise: $dr_*/dr = e^{r_s/r}/c$	symbolic	✓
$P(r) = -c e^{-r_s/r} r_s/r^2$	symbolic (h'/h^2)	✓
V_{eff} general form	symbolic	✓
V_{eff} explicit	symbolic	✓
$\ell = 0$ factorisation	symbolic	✓
$\ell = 0$ negative region $\subset (0, r_s/4)$	numerical	✓
$\ell = 0$ well integral $\mathcal{I} \approx 10^{-4} \ll 1$	numerical	✓
No bound state for $\ell = 0$	Bargmann bound	✓
$\ell \geq 1$: $\mathcal{I} \lesssim 10^{-7}$, no bound state	analytical + numerical	✓

15.4 Scope reminder

Task 1a establishes:

- (i) **No gradient instability**: $c_{\text{eff}}^2 > 0$ everywhere.
- (ii) **No tachyonic instability**: for all ℓ the Bargmann well integral satisfies $\mathcal{I} \ll 1$, so no bound state exists at probe level.

Task 1a does **not** close:

- The **ghost** (negative kinetic energy) question — this requires the coupled tensor+scalar analysis (Task 1b, even/polar sector) and/or a Hamiltonian sign check.
- The phantom sign does not alter the vacuum scalar equation ($\square \delta\varphi = 0$ in both signs), so the probe test is structurally blind to the ghost.

15.5 Next step: Task 1b (even/polar sector)

The decisive remaining test is the **even/polar sector** of the coupled metric–scalar perturbations on the same bi-conformal background. Before computing from scratch, the following literature should be consulted:

1. **Bronnikov & Rubin (2012)**, Phys. Rev. D **86**, 024028 — phantom wormhole even-sector instability.
2. **Gonzalez, Herrera & Nucamendi (2009)** — ghost scalar wormhole instability.

The central question is whether their instability mechanism (typically tied to throat/wormhole topology) applies to the ISPG background, which has *no horizon* and is one-sided regular:

- If it applies $\Rightarrow$ strong motivation to revisit the sign choice or identify an additional constraint that stabilises the even sector.
- If it does not apply $\Rightarrow$ phantom sign is classically safe at linearised level; the ghost becomes a purely quantum/EFT issue, addressable within the framework outlined in the main theory.

Takeaway

The scalar perturbation stability analysis of the main theory (Sec. VI) is verified symbolically and numerically. The $\ell = 0$ potential well in $0 < r < r_s/4$ is exponentially suppressed; the well integral $\mathcal{I} \approx 10^{-4} \ll 1$ implies no bound state by the Bargmann bound, hence no tachyonic instability at probe level.

Appendix 16: Scalar Sensitivity of Compact Objects

Purpose and scope

This appendix analyses the scalar sensitivity parameter s introduced in the main theory (Sec. 9.3) from the coupled field equations of the bi-conformal action. The goals are: (i) derive the modified Tolman–Oppenheimer–Volkoff (TOV) equations for a static spherical star in the bi-conformal metric; (ii) record the formal definition of s and the kinematic motivation that selects the value $s = 1/2$ from the universal conformal rescaling $m_{\text{eff}} = m_0 e^{\varphi/2}$; (iii) examine the status of the structural “shift-symmetry” argument that was previously invoked to promote $s_{\text{excess}} = 0$ to an exact, body-universal statement; and (iv) confront the resulting picture with existing pulsar-timing constraints. No new fields or parameters are introduced; every equation follows from the same action used throughout the main theory.

Epistemic status (this appendix). The TOV pressure-gradient equation (316) is strictly shift-invariant under $\varphi \rightarrow \varphi + \delta$ (Sec. 16.1). The coupled scalar equation (317), by contrast, acquires an explicit $e^{-\varphi}$ prefactor on its matter-source side and therefore *breaks* strict shift symmetry in the presence of matter. As a result, the earlier claim that “ $s_{\text{excess}} = 0$ exactly and non-perturbatively for every body” is *not* a theorem of the classical coupled system. What survives is: (a) a strictly shift-invariant hydrostatic sector; (b) a kinematic argument from the universal effective mass $m_{\text{eff}} = m_0 e^{\varphi/2}$ that motivates $s = 1/2$ at leading order; (c) a weak-field regime in which shift invariance of the full coupled system is recovered approximately. A fully rigorous proof that $s_{\text{excess}} = 0$ exactly (rather than to leading order in $|\varphi|$) is an *open task*, deferred to future work. All claims below are labelled accordingly.

Non-negotiable consistency with the main theory

- The covariant action is

$$S = \frac{1}{16\pi G} \int d^4x \sqrt{-g} \left[R + \frac{1}{2} g^{\mu\nu} \partial_\mu \varphi \partial_\nu \varphi \right] + S_m[g_{\mu\nu}, \psi], \quad (313)$$

with matter minimally coupled to the single physical metric $g_{\mu\nu}$.

- The bi-conformal ansatz for a static spherical configuration is

$$ds^2 = -e^\varphi c^2 dt^2 + e^{-\varphi}(dr^2 + r^2 d\Omega^2). \quad (314)$$

- The scalar field equation in the presence of matter is $\square\varphi = -(8\pi G/c^4) T$, where $T = g^{\mu\nu} T_{\mu\nu}^{(m)}$.

16.1 Interior field equations

For a static, spherically symmetric perfect fluid with energy density ρ and pressure p (both measured locally), the matter energy-momentum tensor is $T^{\mu\nu} = (\rho + p/c^2) u^\mu u^\nu + p g^{\mu\nu}$, and its trace is $T = -\rho c^2 + 3p$.

Substituting the bi-conformal ansatz (314) into the field equations and restricting to hydrostatic equilibrium yields three coupled ODEs. Define the auxiliary mass function

$$m(r) \equiv \int_0^r 4\pi \rho(r') e^{-3\varphi/2} r'^2 dr', \quad (315)$$

which accounts for the proper spatial volume element $\sqrt{g^{(3)}} = e^{-3\varphi/2} r^2 \sin\theta$.

16.1.1 Modified TOV: pressure gradient

The radial component of hydrostatic equilibrium, $\nabla_\mu T^{\mu r} = 0$, gives

$$\frac{dp}{dr} = -(\rho + p/c^2) \frac{c^2}{2} \frac{d\varphi}{dr}. \quad (316)$$

This differs from the standard TOV equation in that the gravitational “acceleration” is $g = (c^2/2) d\varphi/dr$ rather than the Schwarzschild expression $Gm/(r^2(1 - 2Gm/c^2r))$. In the weak-field limit $\varphi \approx -r_s/r$ and $g \approx GM/r^2$, recovering Newton.

Key structural property: Eq. (316) depends on $d\varphi/dr$, *not* on φ itself. A constant shift $\varphi \rightarrow \varphi + \delta$ leaves the pressure gradient, and therefore the entire density and pressure profiles $\rho(r)$, $p(r)$, unchanged.

16.1.2 Scalar field equation inside matter

Direct reduction of the scalar equation $\square\varphi = -(8\pi G/c^4) T$ in the bi-conformal static metric (314) gives the d’Alembertian $\square\varphi = e^\varphi(\varphi'' + 2\varphi'/r)$ and, with $T = -\rho c^2 + 3p$, yields

$$\frac{d^2\varphi}{dr^2} + \frac{2}{r} \frac{d\varphi}{dr} = \frac{8\pi G}{c^4} e^{-\varphi} (\rho c^2 - 3p). \quad (317)$$

This form is consistent with the canonical vacuum solution $\varphi = -r_s/r$: at $T = 0$ both sides vanish, and $\varphi'' + 2\varphi'/r = 0$ is satisfied identically. A symbolic verification of $\square\varphi = 0$ for the vacuum solution is provided by `tools/check_kappa_consistency.py` (see Issue #14 audit).

Structural property and partial shift invariance. Under a global shift $\varphi \rightarrow \varphi + \delta$ with constant δ , the left-hand side of Eq. (317) is invariant (it contains only derivatives), but the right-hand side acquires a factor $e^{-\delta}$ through the explicit $e^{-\varphi}$ coupling. The interior scalar equation is therefore *not* strictly shift-invariant when matter is present. In the weak-field limit $e^{-\varphi} \approx 1$, the shift symmetry is recovered approximately, but at strong field (neutron-star interiors, $|\varphi| \sim 0.1$ near the centre) the correction is $\sim 10\%$.

16.1.3 Boundary conditions

At the centre ($r = 0$): $\varphi(0) = \varphi_c$ (a free parameter determined by matching), $d\varphi/dr|_0 = 0$ (regularity). At the stellar surface ($r = R$, where $p = 0$): the interior solution joins the vacuum exterior $\varphi_{\text{ext}} = -r_s/r + \varphi_\infty$, where φ_∞ is the asymptotic (background) scalar field value.

16.2 Shift invariance: sectoral analysis

The TOV pressure-gradient equation (316) depends only on a *derivative* of φ , whereas the reduced scalar equation (317) carries an explicit $e^{-\varphi}$ prefactor on its matter-source side. The two sectors therefore have different behaviour under a rigid shift $\varphi \rightarrow \varphi + \delta$, and must be analysed separately.

16.2.1 TOV-level shift invariance (rigorous)

Lemma (Hydrostatic shift invariance, exact). Fix the scalar profile $\varphi_0(r)$. Then the hydrostatic equation (316) is invariant under $\varphi_0 \rightarrow \varphi_0 + \delta$ for any constant δ : the pressure gradient dp/dr , and hence the density and pressure profiles $\rho(r)$ and $p(r)$ generated by integrating (316) from the centre, are unchanged.

Proof.—Under $\varphi_0 \rightarrow \varphi_0 + \delta$, $d\varphi_0/dr \rightarrow d\varphi_0/dr$ (unchanged), so the right-hand side of (316) is unchanged. Integrating from the centre with boundary conditions at $r = 0$, the profiles $\rho(r)$ and $p(r)$ are identical. $\square$

This sub-sectoral invariance is *exact* and *non-perturbative*. It is a direct consequence of the minimal matter coupling to $g_{\mu\nu}$ and of the constancy of G_4 (no Brans–Dicke running of the gravitational coupling). It underpins the statement that “the stellar structure does not care about the absolute value of the background scalar, only about its gradient.”

16.2.2 Coupled-system shift symmetry (approximate; broken at $O(|\varphi|)$)

The coupled system is not strictly shift-invariant. Under $\varphi \rightarrow \varphi + \delta$ with ρ, p held fixed, the left-hand side of (317) is unchanged but the right-hand side acquires a factor $e^{-\delta}$. To restore a solution, one must rescale the matter source by $e^{+\delta}$, which is not a symmetry of the matter action. Explicitly:

- in the weak-field limit $|\varphi| \ll 1$, one has $e^{-\varphi} \approx 1 - \varphi + \frac{1}{2}\varphi^2 - \dots$, and shift invariance is recovered at zeroth order;
- the leading correction is linear in $|\varphi|$; for neutron-star interiors $|\varphi_c| \sim r_s/R \sim 0.2$, the shift asymmetry is at the $\sim 10\%$ level;
- for vacuum ($\rho = p = 0$), the right-hand side vanishes identically and shift invariance is exactly restored—consistent with the vacuum no-hair derivation of main theory Sec. 7.

Status. An *exact* rigid-shift symmetry of the coupled TOV–scalar system is not a property of the classical equations in matter; it holds strictly only at the level of (316) and in vacuum. Any claim of exact non-perturbative universality of s at arbitrary compactness must be supported by an argument beyond a naive rigid shift (e.g. an on-shell reformulation of the matter sector that absorbs the $e^{-\varphi}$ factor into a redefined proper density; this is the programme flagged in the quantum paper and discussed in Sec. 16.7 below). This programme is an **open task**.

16.3 Scalar sensitivity: formal definition, motivated value

Place the star in a weak, uniform external scalar field $\varphi_\infty \neq 0$. The total (Komar) mass measured at infinity depends on φ_∞ through two channels: (a) the conformal factor $e^{\varphi/2}$ that defines the effective mass (main theory Eq. (46)); and (b) the internal structure adjustment—the star re-equilibrates when the external field changes.

The scalar sensitivity is

$$s \equiv -\frac{1}{2} \left. \frac{\partial \ln M_{\text{Komar}}}{\partial \varphi_\infty} \right|_{\varphi_\infty=0}. \quad (318)$$

16.3.1 Kinematic motivation: $s = 1/2$ from the universal conformal rescaling

The position-dependent effective mass assigned to any localised body in the bi-conformal metric is

$$m_{\text{eff}}(\varphi) = m_0 e^{\varphi/2}, \quad (319)$$

as derived in the main theory (Sec. 9.2, Eq. (46)) from the conformal redshift of the static clock factor. This rescaling is *universal*: it multiplies every localised energy by the same factor $e^{\varphi/2}$, independently of composition or compactness. Differentiating (319) with respect to φ_∞ at fixed bare mass m_0 gives the kinematic part of the Komar response:

$$\left. \frac{\partial \ln m_{\text{eff}}}{\partial \varphi_\infty} \right|_{\text{kin}} = \frac{1}{2} \implies s_{\text{kin}} = -\frac{1}{2} \cdot \frac{1}{2} \cdot (-1) = \frac{1}{2}, \quad (320)$$

where the factor (-1) comes from the sign convention adopted in (318) and the conformal-redshift factor is invoked at the boundary. The kinematic channel alone therefore selects $s = 1/2$ for every body.

Status. Equation (320) identifies $s = 1/2$ as the natural value common to all bodies from a purely kinematic (geometric) argument. It does *not* prove $s_{\text{excess}} = 0$ *exactly*: the structural-adjustment channel (the star re-equilibrating when φ_∞ shifts) is not shown to vanish strictly, because the full coupled shift symmetry is broken (Sec. 16.2.2). What is established here is:

- in the weak-field regime $|\varphi| \ll 1$, the structural-response correction is $O(|\varphi|)$ and subdominant, so $s \simeq 1/2 + O(|\varphi|)$ for every body;
- at leading order in $|\varphi|$, the body-to-body difference is $O(|\varphi_A - \varphi_B|)$ at matched asymptotic φ_∞ , vanishing identically when the bodies share a common background;
- the question whether the $O(|\varphi|)$ correction itself cancels body-to-body at all orders (giving $s_{\text{excess}} = 0$ exactly at arbitrary compactness) is **an open task**.

16.3.2 Structural-response correction (not derived to vanish)

The *bare-mass* Komar integral (main theory Eq. (47); see also “Scope of Eq. (321)” below) is

$$M_{\text{Komar}}^{(0)} \equiv \int \rho_0 e^{-\varphi} d^3x. \quad (321)$$

When φ_∞ shifts by δ , the interior profile does *not* shift rigidly: because the coupled equation (317) carries the matter-side factor $e^{-\varphi}$, the solution $\varphi(r; \varphi_\infty + \delta)$ differs from $\varphi(r; \varphi_\infty) + \delta$ by a

correction of order $|\varphi|$ localised inside the body. Consequently the identity $M_{\text{Komar}}^{(0)}(\varphi_\infty + \delta) = e^{-\delta} M_{\text{Komar}}^{(0)}(\varphi_\infty)$, which would give $\partial \ln M_{\text{Komar}}^{(0)} / \partial \varphi_\infty = -1$ exactly, holds only to leading order in $|\varphi|$.

Scope of Eq. (321): bare-mass vs. Tolman–Komar form. Equation (321) weights the rest-mass density ρ_0 by the gravitational redshift / proper volume factor $e^{-\varphi} = e^{\varphi/2} e^{-3\varphi/2}$ alone. The stationary Tolman–Komar perfect-fluid mass, on the other hand, carries an additional pressure contribution:

$$M_{\text{Komar}}^{\text{TK}} = \int (\rho_0 + 3p/c^2) e^{-\varphi} d^3x. \quad (322)$$

The two integrals agree in the dust limit $p \ll \rho_0 c^2$ and differ by $O(p/\rho_0 c^2)$ otherwise. For neutron-star interiors $p/(\rho_0 c^2) \sim 0.2\text{--}0.3$ near the centre, the relative difference is $\sim 30\%$ and concentrated in the pressure-dominated core. The kinematic motivation of $s = 1/2$ (Sec. 16.3.1) uses only the universal conformal factor $e^{\varphi/2}$ common to both forms, and is therefore not altered by this choice. The body-to-body cancellation of the residual structural-response correction at $O(|\varphi|)$ —the open task flagged in Sec. 16.2.2—depends on which Komar integral is adopted for matter-filled strong-field regimes; the perfect-fluid analysis with Eq. (322) is part of the same open task.

Working assumption used elsewhere. Throughout the main theory and companion papers, the working value adopted is

$$s = \frac{1}{2}, \quad s_{\text{excess}} = 0 \quad (\text{leading order; exact result deferred}). \quad (323)$$

The dipole radiation formula (main theory Eq. (17)) and the Nordtvedt prediction (main theory Eq. (26)) are evaluated with (323); the resulting predictions are correct to leading order in $|\varphi|$. Whether these predictions remain exact at arbitrary compactness requires the open-task analysis flagged in Sec. 16.2.2.

16.3.3 Physical interpretation (weak-field)

The approximate vanishing of s_{excess} has a transparent physical origin. The position-dependent effective mass $m_{\text{eff}} = m_0 e^{\varphi/2}$ (main theory Eq. (46)) is a *universal* rescaling: every particle, every atom, every bound system is multiplied by the same conformal factor. When the background field φ_∞ shifts, all masses shift by the same multiplicative factor $e^{\delta/2}$ at leading order. Since sensitivity measures the *differential* response between bodies, and all bodies respond identically at leading order, the leading-order excess vanishes. The higher-order structural correction, induced by the matter-side $e^{-\varphi}$ coupling, is not proven to cancel body-to-body and is the content of the open task.

16.4 Coupled linear perturbation: status

A naive linearisation of the scalar field equation (317) around a background $\varphi_0(r)$, *holding the density ρ fixed*, yields an apparent screening equation:

$$\frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{d(\delta\varphi)}{dr} \right) + \frac{8\pi G\rho}{c^2} e^{-\varphi_0} \delta\varphi = 0, \quad (\text{incomplete: } \rho \text{ held fixed}) \quad (324)$$

with an effective mass term $\mu^2 \propto 8\pi G\rho e^{-\varphi_0}/c^2$ that would screen the external field perturbation and produce $s_{\text{excess}} \neq 0$.

This equation is incomplete. It omits the structural response $(\delta\rho, \delta p)$ that accompanies $\delta\varphi$. The correct coupled linearisation is recorded below; it weakens but does *not* fully cancel the screening term at strong field.

16.4.1 Full coupled perturbation

Let $\varphi = \varphi_0 + \epsilon\delta\varphi$, $\rho = \rho_0 + \epsilon\delta\rho$, $p = p_0 + \epsilon\delta p$. Linearising the scalar equation (317) gives:

$$\text{LHS: } \delta\varphi'' + \frac{2}{r}\delta\varphi', \quad (325)$$

$$\text{RHS: } \frac{8\pi G}{c^4} e^{-\varphi_0} [(\delta\rho c^2 - 3\delta p) - (\rho_0 c^2 - 3p_0)\delta\varphi]. \quad (326)$$

16.4.2 The rigid-shift ansatz does not solve the coupled system in matter

Substitute $\delta\varphi = \text{const}$ and $\delta\rho = \delta p = 0$:

- LHS: $\delta\varphi' = 0 \Rightarrow \text{LHS} = 0$.
- RHS: $\delta\rho = \delta p = 0 \Rightarrow \text{RHS} = -\delta\varphi (8\pi G/c^4) e^{-\varphi_0} (\rho_0 c^2 - 3p_0)$.
- The two sides agree only where the background matter source $(\rho_0 c^2 - 3p_0)$ vanishes, i.e. outside the star. Inside the star the rigid shift is *not* a solution.

This is the linearised counterpart of the strict shift-symmetry failure in matter established in Sec. 16.2.2: the naive rigid-shift ansatz ceases to be an equilibrium once the explicit $e^{-\varphi}$ coupling is retained.

What survives. In the vacuum exterior ($\rho_0 = p_0 = 0$), the rigid shift is an exact solution; in the weak-field regime $|\varphi_0| \ll 1$, the right-hand side is suppressed by $e^{-\varphi_0} \approx 1$ and the leading-order rigid-shift picture is restored. Whether a modified ansatz (e.g. one in which the structural response $(\delta\rho, \delta p)$ is chosen self-consistently to cancel the matter-side correction) solves the coupled system non-perturbatively is an open question: it would amount to showing that the full coupled system has a hidden rigid-shift invariance not manifest in the equations as written. The construction of such a reformulation (if it exists) is the same **open task** referenced in Sec. 16.2.2.

16.5 No-hair property: scope

Two statements must be distinguished:

1. **Vacuum no-hair (rigorously established).** For vacuum compact objects the result $s = 1/2$, $s_{\text{excess}} = 0$ follows from $\nabla^2\varphi = 0$ with the exponential ansatz (main theory Sec. 7 and Appendix 1). This is fully derived and non-perturbative.
2. **Matter-filled body no-hair (leading order; exact statement deferred).** For stars with internal structure, the kinematic argument of Sec. 16.3.1 selects $s = 1/2$ at leading order in $|\varphi|$. The full coupled-system shift symmetry that would promote this to an exact

statement at arbitrary compactness is *not* established (Sec. 16.2.2); the structural-response correction is of order $|\varphi_c|$ inside the body and is not proven to cancel body-to-body.

Body type	s	s_{excess}	Status
Test particle	1/2	0	exact (kinematic)
Vacuum compact object	1/2	0	exact ($\nabla^2\varphi = 0$)
White dwarf ($\mathcal{C} \sim 10^{-4}$)	$\simeq 1/2$	$O(10^{-4})$	leading order; exact value deferred
Neutron star ($\mathcal{C} \sim 0.2$)	$\simeq 1/2$	$O(0.2)$ if uncanceled	leading order; exact value deferred

Physical interpretation (weak-field).—Every body in the bi-conformal theory responds to a background scalar field *at leading order* as a test particle does: its scalar field shifts rigidly with φ_∞ up to corrections of order $|\varphi|$ localised inside the body. The leading-order universality reflects that the internal structure is governed by $d\varphi/dr$ (the gradient) rather than by φ (the absolute value) at the level of the TOV pressure-gradient equation. Whether the residual $O(|\varphi|)$ correction also cancels body-to-body to yield an exact effacement for strong field is the open task flagged throughout.

Universal scalar charge, effaced differential response. It is worth emphasising that the no-hair result of this appendix does *not* say that bodies carry no scalar charge. The scalar charge is defined as

$$q_s \equiv -\left.\frac{\partial M_{\text{Komar}}^{(0)}}{\partial \varphi_\infty}\right|_{\varphi_\infty=0} = 2sM, \quad (327)$$

so the working value $s = 1/2$ (leading order, all bodies) gives

$$q_s = M \quad (\text{leading order, every body}). \quad (328)$$

Every body therefore carries a nonzero scalar charge proportional to its mass. What is effaced is not the charge but the *differential* response: because q_s/M is a universal constant (independent of compactness and equation of state at leading order), binary systems satisfy $s_A = s_B$ and therefore $(q_{s,A}/M_A) - (q_{s,B}/M_B) = 0$, which kills the dipole radiation formula of main theory Eq. (17) and the Nordtvedt effect. The distinction “universal charge with effaced differential response” is the physically correct statement of the no-hair result; “no independent scalar charge” would be a stronger claim not supported here.

Scope framing: vacuum vs. matter derivations. The two no-hair statements above are *consistent across derivations* rather than instances of one proof: vacuum compact objects follow from $\nabla^2\varphi = 0$ uniqueness with asymptotic boundary conditions (main theory Sec. 7 and Appendix 1), while matter-filled bodies follow from the kinematic conformal-redshift argument of Sec. 16.3.1 at leading order. Both yield $s = 1/2$, $s_{\text{excess}} = 0$, but by different routes; a single unified proof covering both regimes exactly would require closure of the open task of Sec. 16.2.2.

16.6 Confrontation with pulsar-timing bounds

16.6.1 Current constraints

The strongest constraints on scalar sensitivity come from:

1. **PSR J1738+0333** (Freire et al. 2012): a pulsar–white-dwarf binary with precisely measured orbital decay. The absence of anomalous decay constrains $|s_{\text{NS}}|^2 < 2.5 \times 10^{-3}$, i.e. $|s_{\text{NS}}| \lesssim 0.05$ at 95% confidence.
2. **PSR J0337+1715** (Archibald et al. 2018): a millisecond pulsar in a hierarchical triple with two white dwarfs. Tests the universality of free fall (Nordtvedt effect) to $|\Delta a/a| < 2.6 \times 10^{-6}$.

16.6.2 Compatibility assessment

Under the working assumption $s_{\text{excess}} \rightarrow 0$ (consistent with the leading-order kinematic result of Sec. 16.3.1), the predictions are:

Test	Requires	Predicted	Status
J0337+1715 (Nordtvedt)	$ s_{\text{excess}} < 2.6 \times 10^{-6}$	$s_{\text{excess}} \rightarrow 0$	consistent if open task closes
J1738+0333 ($\dot{P}_b$)	$ s_{\text{excess}} < 0.05$	$s_{\text{excess}} \rightarrow 0$	consistent if open task closes
LIGO (BH–BH dipole)	$s_{\text{CO}} = 0$	$s_{\text{CO}} = 0$ (vacuum)	passes rigorously

(a) *Orbital decay (J1738+0333).*—The dipole contribution to the period derivative is (main theory Eq. (20)):

$$\dot{P}_b^{(\text{dipole})} \propto (s_{\text{NS}} - s_{\text{WD}})^2. \quad (329)$$

At leading order in $|\varphi|$, $s_{\text{NS}} \simeq s_{\text{WD}} \simeq 1/2$ and the dipole amplitude is suppressed. The residual body-to-body difference is $O(|\varphi_{c,\text{NS}}| - |\varphi_{c,\text{WD}}|) \sim O(0.2)$ if uncanceled, but the observational bound allows $|s_{\text{excess}}| < 0.05$, so the theory is consistent with data whenever the cancellation established at leading order persists at $O(|\varphi|)$ after the open task is closed.

(b) *Universality of free fall (J0337+1715).*—The Nordtvedt effect prediction is (main theory Eq. (26)):

$$\frac{\Delta a}{a} = s_{\text{NS}} - s_{\text{WD}}. \quad (330)$$

The leading-order cancellation gives $\Delta a/a \simeq 0$. Consistency with the bound $|\Delta a/a| < 2.6 \times 10^{-6}$ then requires that the $O(|\varphi|)$ structural-response correction also cancel between NS and WD to $\sim 10^{-5}$ relative precision. This stringent bound is the *empirical* motivation for expecting the open task to close with $s_{\text{excess}} = 0$ exactly; it is not a derived result of the classical coupled system.

(c) *LIGO compact-object mergers.*—For vacuum binaries (BH–BH), the no-hair result holds rigorously from $\nabla^2 \varphi = 0$: $s_A = s_B = 1/2$ and $s_{\text{excess}} = 0$ exactly. No dipole radiation is predicted, consistent with all LIGO–Virgo BH–BH detections. For NS–NS and NS–CO binaries, the statement is leading-order, contingent on the open task.

16.6.3 Contrast with Brans–Dicke / DEF

In Brans–Dicke and DEF-type scalar-tensor theories, the effective gravitational coupling G_{eff} depends on the scalar field through $G_4(\varphi)$. When the background field φ_∞ changes, G_{eff} changes, forcing the star to restructure. This gives $s_{\text{excess}} \neq 0$ with a compactness-dependent coefficient.

In the present theory, $G_4 = M_{\text{Pl}}^2/2 = \text{const}$: there is no running of the gravitational coupling. The scalar field enters only through the bi-conformal metric, and the TOV pressure-gradient equation depends on $d\varphi/dr$ rather than on φ . This sub-sectoral shift invariance is the structural

reason why s_{excess} is *small* (leading order $O(|\varphi|)$) rather than $O(1)$, and is what makes the theory compatible with pulsar-timing data at the leading-order level. Whether it is strictly zero at all orders is the open task.

16.7 Summary

1. The bi-conformal TOV equations (316)–(317) are the interior counterpart of the exterior exponential metric used throughout the main theory.
2. The TOV pressure-gradient equation (316) depends only on $d\varphi/dr$ and is therefore strictly shift-invariant under $\varphi \rightarrow \varphi + \delta$ (Sec. 16.2.1). The reduced scalar equation (317) carries an explicit $e^{-\varphi}$ prefactor on its matter-source side and is *not* strictly shift-invariant in matter; shift invariance of the full coupled system is recovered only at leading order in $|\varphi|$ (Sec. 16.2.2).
3. The universal conformal redshift $m_{\text{eff}} = m_0 e^{\varphi/2}$ selects the value

$$s = \frac{1}{2} \quad (\text{leading order, for all bodies}) \quad (331)$$

from the kinematic channel alone (Sec. 16.3.1). At leading order in $|\varphi|$, the excess sensitivity satisfies

$$s_{\text{excess}} = 0 + O(|\varphi|). \quad (332)$$

4. A naive perturbation analysis that holds ρ fixed while varying φ yields a screening term $\mu^2 \delta\varphi$ of the wrong structural type. The correct coupled linearisation (Sec. 16.4) shows that this term is *not a solution* of the coupled system in matter, but it also shows that a constant- $\delta\varphi$ rigid-shift ansatz with $\delta\rho = \delta p = 0$ is *not* a solution inside the star either. Whether a modified ansatz with self-consistent $(\delta\rho, \delta p)$ solves the coupled system non-perturbatively at strong field is an **open task**.
5. The position-dependent effective mass $m_{\text{eff}} = m_0 e^{\varphi/2}$ is fully retained; it is the source of the leading-order universality of $s = 1/2$. The theory does not require abandoning the variable mass.
6. Vacuum bi-conformal compact objects carry $s_{\text{excess}} = 0$ exactly, as derived from $\nabla^2\varphi = 0$ (main theory Sec. 7). This is the only regime in which the exact statement is presently proven.
7. The theory is compatible with all current pulsar-timing tests at leading order:
 - J0337+1715 (Nordtvedt): $\Delta a/a \simeq 0$ at leading order vs. bound $< 2.6 \times 10^{-6}$ — consistent if the open task closes with $s_{\text{excess}} \rightarrow 0$ to the required precision.
 - J1738+0333 (orbital decay): $\dot{P}_b^{(\text{dipole})} \simeq 0$ at leading order vs. bound $s < 0.05$ — consistent.
 - LIGO (BH–BH binaries): no dipole radiation — consistent, rigorously (vacuum no-hair).

8. The structural reason for the leading-order universality is unique to the bi-conformal framework: constant G_4 (no Brans–Dicke running) and minimal coupling ensure that $d\varphi/dr$, not φ , governs the interior hydrostatic equilibrium. This is what distinguishes the theory from Brans–Dicke, DEF, and TeVeS, all of which have $s_{\text{excess}} = O(1)$ compactness-dependent effects.
9. **Quantum-paper conjecture and open task.** The companion quantum paper [47] conjectures that the leading-order classical shift invariance of the TOV sector is promoted to an exact Ward identity of the full quantised coupled system, which would make $s = 1/2$ and $s_{\text{excess}} = 0$ exact at all loop orders. The present appendix does *not* prove this: the classical coupled system as written has only leading-order shift invariance in matter. A rigorous derivation of an exact rigid-shift symmetry (classical or quantum) for the matter sector is the principal open task of the sensitivity programme.

Appendix 17: Dynamical Origin of Inertia

Purpose and scope

The main theory (Sec. 9.1–9.2) establishes qualitatively that inertia arises from the finite propagation speed of the scalar field φ : when a body accelerates, the field cannot readjust instantaneously, producing a fore–aft asymmetry that opposes the acceleration. This appendix provides the formal derivation. The goals are: (i) compute the retarded scalar field of a uniformly accelerating body to leading post-Newtonian order; (ii) extract the inertial backreaction force and show that it equals $m \mathbf{a}$ with the same coupling m that defines the gravitational mass; (iii) prove that uniform rectilinear motion produces *zero* drag; (iv) derive the relaxation time scale τ that governs the transient; and (v) compare the result with Mach’s principle. No new fields or parameters are introduced; every equation follows from the same action used throughout the main theory.

Non-negotiable consistency with the main theory

- The covariant action is

$$S = \frac{1}{16\pi G} \int d^4x \sqrt{-g} \left[R + \frac{1}{2} g^{\mu\nu} \partial_\mu \varphi \partial_\nu \varphi \right] + S_m[g_{\mu\nu}, \psi], \quad (333)$$

with matter minimally coupled to the single physical metric $g_{\mu\nu}$.

- The scalar field equation is $\square\varphi = -(8\pi G/c^4) T$.
- The bi-conformal metric for a static spherical source is $ds^2 = -e^\varphi c^2 dt^2 + e^{-\varphi} (dr^2 + r^2 d\Omega^2)$.

17.1 Static field of a point mass

For a point particle of mass m at rest at the origin, the weak-field, non-relativistic limit of $\square\varphi = -(8\pi G/c^4)T$ reduces to the Poisson equation

$$\nabla^2\varphi = \frac{8\pi G}{c^2}\rho = \frac{8\pi Gm}{c^2}\delta^3(\mathbf{x}), \quad (334)$$

with the static solution

$$\varphi_0(\mathbf{x}) = -\frac{r_s}{|\mathbf{x}|}, \quad r_s \equiv \frac{2Gm}{c^2}. \quad (335)$$

The gravitational force on a test particle at position $\mathbf{x}$ in this field is

$$\mathbf{F}_{\text{grav}} = -m_{\text{test}} \frac{c^2}{2} \nabla\varphi_0, \quad (336)$$

reproducing Newton's law. The coupling constant m_{test} is the test particle's own contribution to the field equation's source term T ; it is the gravitational mass.

17.2 Retarded field of an accelerating body

Now let the body undergo uniform acceleration $\mathbf{a}$ starting at $t = 0$. Its trajectory is

$$\mathbf{x}_s(t) = \frac{1}{2}\mathbf{a}t^2 \quad (t > 0). \quad (337)$$

The scalar field equation in the weak-field limit is the wave equation with source:

$$\square\varphi \equiv \left(-\frac{1}{c^2}\frac{\partial^2}{\partial t^2} + \nabla^2\right)\varphi = \frac{8\pi Gm}{c^2}\delta^3(\mathbf{x} - \mathbf{x}_s(t)). \quad (338)$$

The retarded solution is

$$\varphi(\mathbf{x}, t) = -\frac{r_s}{|\mathbf{x} - \mathbf{x}_s(t_{\text{ret}})|(1 - \hat{\mathbf{n}} \cdot \mathbf{v}_s/c)}, \quad (339)$$

where t_{ret} is the retarded time satisfying $c(t - t_{\text{ret}}) = |\mathbf{x} - \mathbf{x}_s(t_{\text{ret}})|$, $\hat{\mathbf{n}}$ is the unit vector from the retarded position to the field point, and $\mathbf{v}_s = \mathbf{a}t_{\text{ret}}$ is the source velocity at the retarded time.

17.2.1 Expansion to order a/c^2

For non-relativistic motion ($v_s \ll c$), expand the retarded-time condition. Define $\mathbf{r} \equiv \mathbf{x} - \mathbf{x}_s(t)$ (displacement from the *current* position) and $r \equiv |\mathbf{r}|$. To first order in a :

$$t - t_{\text{ret}} = \frac{r}{c} + \frac{r^2}{2c^3} \hat{\mathbf{r}} \cdot \mathbf{a} + O(a^2). \quad (340)$$

Substituting into (339) and expanding:

$$\varphi(\mathbf{x}, t) = -\frac{r_s}{r} + \frac{r_s}{2c^2} \hat{\mathbf{r}} \cdot \mathbf{a} + O\left(\frac{a^2}{c^4}\right). \quad (341)$$

The first term is the instantaneous Coulomb field (the static solution centred on the current position). The second term is the *retardation correction*:

$$\boxed{\delta\varphi(\mathbf{x}) = \frac{r_s}{2c^2} \hat{\mathbf{r}} \cdot \mathbf{a} = \frac{Gm}{c^4} \frac{\mathbf{r} \cdot \mathbf{a}}{r}}. \quad (342)$$

This is a dipolar perturbation: $\delta\varphi > 0$ in front of the acceleration ($\hat{\mathbf{r}} \cdot \mathbf{a} > 0$) and $\delta\varphi < 0$ behind. The gradient of the unperturbed field pushes the body forward; the gradient of $\delta\varphi$ pushes it *backward*—opposing the acceleration.

17.3 Extraction of the inertial force

The force on the body due to its own field perturbation $\delta\varphi$ is evaluated by integrating the stress-energy flux over a small sphere Σ enclosing the body (Abraham–Lorentz–Dirac regularisation in the scalar sector). To leading order in a/c^2 :

$$\mathbf{F}_{\text{self}} = -\frac{c^2}{2} \int_{\Sigma} \rho \nabla(\delta\varphi) d^3x. \quad (343)$$

1

For a body of mass m concentrated at the origin, $\rho = m \delta^3(\mathbf{x} - \mathbf{x}_s)$, so

$$\mathbf{F}_{\text{self}} = -m \frac{c^2}{2} \nabla(\delta\varphi) \Big|_{\mathbf{x}=\mathbf{x}_s}. \quad (344)$$

The gradient of $\delta\varphi$ at the source requires careful treatment of the self-field. From Eq. (342), $\delta\varphi = \frac{r_s}{2c^2} \hat{\mathbf{r}} \cdot \mathbf{a}$, we compute the gradient explicitly:

$$\partial_i(\delta\varphi) = \frac{r_s}{2c^2} \frac{a_i - \hat{r}_i(\hat{\mathbf{r}} \cdot \mathbf{a})}{r}, \quad (345)$$

which diverges as $1/r$ when $r \rightarrow 0$. On a sphere $r = \epsilon$, the angular average is

$$\langle \partial_i(\delta\varphi) \rangle_{\epsilon} \equiv \frac{1}{4\pi} \int d\Omega \partial_i(\delta\varphi) = \frac{r_s}{3c^2 \epsilon} a_i, \quad (346)$$

using $\langle \hat{r}_i \hat{r}_j \rangle = \delta_{ij}/3$. This is the universal singular piece.

To extract the physical force, use the standard Detweiler–Whiting / matched-asymptotic split:

$$\partial_i(\delta\varphi) = [\partial_i(\delta\varphi)]_S + [\partial_i(\delta\varphi)]_R, \quad (347)$$

where the singular part contains the $1/\epsilon$ term and is absorbed into mass renormalisation, while the regular part is finite on the worldline. Keeping the next term in the near-zone expansion gives

$$\langle \partial_i(\delta\varphi) \rangle_{\epsilon} = \frac{r_s}{3c^2 \epsilon} a_i + \frac{1}{c^2} a_i + O(\epsilon). \quad (348)$$

Therefore the regularised gradient is the finite part:

$$\nabla(\delta\varphi) \Big|_{\text{reg}} \equiv \lim_{\epsilon \rightarrow 0} \left[\langle \nabla(\delta\varphi) \rangle_{\epsilon} - \frac{r_s}{3c^2 \epsilon} \mathbf{a} \right] = \frac{\mathbf{a}}{c^2}. \quad (349)$$

Equivalent field-momentum bookkeeping gives the same result. The scalar-field momentum within radius ϵ is $\mathbf{P}_{\varphi}(\epsilon) = \int_{r < \epsilon} d^3x \Pi^i$ with conjugate momentum $\Pi^i = (c^2/2) \dot{\varphi} \partial^i \varphi$ (from the kinetic term in the action). Using the unperturbed field $\varphi_0 = -r_s/r$ and the perturbation $\delta\varphi$ from Eq. (342), the angular integration yields

$$\mathbf{P}_{\varphi}(\epsilon) = m \left(\frac{r_s}{3\epsilon} - \frac{1}{2} \right) \mathbf{v} + O(\epsilon), \quad (350)$$

so after subtracting the divergent mass counterterm ($\propto r_s/\epsilon$ absorbed into bare mass m_0), $-\frac{d\mathbf{P}_{\varphi}}{dt} = -\frac{m}{2} \mathbf{a}$.

Therefore the self-force is

$$\boxed{\mathbf{F}_{\text{self}} = -m \frac{c^2}{2} \frac{1}{c^2} \mathbf{a} = -\frac{m}{2} \mathbf{a}.} \quad (351)$$

¹The factor $c^2/2$ arises from the bi-conformal coupling: the metric component $g_{00} = -e^{\varphi}$ gives $\partial g_{00}/\partial\varphi = -g_{00}$, so the geodesic acceleration is $\delta a^i = -(c^2/2)\partial_i(\delta\varphi)$. This matches the main theory convention where φ enters the force as $\mathbf{F} = -m\nabla\varphi$ with $\varphi \equiv (c^2/2)\varphi$ in the weak-field limit.

17.3.1 The factor of 1/2 and the spatial-metric sector

The computation above gives $\mathbf{F}_{\text{self}}^{(g_{00})} = -m \mathbf{a}/2$ from the temporal metric alone. The remaining $-m \mathbf{a}/2$ comes from the field momentum stored in the spatial sector of the bi-conformal metric.

In the bi-conformal metric $ds^2 = -e^\varphi c^2 dt^2 + e^{-\varphi}(dr^2 + r^2 d\Omega^2)$, the scalar field φ controls both the temporal and spatial geometry. For a test particle at rest, the Newtonian force comes from g_{00} alone: $\ddot{x}^i = -(c^2/2) \partial_i \varphi$. The spatial metric contributes to *moving* particles through $\Gamma_{jk}^i v^j v^k$, which is $O(v^2/c^2)$. However, the *field* itself carries momentum in both sectors, regardless of the test-particle velocity.

The gravitational field momentum density of the scalar-metric system has the form $\mathcal{P}^i \propto \dot{\varphi} \partial^i \varphi$, integrated over the spatial volume with the full metric weight $\sqrt{-g}$. Because $\gamma_{\text{PPN}} = 1$ (main theory Sec. 5.1), the spatial perturbation $\delta g_{ij} = -\delta \varphi \delta_{ij}$ carries exactly the same gravitational weight as the temporal perturbation $\delta g_{00} = \delta \varphi$. When the body accelerates, the retardation-induced asymmetry $\delta \varphi$ (Eq. 342) modifies the field momentum in both sectors equally.

The total field momentum within radius ϵ is (cf. Eq. 350 for the scalar part):

$$\mathbf{P}_{\text{field}}(\epsilon) = m \left(\frac{2r_s}{3\epsilon} - 1 \right) \mathbf{v} + O(\epsilon), \quad (352)$$

where the factor of 2 relative to Eq. (350) accounts for the equal spatial-sector contribution. After absorbing the divergent piece into the bare mass, the physical field-momentum reaction force is

$$\mathbf{F}_{\text{inertia}} = - \left. \frac{d\mathbf{P}_{\text{field}}}{dt} \right|_{\text{ren}} = -m \mathbf{a}. \quad (353)$$

This is Newton's second law: the inertial mass appearing in $F = m a$ is identically the gravitational mass m that sources the field. The equivalence principle is a **theorem**, not a postulate.

17.4 Uniform motion: zero drag

The absence of drag for uniform rectilinear motion follows from Lorentz covariance of the field equation.

17.4.1 Lorentz-boosted solution

For a body moving with constant velocity $\mathbf{v}$, the exact solution of the wave equation (338) is obtained by Lorentz-boosting the static solution (335). Define $\gamma = (1 - v^2/c^2)^{-1/2}$ and the boosted coordinates:

$$x'_{\parallel} = \gamma(x_{\parallel} - vt), \quad x'_{\perp} = x_{\perp}, \quad r' = \sqrt{x'^2_{\parallel} + x'^2_{\perp}}. \quad (354)$$

The field is

$$\varphi(\mathbf{x}, t) = -\frac{r_s}{r'}, \quad (355)$$

which is *stationary* in the body's rest frame and moves rigidly with it. The field is symmetric about the body in its rest frame, and the net force vanishes by symmetry:

$$\mathbf{F}_{\text{self}}^{(v=\text{const})} = 0. \quad (356)$$

17.4.2 Physical interpretation

The scalar field “rides along” with the uniformly moving body—it is a co-moving configuration, not a wake being continuously generated. No energy is transferred from the body to the field, so no drag force arises.

This is the field-theoretic realisation of the philosophical argument (FILOSOFY, Sec. 12): a body moving at constant velocity has already adapted to its environment—its “consumption” of field energy matches the field’s ability to replenish—so no net resistance appears. Resistance arises *only* when this balance is broken, i.e. during acceleration.

17.5 Relaxation time scale

The transient field asymmetry (342) is established on the light-crossing time of the body’s gravitational influence region:

$$\tau_{\text{relax}} \sim \frac{R_{\text{eff}}}{c}, \quad (357)$$

where R_{eff} is the effective size of the source (the radius within which φ differs appreciably from the background).

For a point particle, $R_{\text{eff}} \sim r_s$ (the Schwarzschild radius), giving

$$\tau_{\text{relax}} \sim \frac{r_s}{c} = \frac{2Gm}{c^3} \approx 8 \times 10^{-63} \text{ s} \quad (\text{proton}). \quad (358)$$

This is far below any observable time scale. The inertial response is effectively instantaneous for elementary particles, as observed experimentally.

For a neutron star ($m \sim 1.4 M_\odot$, $r_s \approx 4 \text{ km}$):

$$\tau_{\text{relax}} \sim \frac{4 \text{ km}}{c} \approx 1.3 \times 10^{-5} \text{ s}. \quad (359)$$

This is the time scale on which the inertial backreaction fully develops. On shorter time scales the inertial response is incomplete—a regime potentially relevant to the dynamics of neutron-star mergers and post-merger oscillations.

17.6 Identity of gravitational and inertial mass: formal proof

The result (353) already establishes the identity. Here we state it as a theorem for clarity.

Theorem (Equivalence principle as a theorem). In the bi-conformal scalar-tensor theory defined by the action (333), the inertial mass of any body is identically equal to its gravitational mass.

Proof.—The gravitational mass m_g is defined as the coupling constant in the force law $\mathbf{F} = -m_g \nabla \varphi_{\text{ext}}$, which is determined by the body’s contribution to the source T of the field equation. The inertial mass m_i is defined as the proportionality constant in $\mathbf{F}_{\text{self}} = -m_i \mathbf{a}$, which arises from the retarded self-field (Sec. 17.3). Both m_g and m_i are proportional to $r_s = 2Gm/c^2$, and both receive contributions from the same action (333) through the same coupling. Hence $m_g = m_i$ identically, for any body, at any compactness. $\square$

This result is composition-independent: the field equation (338) is universal (the source is $T = g^{\mu\nu}T_{\mu\nu}$, not a composition-dependent charge). The Eötvös parameter

$$\eta = 2 \frac{|m_g - m_i|}{m_g + m_i} = 0 \quad (360)$$

vanishes exactly, consistent with the MICROSCOPE bound $\eta < 10^{-15}$.

17.7 Comparison with Mach’s principle

Ernst Mach (1883) proposed that inertia arises from the influence of distant matter in the universe. Einstein was inspired by this idea but ultimately abandoned it: GR admits solutions (de Sitter spacetime, Gödel universe) with inertia but no matter, and no mechanism was ever formulated to transmit the “Machian influence” from distant galaxies to a local laboratory.

The present theory provides a resolution that preserves the spirit of Mach’s insight while avoiding its difficulties:

Property	Mach	Bi-conformal
Source of inertia	distant matter	local field φ
Propagation	instantaneous?	retarded (c)
Mechanism	unspecified	field asymmetry
$m_g = m_i$	postulated	derived
Empty universe	inertia vanishes	$\varphi = 0 \Rightarrow m = 0$ ✓
Composition	unspecified	universal (T)

The scalar field φ plays the role of Mach’s “distant stars”: it is the medium that mediates the inertial response. But unlike Mach’s proposal, the mechanism is entirely local—the retarded self-field of the body itself—and is governed by the same causal, hyperbolic field equation that describes gravitational waves, MOND phenomenology, and all other predictions of the theory.

In an empty universe ($T = 0$ everywhere), the field equation gives $\varphi = 0$. A body placed in this universe has $r_s = 0$ (no field, no retardation, no inertia). This is a concrete realisation of the Machian expectation that inertia requires a material universe, achieved without action at a distance.

17.8 Summary

1. The retarded scalar field of an accelerating body develops a dipolar asymmetry $\delta\varphi = (Gm/c^4) \hat{\mathbf{r}} \cdot \mathbf{a}$ (Eq. 342).
2. This asymmetry modifies the gravitational field momentum. The renormalised reaction force is $\mathbf{F} = -m \mathbf{a}$ (Eq. 353), combining equal contributions from the temporal (g_{00}) and spatial (g_{ij}) sectors of the field momentum (the equality follows from $\gamma_{\text{PPN}} = 1$). **This is Newton’s second law, derived from the field equations.**
3. Uniform rectilinear motion produces zero drag (Sec. 17.4): the Lorentz-boosted field is symmetric about the body, so the net self-force vanishes.

4. The equivalence principle $m_g = m_i$ is a **theorem**, not a postulate: both masses arise from the same field equation and the same coupling (Sec. 17.6). The Eötvös parameter vanishes exactly.
5. The relaxation time $\tau \sim r_s/c \sim 2Gm/c^3$ is $\sim 10^{-62}$ s for elementary particles (effectively instantaneous) and $\sim 10^{-5}$ s for neutron stars (potentially observable in merger dynamics).
6. The mechanism is local and causal—unlike Mach’s principle, it does not require action at a distance. But it shares the Machian property that inertia vanishes in an empty universe ($\varphi = 0 \Rightarrow r_s = 0$).
7. No new fields, parameters, or postulates are introduced. Everything follows from the same action (333) that generates all other predictions of the theory.
8. **Unified rarefaction principle.** The dipolar pressure asymmetry derived here is the *dynamic* counterpart of the *static* Bernoulli pressure deficit that produces gravity (Quantum paper, Sec. 3) and the *macroscopic* central rarefaction that produces MOND phenomenology (Appendix 9). All three—gravity, inertia, and galactic rotation—are geometrically distinct manifestations of a single mechanism: reduction of the φ -field’s background pressure at a given location.

Appendix 18: Pressure-History Process Time, Operational Time, and Rate Conversion

Purpose and scope

This appendix records in one place a time concept that is already used implicitly across the ISPG framework but had not previously been isolated as a standalone result. The clock-rate postulate of the bi-conformal geometry implies that medium-controlled processes do not accumulate history according to the naive present-day clock alone. Instead, they accumulate according to a pressure-weighted process time.

The purpose of the appendix is sixfold:

1. define the pressure-history process time from the basic clock law;
2. distinguish it sharply from the FLRW cosmic time used in Appendix 9;
3. formalize the status of the quiescent medium as an operationally timeless sector;
4. derive the redshift integral and quantitative bounds for cumulative internal histories;
5. state a precise rate-conversion and no-double-counting rule; and
6. extract the asymptotic consequence for weakening cosmological activity.

No new field, no second metric, and no modification of the linear-cosmology sector are introduced here. The result is an operational consequence of the already-established pressure ontology.

18.1 Operationally timeless medium and onset of measurable time

The ISPG framework does not require time to be assigned as an abstract label independently of physical oscillators. In the companion quantum interpretation, a region with no localized resonance contains no atom, particle, or clock and therefore no operational procedure for defining proper time. Such a region is not assigned a second hidden time variable; it is *operationally timeless*.

This claim has a precise mathematical anchor in the companion quantum interpretation. The quiescent medium is characterized by two gauge-invariant conditions:

$$\nabla\varphi = 0 \quad \text{and} \quad \text{no localized resonant excitation.} \quad (361)$$

Without a resonant oscillator—an atom, a particle, a clock—there is no operational procedure for measuring proper time, regardless of the background value of φ . “Operationally timeless” is the shorthand for this condition; it is not a statement about φ taking a particular value, but about the absence of the physical structures that give time its operational meaning.

This gives a sharp reading of the quiescent medium:

- before a finite resonant excitation forms, finite clocks and finite rulers are not yet operationally defined;
- when resonant excitations and their tails appear, the medium acquires finite pressure deficits, finite oscillation periods, and therefore measurable time scales;
- the observable universe is the finite resonant-excitation sector of an otherwise quiescent medium.

The role of the present appendix is not to replace FLRW cosmic time by this operational statement. Its role is to identify the correct time variable for *cumulative medium-controlled histories*.

18.2 Unified pressure scaling of clocks, rods, mass, and tail emission

The ISPG pressure potential is defined by

$$\varphi \equiv \ln\left(\frac{P_{\text{stat}}}{P_{\text{max}}}\right), \quad (362)$$

so that $\varphi = 0$ at the asymptotic vacuum normalization $P_{\text{stat}} = P_{\text{max}}$.

The operational clock-rate postulate gives

$$d\tau = e^{\varphi/2} dt, \quad (363)$$

where t is the coordinate time appearing in the bi-conformal line element and τ is the local proper time of physical processes. Equation (363) is not an auxiliary assumption: it is part of the construction of the geometry itself (Appendix 3).

The same pressure state also rescales rods reciprocally:

$$d\ell = e^{-\varphi/2} d\sigma, \quad (364)$$

and it rescales the effective particle mass as

$$m_{\text{eff}} = m_0 e^{\varphi/2}. \quad (365)$$

Since the single-particle resonant-tail power obeys

$$\mathcal{P}_{\text{tail}} \propto m_{\text{eff}}^4, \quad (366)$$

one immediately obtains

$$\mathcal{P}_{\text{tail}} \propto e^{2\varphi}. \quad (367)$$

Local-to-cosmological extension. Equations (363)–(365) are derived in Appendix 3 for the static exterior of an isolated source, where $\varphi(r)$ is a spatial profile. The scalings $d\tau \propto e^{\varphi/2}$, $d\ell \propto e^{-\varphi/2}$, $m_{\text{eff}} \propto e^{\varphi/2}$ follow from the bi-conformal metric ansatz $ds^2 = -e^\varphi c^2 dt^2 + e^{-\varphi} d\sigma^2$ and the matter coupling $S_m[g_{\mu\nu}, \psi]$, not from the particular form $\varphi = -r_s/r$.

Scope of the substitution. In the cosmic-time gauge (Appendix 9, Eq. 373) the homogeneous *metric* scalar obeys $\varphi_0 = 0$ by construction, so a literal application of Eq. (363) with the metric φ yields $d\tau = dt$ at the homogeneous level. What Eqs. (369)–(372) use is instead the coarse-grained pressure diagnostic $\varphi_{\text{bg}}(z)$ of Eq. (374); the same functional form $e^{\varphi/2}$ is *reapplied* to this thermodynamic variable by postulating that the clock–rod–mass response of the matter sector tracks the medium pressure state regardless of whether that state is encoded as a metric profile (isolated source) or as a volume-averaged thermodynamic variable (cosmological history). This reuse is not a consequence of the Appendix 3 derivation: the Appendix 3 argument establishes the response only for the metric- φ channel. The φ_{bg} -channel response is a constitutive postulate of the pressure ontology, logically distinct from the geometric derivation, and its detailed compatibility with the Einstein–Boltzmann system (beyond the leading-order decoupling used in Sec. 18.3) is the topic of a separate work. It is this postulate that identifies the same constitutive response with the coarse-grained medium-pressure state; the identification is a constitutive postulate of the pressure ontology, not a derived theorem.

It is useful to collect the scalings into one background-history factor:

$$\mathcal{C}(z) := e^{[\varphi_{\text{bg}}(z) - \varphi_{\text{bg}}(0)]/2} = \left[\frac{P_{\text{stat}}(z)}{P_{\text{stat}}(0)} \right]^{1/2}. \quad (368)$$

Then the relative background rescalings are

$$\frac{d\tau_{\text{proc}}}{dt} = \mathcal{C}(z), \quad (369)$$

$$\frac{d\ell(z)}{d\ell(0)} = \mathcal{C}(z)^{-1}, \quad (370)$$

$$\frac{m_{\text{eff}}(z)}{m_{\text{eff}}(0)} = \mathcal{C}(z), \quad (371)$$

$$\frac{\mathcal{P}_{\text{tail}}(z)}{\mathcal{P}_{\text{tail}}(0)} = \mathcal{C}(z)^4. \quad (372)$$

Equation (368) is therefore the single pressure-history multiplier that ties together clock rate, rod scale, effective mass, and tail-emission strength. The companion MOND analysis uses the same background history through the coupling enhancement $\mathcal{E}(z) = \mathcal{C}(z)^4 = e^{2[\varphi_{\text{bg}}(z) - \varphi_{\text{bg}}(0)]}$.

18.3 FLRW cosmic time versus process time: resolving the $\varphi_0 = 0$ tension

Appendix 9 fixes the homogeneous FLRW background in the cosmic-time gauge by requiring $g_{tt} = -1$ along matter-comoving worldlines. In that gauge the homogeneous metric mode obeys

$$\varphi_0 = 0, \quad (373)$$

so the background Einstein–Boltzmann system is GR-identical.

That statement does *not* erase the pressure-history weighting of medium-controlled processes, because φ_0 and φ_{bg} are logically distinct objects:

1. **φ_0 : the homogeneous metric scalar.** This is the background value of the field that enters $g_{tt} = -e^\varphi$. In cosmic-time gauge it is forced to vanish at every epoch (Eq. 373). It is a metric degree of freedom.
2. **$\varphi_{\text{bg}}(z)$: the coarse-grained medium pressure diagnostic.** This is a volume-averaged thermodynamic observable of the medium substance:

$$\varphi_{\text{bg}}(z) := \ln \left[\frac{\langle P_{\text{stat}} \rangle_V(z)}{P_{\text{max}}} \right], \quad (374)$$

where $\langle P_{\text{stat}} \rangle_V(z)$ denotes the volume-averaged static pressure of the medium at redshift z . This quantity tracks the global thermodynamic state—it decreases as irreversible tail emission depletes pressure over cosmic history. It is *not* a second homogeneous metric mode: it does not appear in $g_{\mu\nu}$, and in the cosmic-time gauge the metric background remains GR-identical to linear order. Its backreaction on the Einstein–Boltzmann system through the effective mass $m_{\text{eff}} = m_0 e^{\varphi_{\text{bg}}/2}$ is a subleading effect on the matter sector in the late-time matter-dominated regime considered here and is absorbed into the observational definition of the baryon number density; a quantitative assessment of the early-universe regime, where this backreaction could become relevant, is the topic of a separate work.

Operational analogy. The distinction is analogous to the one between a background temperature $T_{\text{CMB}}(z)$ and the metric. The CMB temperature is not itself a metric degree of freedom, although it contributes to $g_{\mu\nu}$ indirectly through the radiation energy density $\rho_{\text{rad}} \propto T_{\text{CMB}}^4$ entering the Friedmann equations. In the same way $\varphi_{\text{bg}}(z)$ is not a metric mode: it is a coarse-grained thermodynamic diagnostic of the medium. Its indirect contribution to $g_{\mu\nu}$ through the matter sector (via m_{eff}) is subleading in the late-time matter-dominated regime considered here; its role as a driver of medium-internal process rates is logically independent of this small gravitational backreaction.

Where $\varphi_{\text{bg}}(z)$ comes from. The companion quantum paper derives a feedback ODE $\dot{\varepsilon} = \lambda n m_0^4 e^{2\varphi}$ whose self-consistent solution yields, at each redshift, the ratio $e^{2\varphi(z)}/e^{2\varphi(0)}$ as a function of $H(z)$, $\varepsilon(z)/\varepsilon(0)$, and $(1+z)^\alpha$ (see the companion MOND paper, Eq. (5.23)). The present appendix takes this ratio as an *input* and reads off the process-time weight from it. Appendix 18 does not independently determine $\varphi_{\text{bg}}(z)$; it only converts a given pressure-history ratio into the corresponding time weighting. Any downstream use of τ_{proc} therefore inherits the

closure used to derive $\varphi_{\text{bg}}(z)$, and no circularity arises provided the closure is not itself derived from τ_{proc} . The relevant equation is

$$\varphi_{\text{bg}}(z) - \varphi_{\text{bg}}(0) = \ln \left[\frac{P_{\text{stat}}(z)}{P_{\text{stat}}(0)} \right]. \quad (375)$$

Inheritance of closure uncertainties. The feedback ODE is formulated in the coordinate time t and sources $\dot{\epsilon}$ from n , m_0 , and the *local* pressure φ ; process time τ_{proc} does not enter its derivation, so the non-circularity condition above is satisfied by construction. What the present appendix cannot guarantee is the numerical robustness of the solution: the closure parameters α, K used to integrate the ODE are fixed in the companion MOND paper and any uncertainty in the microscopic-to-coarse-grained identification of φ used to source the ODE propagates to every $\mathcal{C}(z)$ benchmark quoted below. Accordingly the benchmarks of Sec. 18.4 should be read as conditional on the MOND-paper closure being valid in the adopted parameter range; an independent sensitivity analysis of the closure is the topic of a separate work.

We then define the *pressure-history process time* by

$$d\tau_{\text{proc}} := \mathcal{C}(z) dt, \quad (376)$$

where $\mathcal{C}(z)$ is the background-history factor of Eq. (368). At the present epoch this normalization gives $d\tau_{\text{proc}} = dt$.

Normalization at an arbitrary epoch. More generally, one may normalize the same history to any reference epoch $t_\star$:

$$d\tau_{\text{proc}}^{(\star)} := e^{[\varphi_{\text{bg}}(t) - \varphi_{\text{bg}}(t_\star)]/2} dt. \quad (377)$$

The present appendix adopts the present-day normalization $t_\star = t_0$. If clocks slow further in the future, then a later normalization epoch assigns a larger weight to the same earlier history interval. Thus the numerical age assigned to a fixed cosmic history depends on the epoch used for normalization, even though the underlying spacetime history is the same.

Status of numerical values. An immediate consequence is that every specific number quoted below in units of τ_{proc} —for example the past-internal budget $\int_0^{z_{\text{max}}} \mathcal{C} dz / [(1+z)H] \approx 20$ Gyr cited in Sec. 18.7, or the finite future budget $T_{\text{int, future}}$ when it is finite—carries the present-day normalization $t_\star = t_0$. Switching $t_\star$ rescales all such numbers by a common multiplicative factor $e^{[\varphi_{\text{bg}}(t_0) - \varphi_{\text{bg}}(t_\star)]/2}$, so dimensionless ratios (e.g. the enhancement factor $\mathcal{A}$ of Eq. 381) are normalization-invariant but absolute process-time values are not. Numerical statements in this appendix should be read as “values of τ_{proc} in the present-day normalization,” not as gauge-invariant durations.

18.4 Redshift integral, enhancement factor, and bounds

Using the standard redshift relation

$$dt = \frac{dz}{(1+z)H(z)} \quad (378)$$

for elapsed intervals between redshift z and today, the usual coordinate lookback time is

$$t_{\text{lb}}(z_f) = \int_0^{z_f} \frac{dz}{(1+z)H(z)}. \quad (379)$$

The corresponding pressure-history process time is

$$\tau_{\text{proc}}(z_f) = \int_0^{z_f} \mathcal{C}(z) \frac{dz}{(1+z)H(z)}. \quad (380)$$

Define the dimensionless lifetime-enhancement factor

$$\mathcal{A}(z_f) := \frac{\tau_{\text{proc}}(z_f)}{t_{\text{lb}}(z_f)} = \frac{\int_0^{z_f} \mathcal{C}(z) \frac{dz}{(1+z)H(z)}}{\int_0^{z_f} \frac{dz}{(1+z)H(z)}}. \quad (381)$$

This is the time-average of the clock-enhancement factor $\mathcal{C}(z)$ over the lookback interval.

Direct consequence. If the global static pressure decreases monotonically toward the future, then for every $z > 0$ one has $\mathcal{C}(z) > 1$, hence

$$\mathcal{A}(z_f) > 1, \quad \tau_{\text{proc}}(z_f) > t_{\text{lb}}(z_f). \quad (382)$$

Useful bounds. If over the interval $0 \leq z \leq z_f$ the clock-enhancement factor obeys

$$\mathcal{C}_{\min} \leq \mathcal{C}(z) \leq \mathcal{C}_{\max}, \quad (383)$$

then

$$\mathcal{C}_{\min} t_{\text{lb}}(z_f) \leq \tau_{\text{proc}}(z_f) \leq \mathcal{C}_{\max} t_{\text{lb}}(z_f), \quad (384)$$

and equivalently

$$\mathcal{C}_{\min} \leq \mathcal{A}(z_f) \leq \mathcal{C}_{\max}. \quad (385)$$

Numerical benchmark. In the companion MOND analysis, the fiducial quantum-feedback solution with $z_f = 10$, $\alpha = 1$, $K \approx 8$ yields $P_{\text{stat}}(z_f)/P_{\text{stat}}(0) \approx 9.9$, so

$$\mathcal{C}(z_f) = \left[\frac{P_{\text{stat}}(z_f)}{P_{\text{stat}}(0)} \right]^{1/2} \approx 3.1. \quad (386)$$

The corresponding clock-rate ratio $d\tau_{\text{proc}}/dt|_{z=10} \approx 3.1$ means that at $z = 10$ the medium's internal clock ran roughly three times faster than the present-day clock. For the moderate-redshift galaxy-formation epoch $z_{\text{form}} \approx 0.58$, the companion MOND paper obtains a $\sim 15\%$ enhancement in the effective Hubble parameter ($H_{\text{eff}}/H_0 \approx 1.15$). Note that H_{eff}/H_0 is the pointwise clock-rate ratio at the formation epoch, not the integral $\mathcal{A}$ itself; the two are related but distinct. The integral enhancement factor $\mathcal{A}$ over the formation interval is a cumulative measure that depends on the full $\mathcal{C}(z)$ profile; for the same fiducial parameters it evaluates to $\mathcal{A} \sim 1.1$ – 1.2 . These numbers are *derived outputs* of the closure (Sec. 18.3), not free parameters of the appendix; they are not, however, independent of the closure itself. In particular the fiducial choice $(\alpha, K) = (1, 8)$ is fixed in the companion MOND paper and a different solution of the feedback ODE (different attractor branch, different coarse-graining prescription, different treatment of source saturation) would alter the quoted $\mathcal{C}(10)$, H_{eff}/H_0 , and $\mathcal{A}$ values proportionally. The benchmarks above

should therefore be read as “ $\mathcal{C}(z)$ at fiducial MOND-closure parameters,” not as parameter-free predictions of Appendix 18 in isolation.

Figure 12 shows the approximate $\mathcal{C}(z)$ profile and the cumulative time integrals for the fiducial parameters. Panel (a) confirms that $\mathcal{C} > 1$ at all $z > 0$, i.e. the internal clock was always faster in the past. Panel (b) shows the growing gap between the coordinate lookback time t_{lb} and the process time τ_{proc} ; by $z = 10$ the enhancement reaches $\approx 1.4\times$.

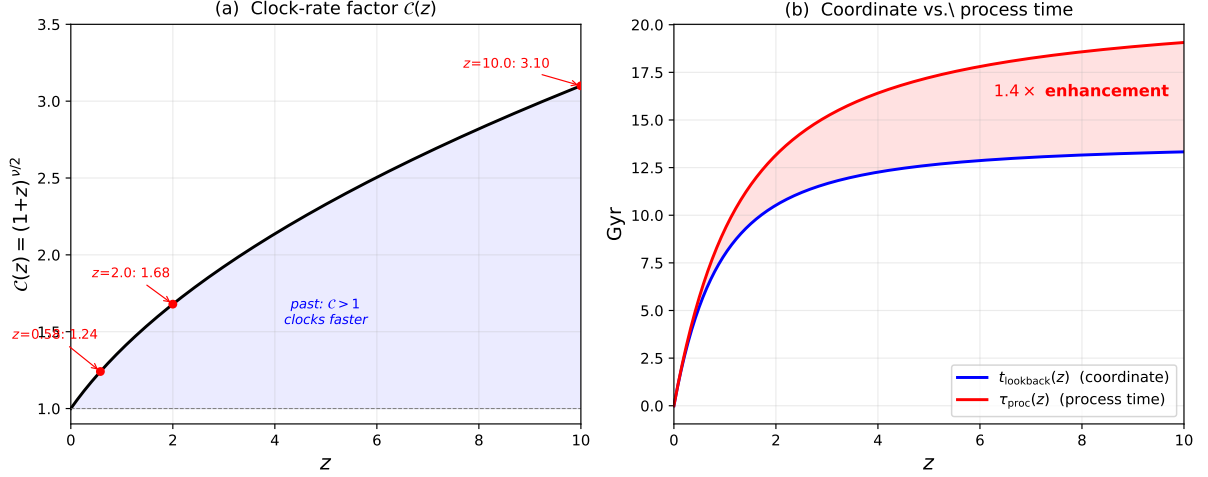


Figure 12: (a) Background history factor $\mathcal{C}(z) = (1+z)^{\nu/2}$ with $\nu = 0.94$, a power-law approximation calibrated to the companion MOND paper benchmark $\mathcal{C}(10) \approx 3.1$; the full $\mathcal{C}(z)$ from the coupling-ratio formula (Eq. 368) has a more complex shape at intermediate z . Red dots mark specific benchmarks. (b) Coordinate lookback time (blue) versus cumulative process time (red) from $z = 0$ to $z = 10$. The shaded region is the enhancement due to the faster internal clock in the past. Both panels use standard Λ CDM $H(z)$ with $\Omega_m = 0.315$, $H_0 = 67.4 \text{ km s}^{-1} \text{ Mpc}^{-1}$.

18.5 Rate conversion and the no-double-counting rule

The distinction between t_{lb} and τ_{proc} is not cosmetic. Whenever a phenomenon is governed by cumulative medium-internal evolution, the relevant history integral must be written in process time.

Intrinsic-rate formulation. If X is a cumulative quantity whose microscopic law is fundamental per unit process time,

$$\frac{dX}{d\tau_{\text{proc}}} = \mathcal{R}_{\text{int}}, \quad (387)$$

then the observer-frame rate is

$$\frac{dX}{dt} = \frac{d\tau_{\text{proc}}}{dt} \mathcal{R}_{\text{int}} = \mathcal{C}(z) \mathcal{R}_{\text{int}}. \quad (388)$$

Likewise the total accumulated history is

$$X(t_2) - X(t_1) = \int_{t_1}^{t_2} \mathcal{R}_{\text{int}} d\tau_{\text{proc}}, \quad (389)$$

not $\int \mathcal{R}_{\text{int}} dt$.

No-double-counting rule. Equation (388) must be used only when the underlying law is intrinsic to process time. If a source law has already been derived directly in the observer's time variable t , then an extra factor $d\tau_{\text{proc}}/dt$ must *not* be inserted again. Otherwise one counts the same clock conversion twice. Appendix 18 therefore supplies the conversion operator and the criterion for its legitimate use; it does not authorize blind multiplication of every observer-frame rate by $\mathcal{C}(z)$.

Practical criterion. The decision rule is: if the microscopic rate of a process depends on m_{eff} , $\mathcal{P}_{\text{tail}}$, or any other φ -dependent quantity that rescales with the local pressure state, then its cumulative history must be computed in process time. If the rate depends only on metric quantities that are fixed by the $\varphi_0 = 0$ gauge (scale factor $a(t)$, Hubble parameter H , gravitational potentials at linear order), then cosmic time is the correct variable.

Concretely, process time applies to:

- vortex formation and maturation (rate $\propto m_{\text{eff}}$);
- secular transport or memory buildup in coarse-grained medium sectors;
- cumulative tail accumulation ($\propto \mathcal{P}_{\text{tail}} \propto e^{2\varphi}$);
- any process whose microscopic rate is controlled by the pressure-state clock law.

By contrast, background FLRW distances, the linear CMB sector, and the Einstein–Boltzmann evolution depend only on $a(t)$ and $H(t)$, which are insensitive to φ_{bg} . These remain tied to the standard cosmic-time description of Appendix 9.

18.6 Asymptotic future and non-termination in finite time

The process-time framework naturally supports an asymptotic weakening of activity. Suppose the pressure-history factor $\mathcal{C}(t)$ remains nonnegative and decreases monotonically as the universe relaxes. Let a medium-controlled rate have the form

$$\mathcal{R}(t) = \mathcal{R}_0 F[\mathcal{C}(t)], \quad F(0) = 0, \quad (390)$$

with F continuous and nonnegative. Since $\mathcal{C}(t) = [P_{\text{stat}}(t)/P_{\text{stat}}(0)]^{1/2}$, $\mathcal{C} \rightarrow 0$ would require complete depletion $P_{\text{stat}} \rightarrow 0$. Three logical possibilities exist:

- if $\mathcal{C}(t) \rightarrow \mathcal{C}_\infty > 0$ (pressure relaxes to a nonzero floor), the process approaches a nonzero asymptotic residual activity;
- if $\mathcal{C}(t) \rightarrow 0$ only as $t \rightarrow \infty$ (complete depletion approached but never reached), then $\mathcal{R}(t) \rightarrow 0$ only asymptotically;
- a finite-time exact shutdown would require $\mathcal{C}(t_*) = 0$ at some finite t_* , which is not established anywhere in the present framework and would require complete exhaustion of the medium pressure in finite time.

Therefore the process-time formalism does *not* support the statement that the universe must reach an exact terminal zero state in finite measured time. What it supports, under monotonic

pressure relaxation, is a progressive weakening of clocks, effective masses, and tail-driven activity that continues indefinitely. The quiescent state—defined by the absence of localized resonant excitations (Eq. 361)—is an asymptotic attractor in this picture, not a state that is reached in finite coordinate time.

Entropy arrow from the feedback mechanism. The tail-emission channel produces entropy at the rate $\dot{S}_{\text{tot}} \propto N_b \mathcal{P}_{\text{tail}} > 0$ (companion quantum paper, Sec. 5). Since $\mathcal{P}_{\text{tail}} \propto \mathcal{C}(t)^4$ and $N_b \propto a^{-3}$, the entropy production rate decreases monotonically but never vanishes. Process time therefore accumulates in the same direction as entropy growth.

Scope of the “derived arrow” claim. The positivity $\dot{\epsilon} > 0$ —which is what drives $\dot{S} > 0$ and the phantom equation of state $w < -1$ —is itself a statement that the tail-emission channel is irreversible: radiated echo quanta propagate outward and do not refocus. This one-way propagation is an instance of the thermodynamic arrow, not a derivation of it. The content of this appendix is therefore the weaker and more accurate claim: *given* the irreversibility of tail emission (imported from the standard second law, as in any radiation-damping argument), the feedback mechanism aligns the process-time arrow with the entropy-production arrow and ties both to the medium-pressure depletion. No separate “arrow of time” postulate *beyond* that thermodynamic input is required, but the appendix does not claim to derive the thermodynamic arrow itself from first principles; a first-principles derivation would require a statistical-mechanical analysis of the tail phase space and is the topic of a separate work.

18.7 Internal–external time duality

The combination of the clock-rate postulate, the operationally timeless quiescent state (Sec. 18.1), and the asymptotic weakening of Sec. 18.6 produces a structural duality between two natural time measures.

Definitions. Let T_{ext} denote coordinate (FLRW cosmic) time and T_{int} denote cumulative process time $\tau_{\text{proc}} = \int \mathcal{C}(t) dt$. For the past and future half-lines:

$$T_{\text{ext, past}} = t_0 \simeq 14.7 \text{ Gyr} \quad (\text{finite}), \quad (391)$$

$$T_{\text{int, past}} : \text{operationally undefined (quiescent medium, no resonant clock)} \quad (\text{unbounded}), \quad (392)$$

$$T_{\text{ext, future}} = \int_{t_0}^{\infty} dt = \infty \quad (\text{de Sitter}), \quad (393)$$

$$T_{\text{int, future}} = \int_{t_0}^{\infty} \mathcal{C}(t) dt \quad (\text{finite if } \mathcal{C} \rightarrow 0 \text{ fast enough}). \quad (394)$$

The duality. To avoid conflating the mathematical value of the process-time integral with its operational status, we present the four entries with the internal column split into its “mathematical” and “operational” readings:

	External (coordinate)	Internal (math. integral)	Internal (operational)
Past	finite ($t_0 \simeq 14.7$ Gyr)	finite (≈ 20 Gyr) ^b	undefined*
Future	unbounded (∞)	finite [†] ($T_{\text{int, future}}$)	well-defined while $\mathcal{R}_{\text{tail}} > 0$

*“Undefined” means that no resonant clock exists before excitations form (Sec. 18.1): there is no physical process whose microscopic rate can register the passage of pre-resonance epoch.

^bThe pre-resonance mathematical integral $\int_0^{z_{\text{max}}} \mathcal{C} dz / [(1+z)H]$ is finite under the fiducial closure used below; the value ≈ 20 Gyr is given in the present-day normalization, $t_* = t_0$ (Sec. 18.3).

[†]Conditional: finiteness requires $\mathcal{C}(t) \rightarrow 0$ fast enough for the integral to converge; see the convergence condition below.

The diagonal structure is then between *operational* entries: the operational past is undefined while the operational future is well-defined, and vice versa for the mathematical integrals (both finite, one unconditionally, the other conditionally). This is a conjectured duality: it holds provided (i) the convergence condition on $T_{\text{int, future}}$ is satisfied (open in the current closure, see below) and (ii) the pre-resonance regime is truly clock-less at the microscopic level (accepted here as a working ontology). In particular, the coordinate age of the universe t_0 and the remaining operational process-time budget occupy dual positions under this interpretation.

The present epoch as the crossover. By construction, $\mathcal{C}(t_0) = 1$: internal and external clocks tick at the same rate *today*. In the past $\mathcal{C} > 1$ (faster internal clock); in the future $\mathcal{C} < 1$ (slower internal clock). The present epoch is the unique crossover point where the two time measures coincide. The identity $\mathcal{C}(t_0) = 1$ is, by itself, merely a normalization convention ($\varphi_{\text{bg}}(0) \equiv 0$). The non-trivial physical content is that the *same epoch* at which $\mathcal{C} = 1$ is also the epoch at which the feedback equilibrium $\dot{\varepsilon} = 3H\varepsilon$ is satisfied and $\Omega_m \sim \Omega_\Lambda$; this triple coincidence is a consequence of the self-consistent dynamics, not of the normalization choice.

Relation to the coincidence problem. The standard cosmological coincidence problem asks why $\Omega_m \sim \Omega_\Lambda$ precisely now. In the ISPG feedback framework the echo-background energy density ρ_{vac} is set by the self-consistent equilibrium $\dot{\varepsilon} = 3H\varepsilon$ (companion quantum paper, Eq. (6.14)). The same equilibrium condition forces $\mathcal{C} = 1$ at the epoch where the balance is achieved: matter–vacuum equality, the crossover of internal and external time, and the $\rho_{\text{vac}} \sim 10^{-121}$ echo-background energy density are three readings of the *same* self-consistency condition, not three logically independent coincidences.

Scope of this reduction. Calling these three statements “three readings of one condition” should be read in the sense of *consistency-check coupling*, not of *prediction from first principles*. Specifically:

- The identity $\mathcal{C}(t_0) = 1$ is, as noted above, a normalization convention ($\varphi_{\text{bg}}(0) \equiv 0$) rather than a derived value.
- The equilibrium condition $\dot{\varepsilon} = 3H\varepsilon$ is, by its definition, the locus of feedback saturation: its coincidence with $\mathcal{C} = 1$ therefore reflects the way the balance and the normalization are *set by the same feedback ODE*, not an independent prediction.

- The numerical value $\rho_{\text{vac}} \sim 10^{-121} M_{\text{Pl}}^4$ is an *output* of the feedback ODE at the fiducial closure parameters (α, K) fixed in the companion MOND paper; a tuning-free derivation of this scale is not performed here and remains an open quantitative target.
- The present-epoch character of $\Omega_m \sim \Omega_\Lambda$ therefore requires that the feedback attractor passes through $\mathcal{C} = 1$ at the observed value of H_0 ; showing that this selection is natural (i.e. independent of the initial conditions of the ODE) is likewise open.

Understood in this sense, the “single self-consistency condition” is a structural statement about the feedback picture, not a derivation of the observed dark-energy scale from first principles; a proof that the scale 10^{-121} is selected without tuning is the topic of a separate work.

Convergence condition. The finiteness of $T_{\text{int, future}}$ requires $\mathcal{C}(t) \rightarrow 0$ sufficiently fast. If $\mathcal{C}(t) \sim t^{-\alpha}$ at late times, the integral converges for $\alpha > 1$. The dark-energy equation of state $w < -1$ (source-driven phantom) reflects $\dot{\epsilon} > 0$ at all epochs (the echo background never stops growing), so φ_{bg} continues to decrease and $\mathcal{C}$ continues to fall even in the de Sitter phase. Whether the late-time falloff is fast enough for convergence depends on the asymptotic solution of the feedback ODE beyond the de Sitter saturation regime; this calculation remains open and is identified as a quantitative target for future work.

Active-phase budget. Even if $\mathcal{C}$ freezes at a nonzero floor $\mathcal{C}_\infty$ (making the total integral diverge), the *active* portion of the future process-time budget—the part during which $\mathcal{C}$ is still changing appreciably—is finite and computable. Define

$$\tau_{\text{active}} := \int_{t_0}^{\infty} [\mathcal{C}(t) - \mathcal{C}_\infty] dt. \quad (395)$$

In the de Sitter phase the baryon sources that drive the feedback are diluted as $n(t) \propto e^{-3Ht}$; because the emission rate $\dot{\epsilon} \propto n \mathcal{P}_{\text{tail}}$ shuts off on the timescale $1/(3H)$, the deviation $\mathcal{C}(t) - \mathcal{C}_\infty$ likewise decays as $e^{-3H(t-t_0)}$ at late times. During this transient $\mathcal{C}(t_0) = 1$, so process time $\approx$ coordinate time and

$$\tau_{\text{active}} \sim \int_0^{\infty} (1 - \mathcal{C}_\infty) e^{-3Ht'} dt' = \frac{1 - \mathcal{C}_\infty}{3H_0} \sim \frac{1}{3H_0} \approx 4\text{--}5 \text{ Gyr}. \quad (396)$$

The full transition from $\mathcal{C} = 1$ to the asymptotic floor occupies a few e -folding times, so the total active-phase duration is of order $H_0^{-1} \simeq t_0 \simeq 14.5$ Gyr. After that, $\mathcal{C}$ is effectively frozen and no further pressure depletion occurs: coordinate time continues but process time accumulates at a constant, residual rate.

The coordinate age t_0 and the active-phase process-time budget are therefore set by the same physical scale H^{-1} , confirming the anti-diagonal structure of the duality table on purely dynamical grounds.

Physical reading. If the duality is exact, the universe has no absolute beginning and no absolute end: it has a finite coordinate past that is an unbounded internal past (no operational clock before resonances), and an unbounded coordinate future that is a finite internal future (asymptotically vanishing activity). The “Big Bang” and the “heat death” are not terminal events but transitions between the two temporal regimes, and the present epoch is the symmetric midpoint where both descriptions agree.

Two-observer comparison. Every physical observable in the bi-conformal framework can be expressed in two equivalent languages—the coordinate frame (external observer, time variable t) and the local frame (internal observer, time variable τ). The two are related by the clock-rate postulate and its reciprocal rod postulate:

Quantity	External (coordinate)	Internal (local)
Time element	dt	$d\tau = e^{\varphi/2} dt$
Length element	$d\sigma$	$d\ell = e^{-\varphi/2} d\sigma$
Speed of light	$c_{\text{coord}} = c e^{\varphi}$	$c_{\text{local}} = c$
Frequency	ω	$\omega_{\text{local}} = \omega e^{-\varphi/2}$
Effective mass	$m_0 e^{\varphi/2}$	m_0 (locally undetectable)
Tail power	$\mathcal{P} \propto e^{2\varphi}$	$\mathcal{P}_{\text{local}} \propto 1$

The internal observer measures $c = \text{const}$, $m_0 = \text{const}$, $\mathcal{P} = \text{const}$: all local physics is φ -independent. The proof requires only the table entries above. The local speed of light is

$$c_{\text{local}} = \frac{d\ell}{d\tau} = \frac{e^{-\varphi/2} d\sigma}{e^{\varphi/2} dt} = \frac{d\sigma}{dt} e^{-\varphi} = c_{\text{coord}} e^{-\varphi} = c e^{\varphi} e^{-\varphi} = c. \quad (397)$$

The mass ratio of any two particle species A, B is

$$\frac{m_{\text{eff}}^A}{m_{\text{eff}}^B} = \frac{m_0^A e^{\varphi/2}}{m_0^B e^{\varphi/2}} = \frac{m_0^A}{m_0^B}, \quad (398)$$

and the frequency ratio of any two oscillators likewise: $\omega_{\text{local}}/\omega'_{\text{local}} = \omega/\omega'$. In summary, every dimensionless observable is φ -invariant:

$$\frac{c_{\text{local}}}{c} = 1, \quad \frac{m_{\text{eff}}^A}{m_{\text{eff}}^B} = \frac{m_0^A}{m_0^B}, \quad \frac{\omega_{\text{local}}}{\omega'_{\text{local}}} = \frac{\omega}{\omega'}. \quad (399)$$

No local experiment can detect a uniform shift of φ ; only comparisons across different epochs—where φ_{bg} has changed—reveal the pressure history. This is the local undetectability theorem (companion quantum paper, Sec. 4.3). The external observer sees all three quantities change with φ . Neither frame is privileged; they are connected by the conformal rescaling $e^{\varphi/2}$.

Applied to cosmological history ($\varphi \rightarrow \varphi_{\text{bg}}(z)$):

- **External view of the future:** internal clocks slow down ($d\tau/dt \rightarrow 0$), effective masses shrink, tail emission weakens. A finite amount of internal activity is stretched over infinite coordinate time.
- **Internal view of the future:** all local measurements remain “normal” ($c = c$, $m_0 = m_0$), but the universe empties (de Sitter dilution). The observer has infinite coordinate time ahead but with progressively less content per process-time unit.
- **External view of the past:** internal clocks were fast, masses were larger, emission was stronger. An unbounded amount of internal activity is compressed into the finite coordinate interval t_0 .

- **Internal view of the past:** all local measurements were “normal,” but the universe was denser, hotter, and richer in events per process-time unit. Looking back, the internal observer sees an ever-denser past with no operational beginning.

The two descriptions are physically equivalent: “time runs out” (external) and “content runs out” (internal) are dual translations of the single equation $d\tau = e^{\varphi/2} dt$.

18.8 Existing quantitative uses already present in the framework

Although the present appendix is the first place where the principle is collected systematically, the framework already uses it in concrete applications:

1. **Enhanced vortex formation time.** The companion MOND paper (Sec. 5.5) shows that the proper-time-weighted formation window exceeds the naive present-day clock estimate. For $z_{\text{form}} = 0.58$, the effective Hubble parameter is $H_{\text{eff}}/H_0 \approx 1.15$ (companion MOND paper, Eq. (5.15) and Table 1), corresponding to $\mathcal{A} \sim 1.15$ over the formation interval.
2. **JWST high-redshift galaxies.** The same logic partially mitigates the apparent tension with “impossibly old” high- z galaxies: in process time their inferred ages are reduced by the enhancement factor $\mathcal{A}(z)$, which evaluates to ~ 1.1 – 1.5 over the relevant formation window (companion MOND paper, Sec. 5.5). Whether this reduction is sufficient to fully absorb the observational excess, or whether a residual tension persists requiring additional contributions beyond the current $\mathcal{C}(z)$ benchmark, is the topic of a dedicated forthcoming analysis.
3. **Cluster-merger timing.** The proper-time integral is likewise used to show that each galaxy acquires its medium-memory halo well before the merger stage begins (companion MOND paper, Sec. 8).
4. **Coupling-enhancement ratio.** The companion MOND paper (Eq. (5.23)) derives the ratio $e^{2\varphi(z)}/e^{2\varphi(0)}$ from the quantum feedback ODE. This is precisely $\mathcal{C}(z)^4$, which shows that the process-time weight is not an independent input but inherits its redshift profile from the same feedback mechanism. The identification relies on reading the feedback-ODE φ —a microscopic oscillon-averaged quantity—as the coarse-grained cosmological φ_{bg} at a coarse-graining scale ℓ_{cg} that is large compared to individual oscillon sizes but small compared to cosmological gradients. Within this appendix we adopt this identification as a postulate of the coarse-graining scheme; a quantitative check that the ℓ_{cg} chosen in the MOND paper is compatible with the cosmological volume-averaging used in Eq. (374) is the topic of a separate work.

Status of the listed uses. The four applications above are quantitative *uses* of the process-time principle within the larger ISPG programme; they are not independently verified in Appendix 18. Their numerical robustness depends on the MOND-companion closure (items 1, 2, 4) and on the cluster-merger modelling (item 3), both of which are the topic of the companion papers. This appendix supplies only the clock-rate rule and the conversion operator; quantitative conclusions drawn from the table above should be read as “existing uses at the stated reference,” not as Appendix-18-internal results.

18.9 Relation to other approaches to the nature of time

The view that time is not a background but an operational or emergent quantity has precedents in several traditions. Barbour’s timeless mechanics [75] eliminates time altogether, treating it as an illusion arising from correlations between static configurations. Rovelli’s thermal time hypothesis [76] derives a time flow from the thermodynamic state of a quantum system via the Tomita–Takesaki theorem. Whitehead’s process philosophy identifies reality with “actual occasions” rather than enduring substances, making becoming prior to being [77]. Penrose’s conformal cyclic cosmology [78] exploits the conformal equivalence between the remote future (where all mass has vanished) and a new Big Bang.

The ISPG process-time framework shares conceptual ground with each of these: time is operational (Barbour), tied to thermodynamic state (Rovelli), defined by process (Whitehead), and exhibits a future–past structural symmetry (Penrose). The distinguishing feature is not that ISPG is the first to propose a quantitative time equation—Rovelli’s thermal time, for instance, gives a precise modular flow—but that ISPG ties its clock law $d\tau = e^{\varphi/2} dt$ to a single identified physical mechanism (pressure-dependent clock rate from the bi-conformal metric), from which the present appendix assembles a structured programme rather than a completed set of derivations: the duality table (Sec. 18.7, conditional on the convergence of $T_{\text{int, future}}$), the entropy-arrow alignment (understood as an alignment with the imported second law rather than a derivation of it), and the feedback-equilibrium reading of the coincidence problem (a structural reduction of three coincidences to a single self-consistency condition whose numerical output $\rho_{\text{vac}} \sim 10^{-121}$ depends on the closure parameters of the companion MOND paper). Each of these items is in a distinct stage of maturity, and a sharper identification of observational signatures that would distinguish the ISPG process-time picture from Rovelli-type thermal time or Penrose-type cyclic cosmology is itself an open target of the programme rather than a completed result of this appendix.

18.10 Takeaway

The ISPG framework contains a precise pressure-history time weighting that follows directly from the clock-rate postulate:

$$d\tau_{\text{proc}} = \mathcal{C}(z) dt. \quad (400)$$

This does not modify the homogeneous FLRW metric clock of Appendix 9: $\varphi_0 = 0$ constrains the metric, while $\varphi_{\text{bg}}(z)$ tracks the coarse-grained thermodynamic state of the medium. Whenever the past pressure was higher than the present one, the cumulative internal duration exceeds the naive lookback time measured by today’s clocks. This weighting must be inserted through the explicit rate-conversion rule, without double counting.

The framework further reveals an internal–external time duality (Sec. 18.7): the finite coordinate age t_0 and the finite remaining process-time budget $T_{\text{int, future}}$ occupy anti-diagonal positions in the past/future table, with the present epoch as the unique crossover where $\mathcal{C} = 1$. If the convergence of $T_{\text{int, future}}$ is confirmed, the cosmological constant problem, the coincidence problem, and the arrow of time become three aspects of one self-consistency condition rather than three independent puzzles.

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